View A

Section Plate

CoDa-Standard reading · bars + plan-view scatter · magnitude index

§ DEU Germany

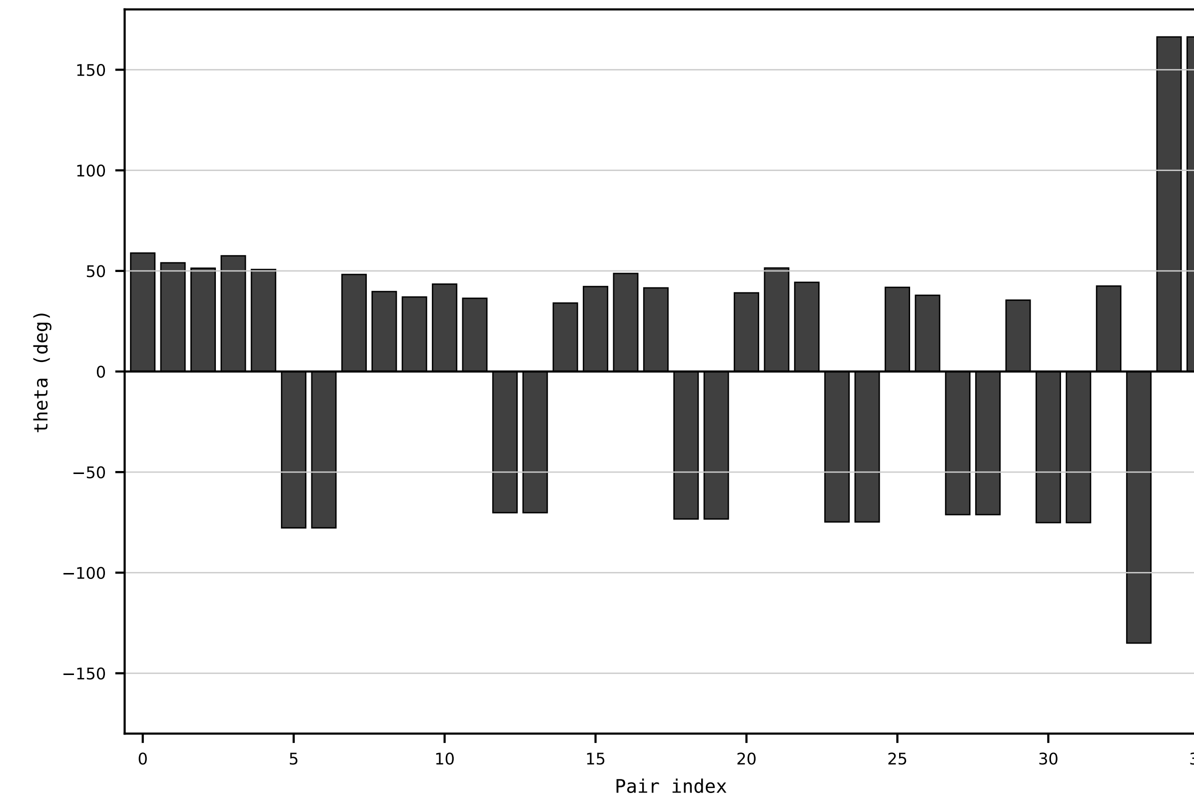
SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2000
t       : 0 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.430719
Ring    = Hs-3
E_metric= 1575.1016
kappa_HS= 296680000000000000.00
omega   = 0.0000 deg
d_A     = 0.000000
Helm    = ---
Helm d  = ---
DR      = 40.231 HLR
DR ratio= 296679999472192320.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

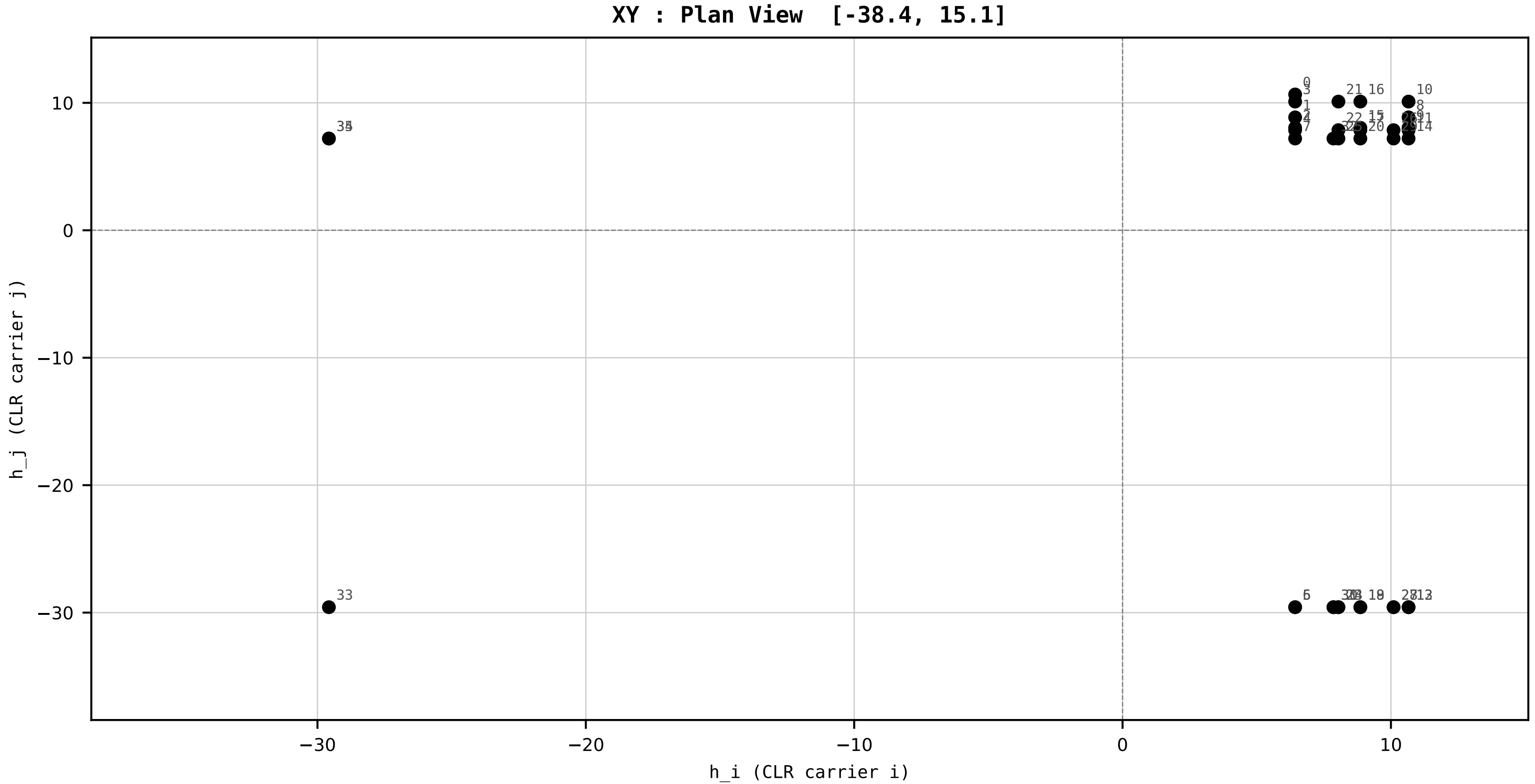
PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind

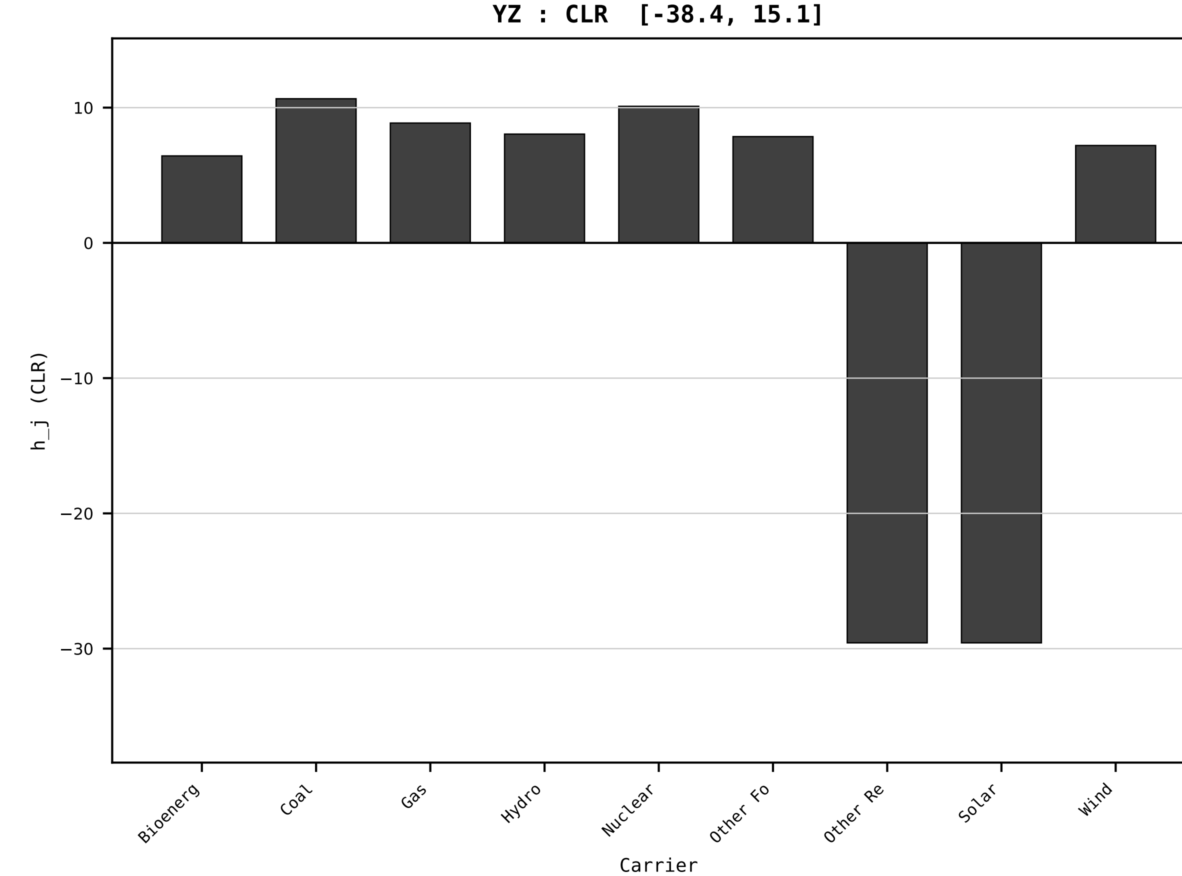
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-38.4, 15.1]



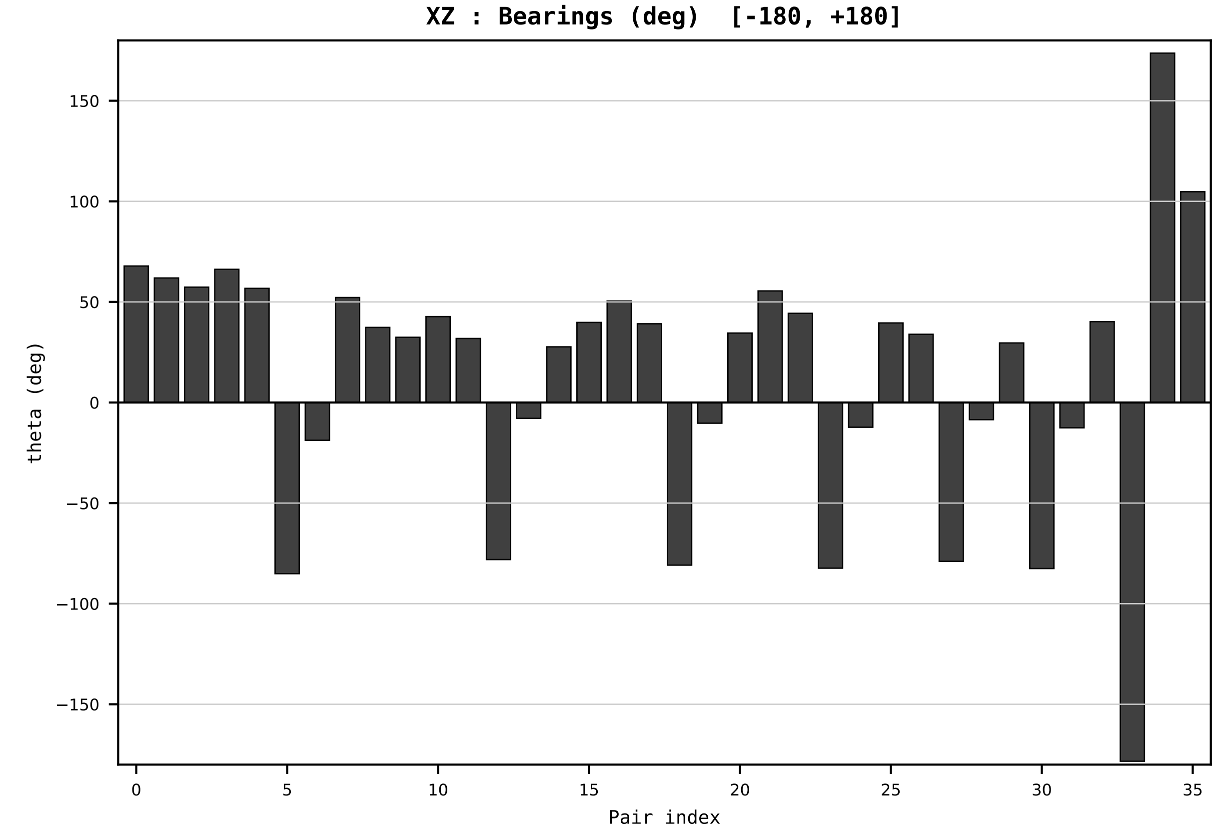
YZ : CLR [-38.4, 15.1]

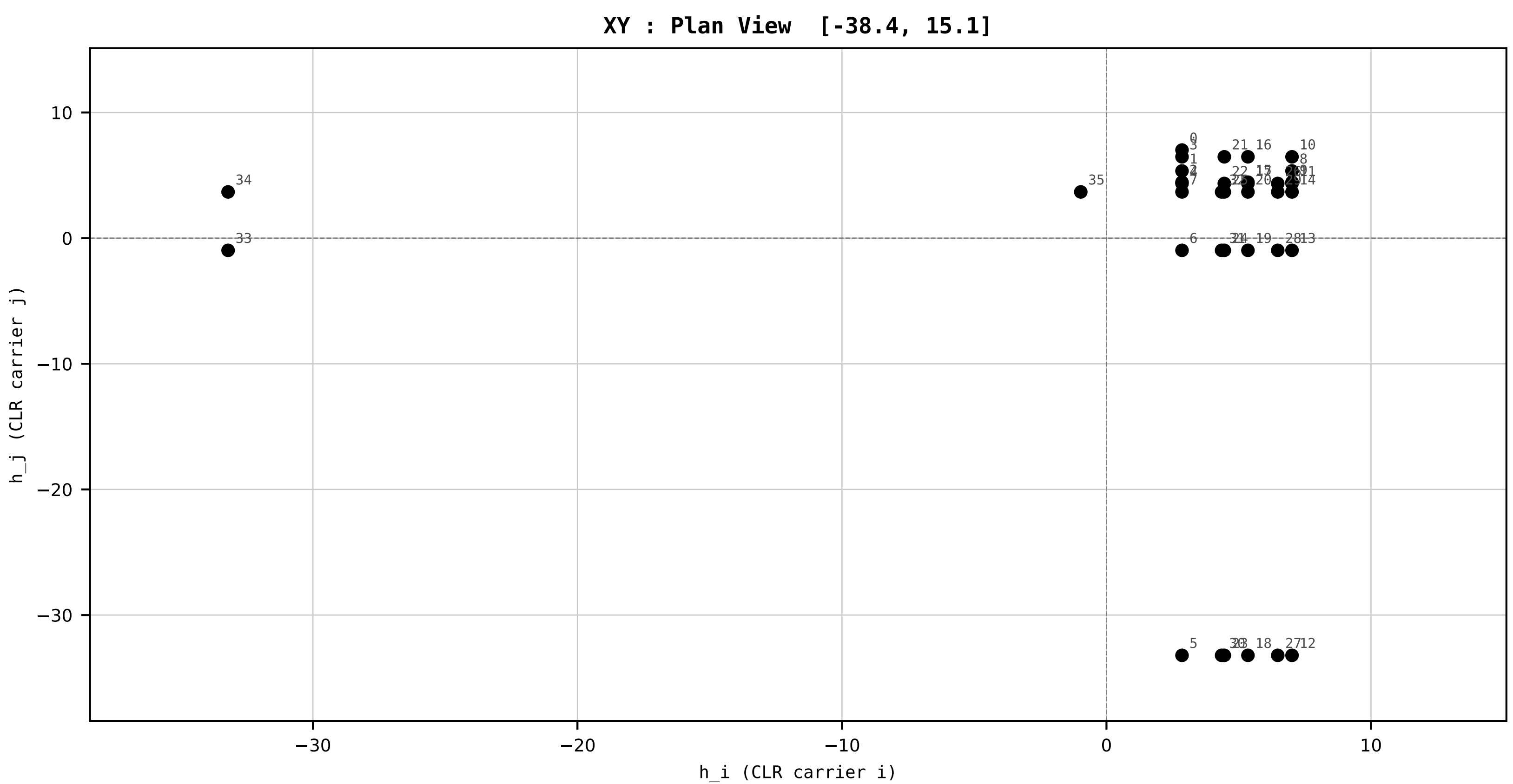


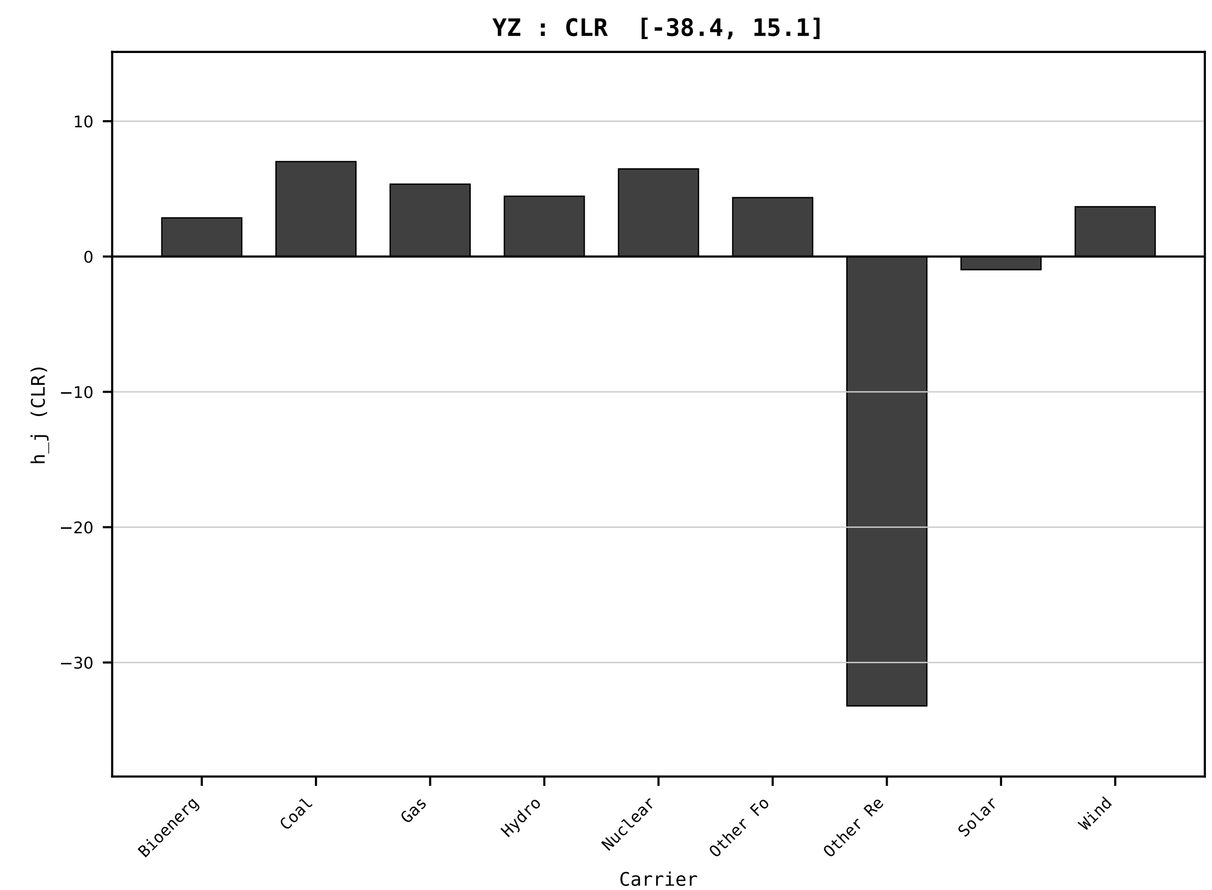
SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2001
t       : 1 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.414163
Ring    = Hs-3
E_metric= 897.3656
kappa_HS= 293740000000000000.00
omega   = 39.6073 deg
d_A     = 30.338234
Helm    = Solar
Helm d  = +28.602816
DR      = 40.221 HLR
DR ratio= 293740000811779456.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal
 1=Bioenergy-Gas
 2=Bioenergy-Hydro
 3=Bioenergy-Nuclear
 4=Bioenergy-Other Fossil
 5=Bioenergy-Other Renewables
 6=Bioenergy-Solar
 7=Bioenergy-Wind
 8=Coal-Gas
 9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind
```



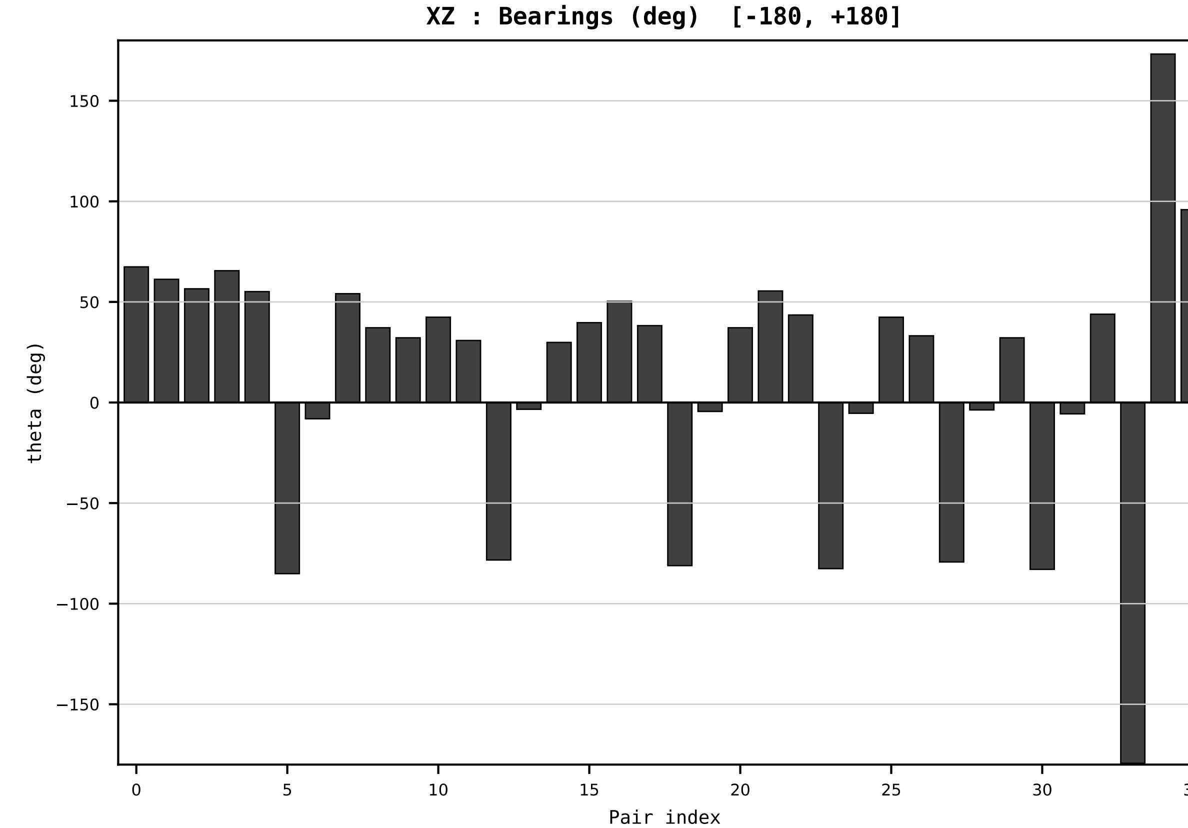


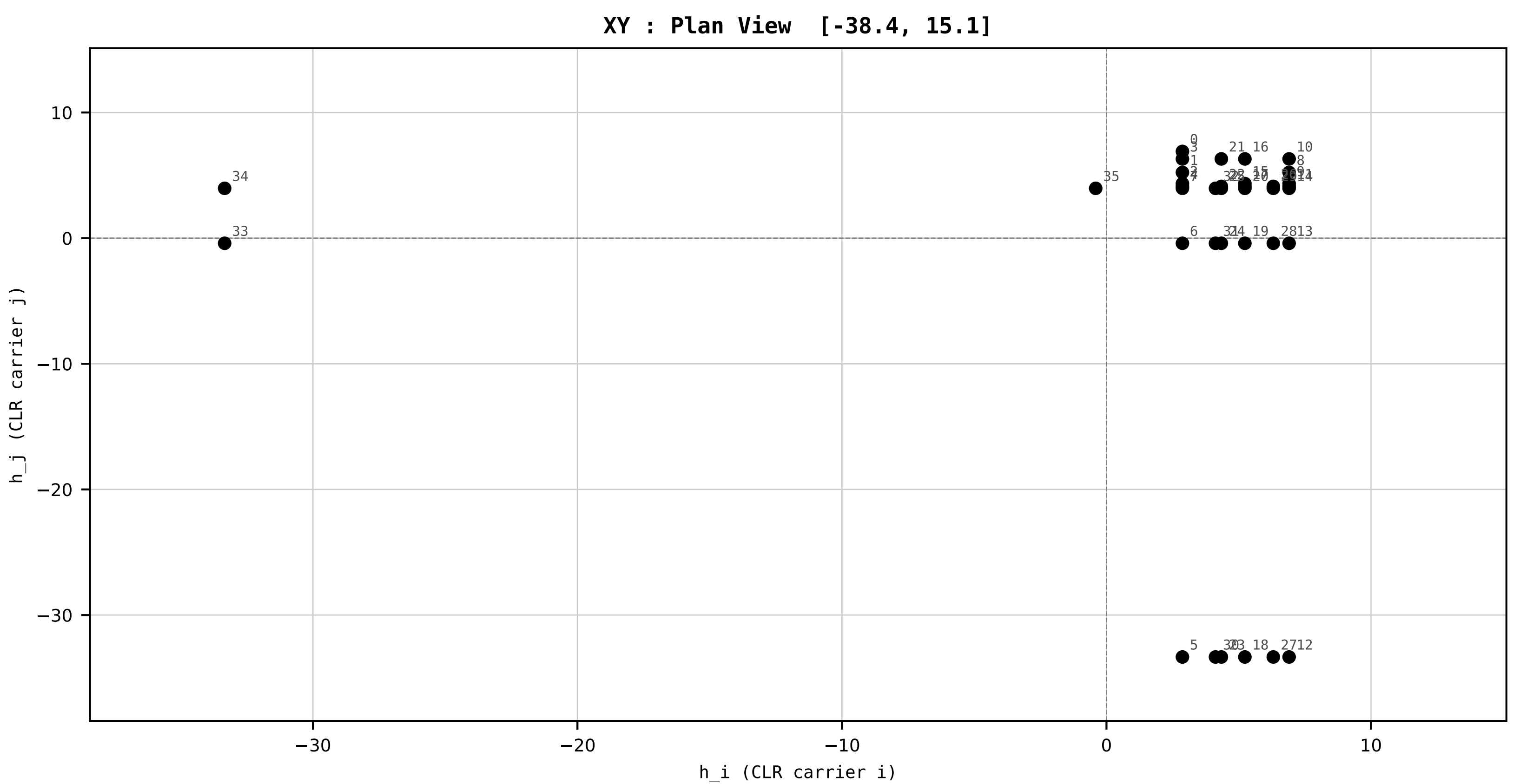


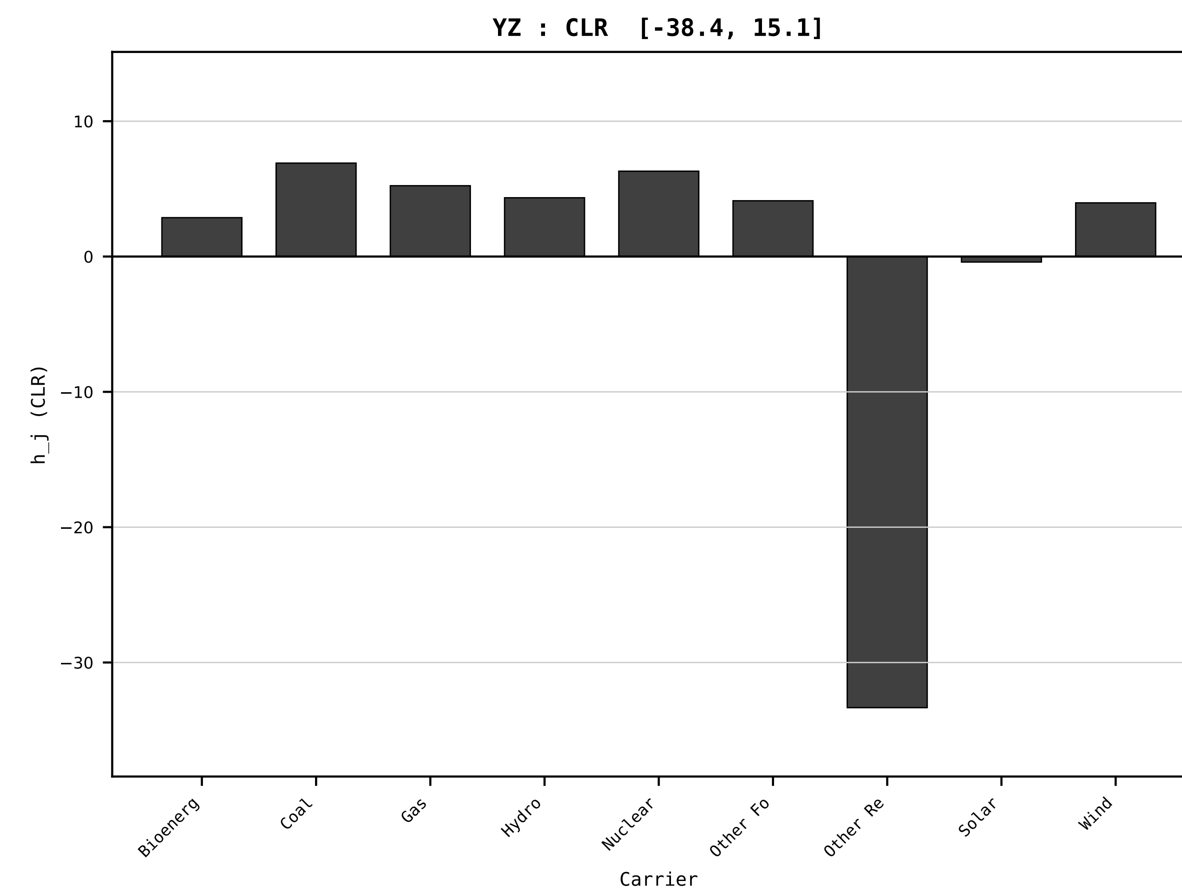
SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2002
t       : 2 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.405699
Ring    = Hs-3
E_metric= 897.5397
kappa_HS= 2996000000000000000.00
omega   = 1.1699 deg
d_A     = 0.732611
Helm    = Solar
Helm d  = +0.563622
DR      = 40.241 HLR
DR ratio= 2996000001032496448.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



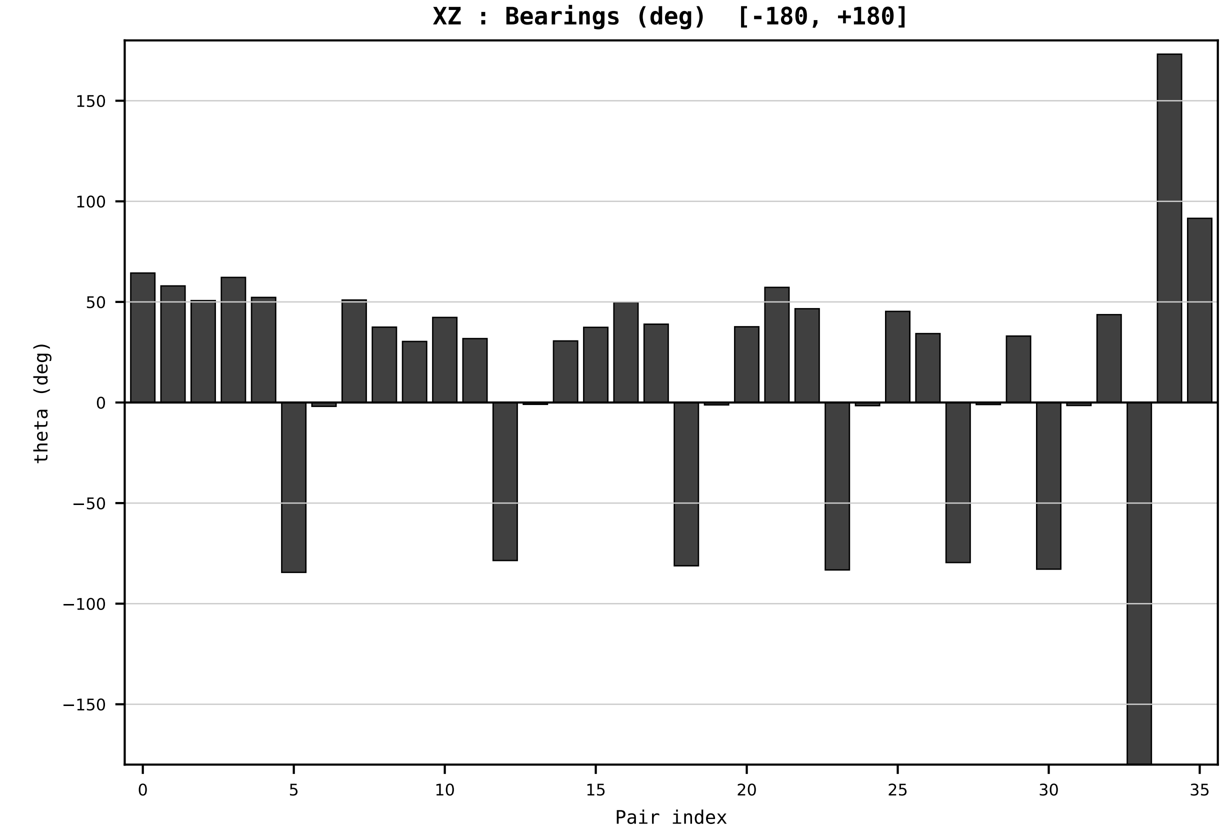


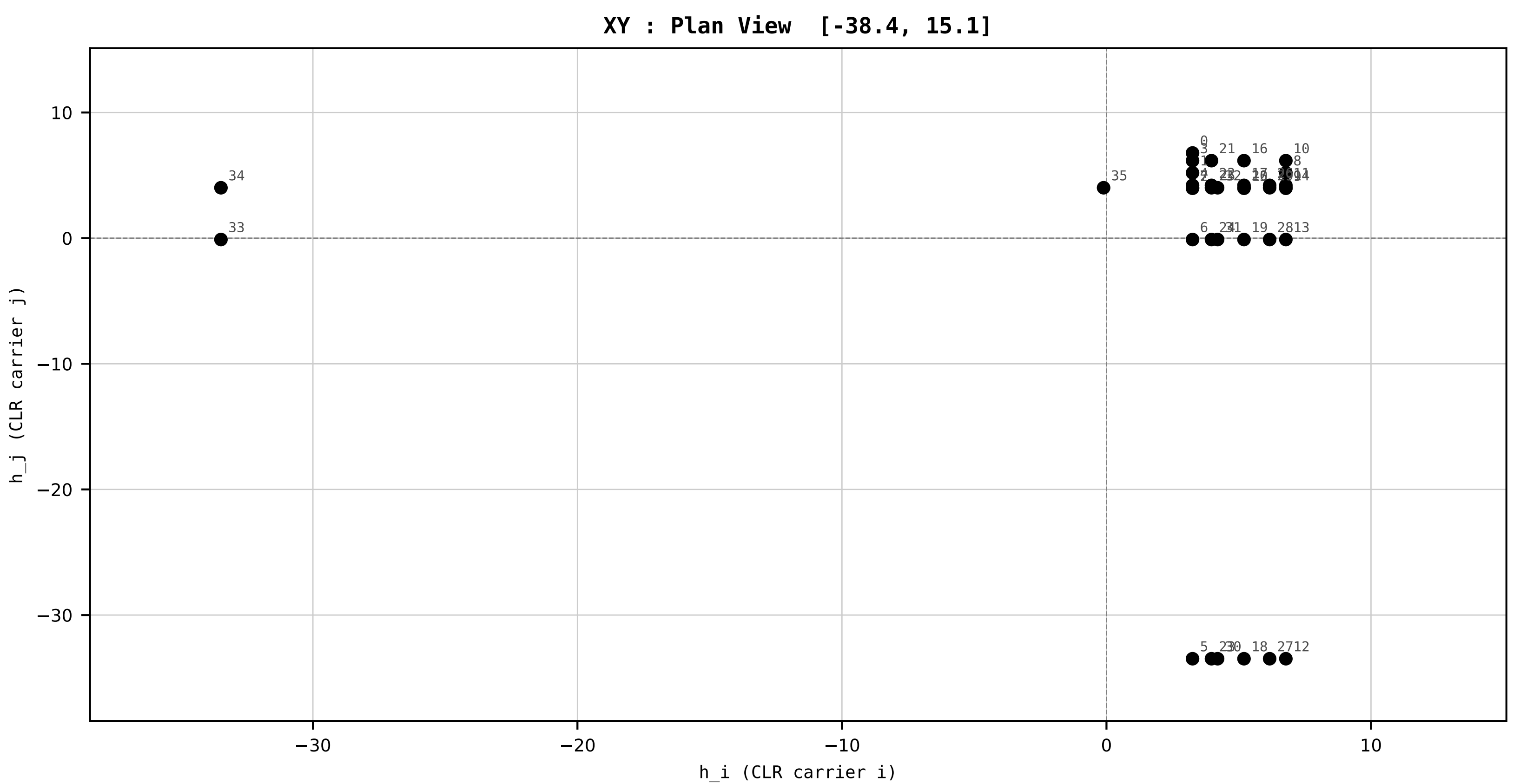


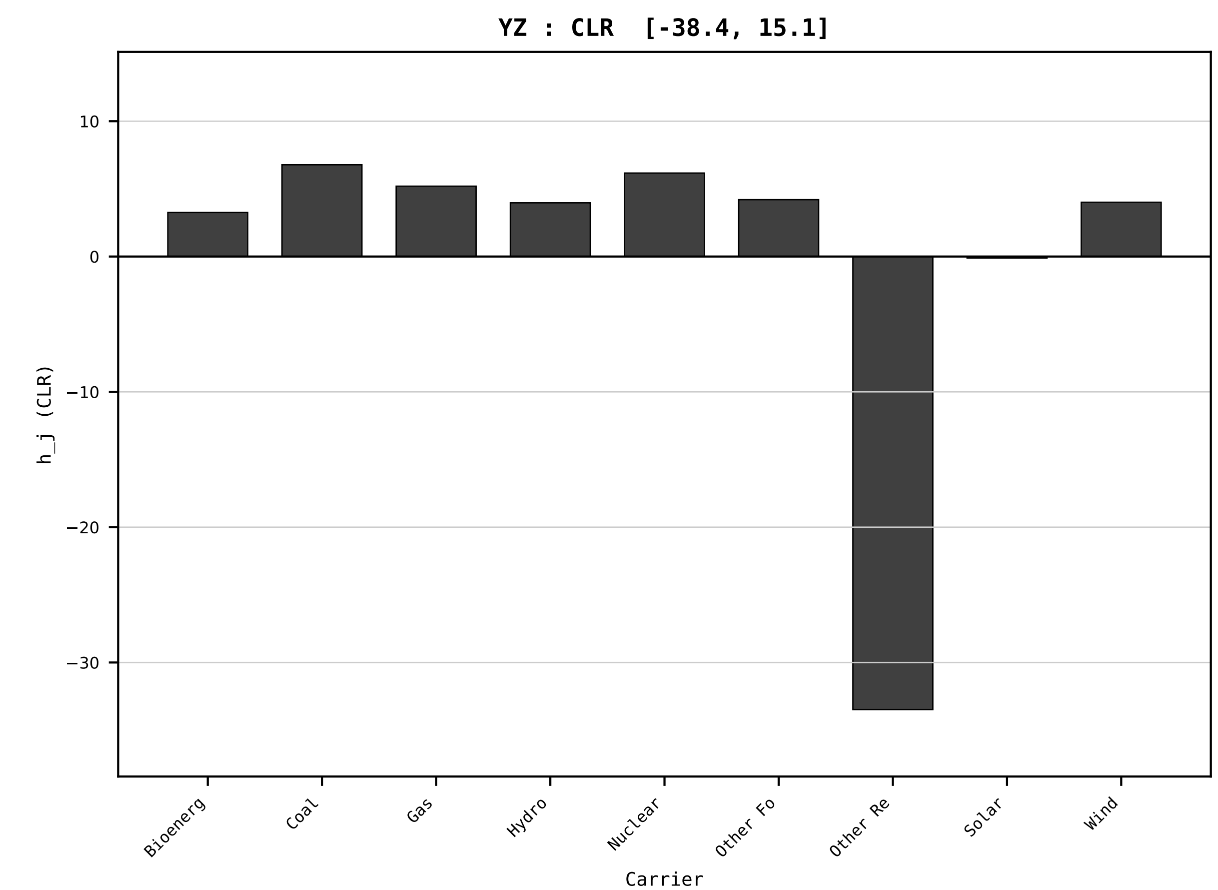
SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2003
t       : 3 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.389220
Ring    = Hs-3
E_metric= 899.7401
kappa_HS= 304629999999999936.00
omega   = 1.0479 deg
d_A     = 0.661385
Helm    = Bioenergy
Helm d  = +0.382457
DR      = 40.258 HLR
DR ratio= 304629999895005696.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



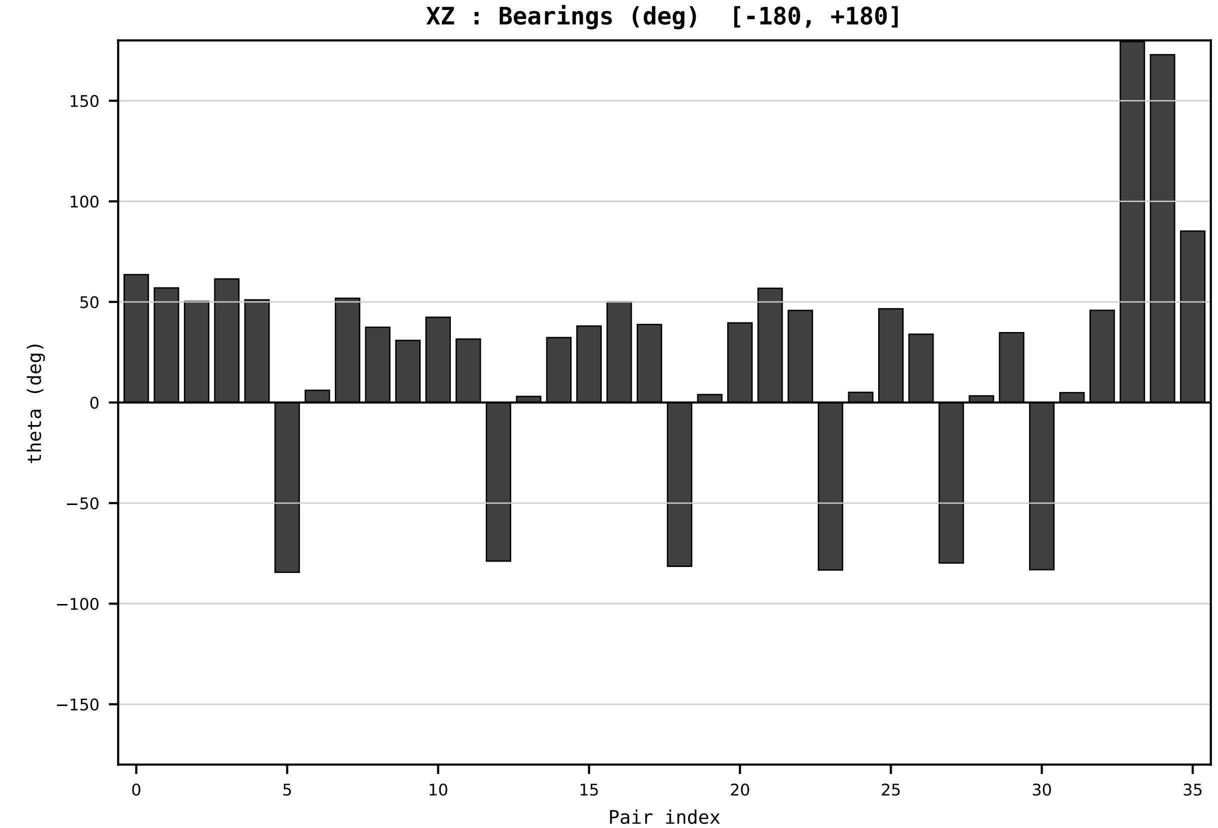


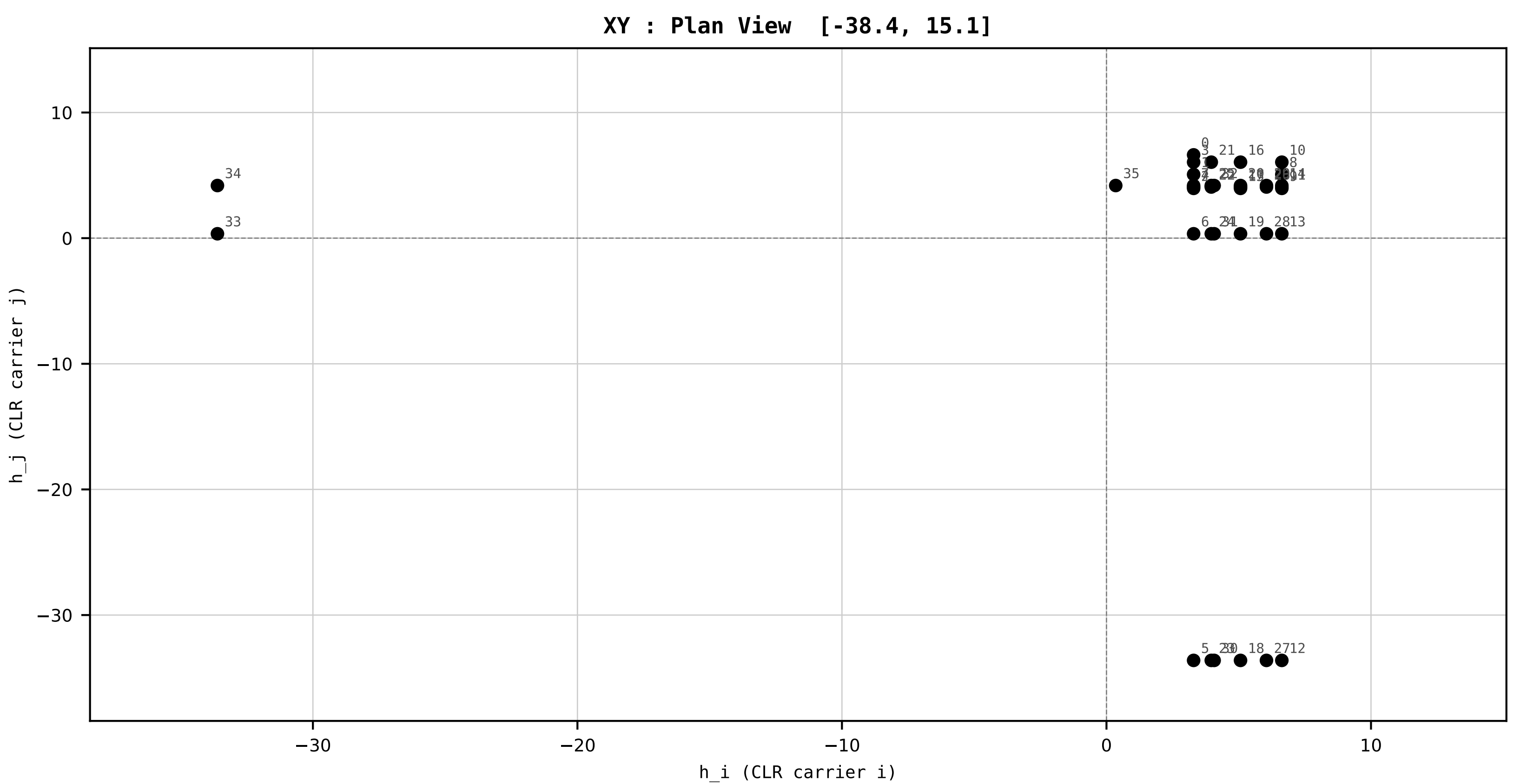


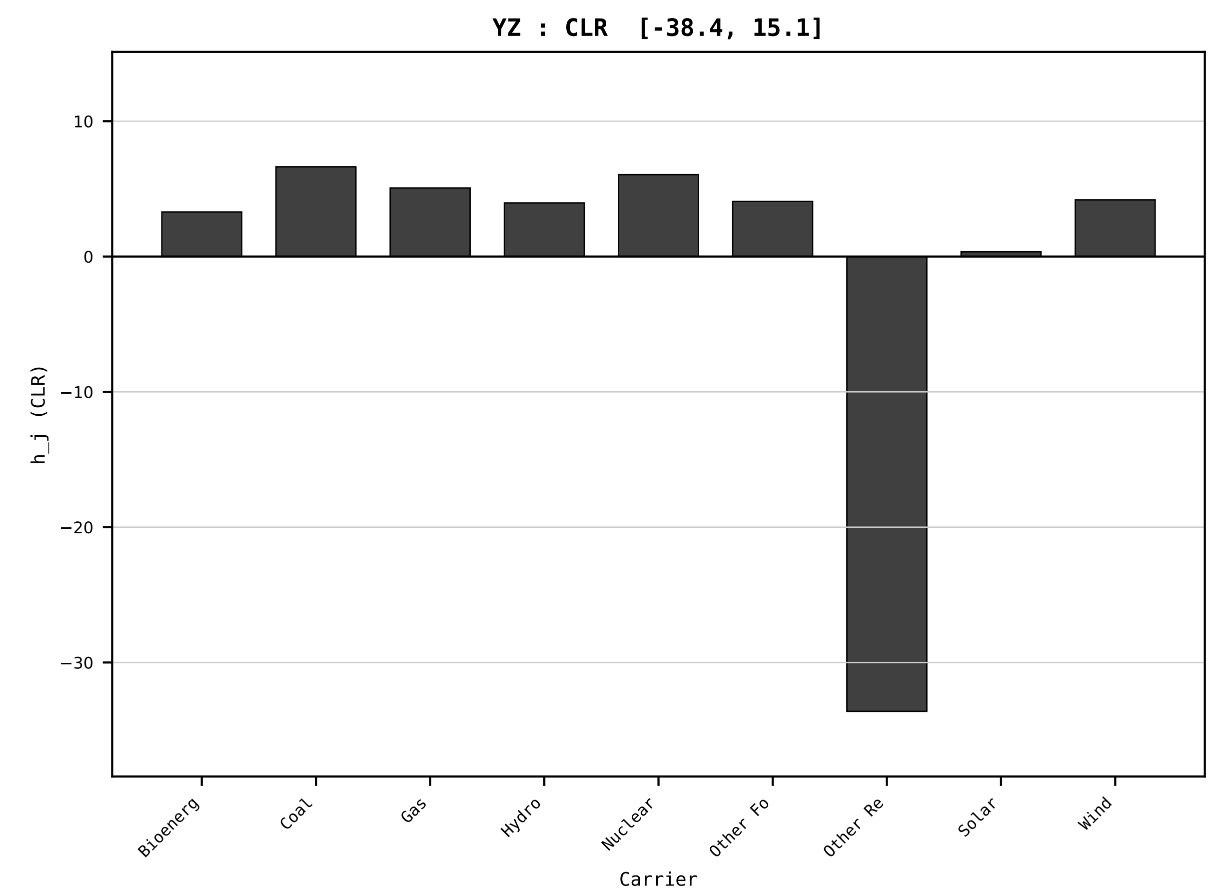
SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2004
t       : 4 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.367842
Ring    = Hs-3
E_metric= 902.0612
kappa_HS= 298769999999999936.00
omega   = 0.9133 deg
d_A     = 0.577156
Helm    = Solar
Helm d  = +0.458667
DR      = 40.238 HLR
DR ratio= 298769999545169920.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



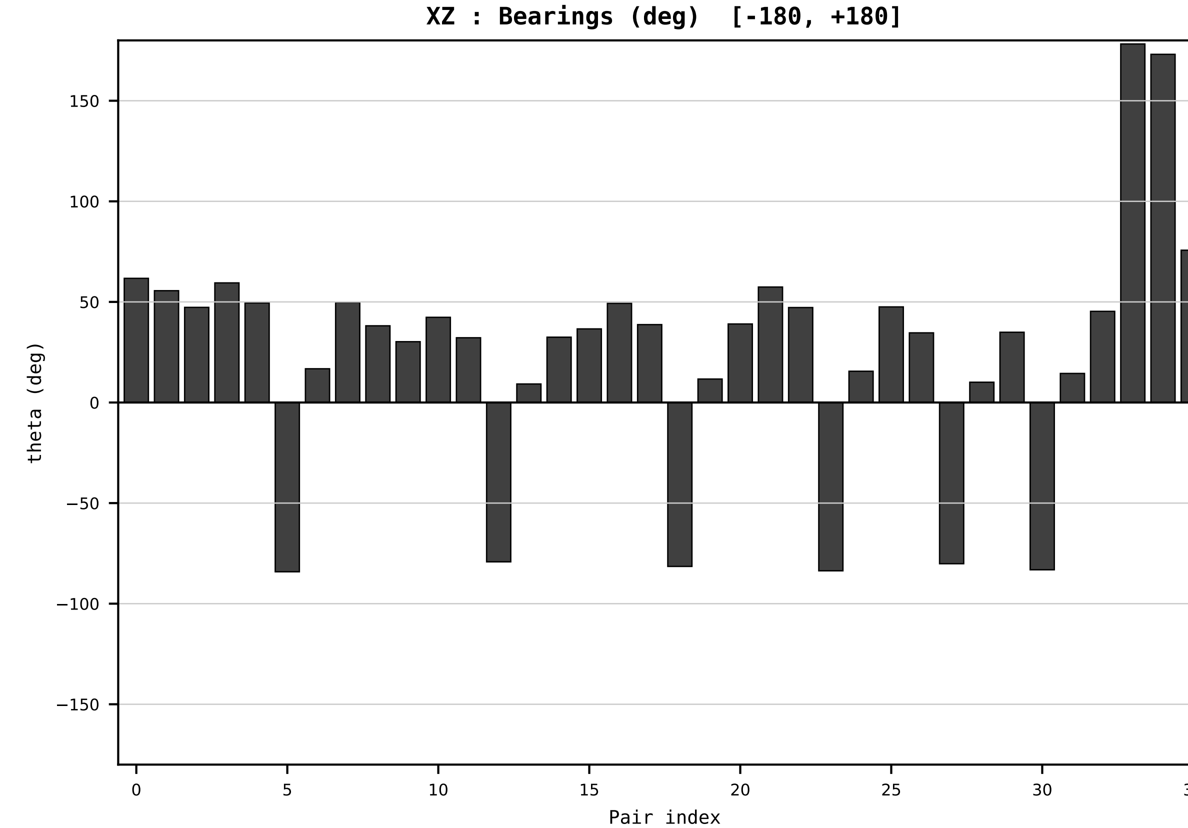


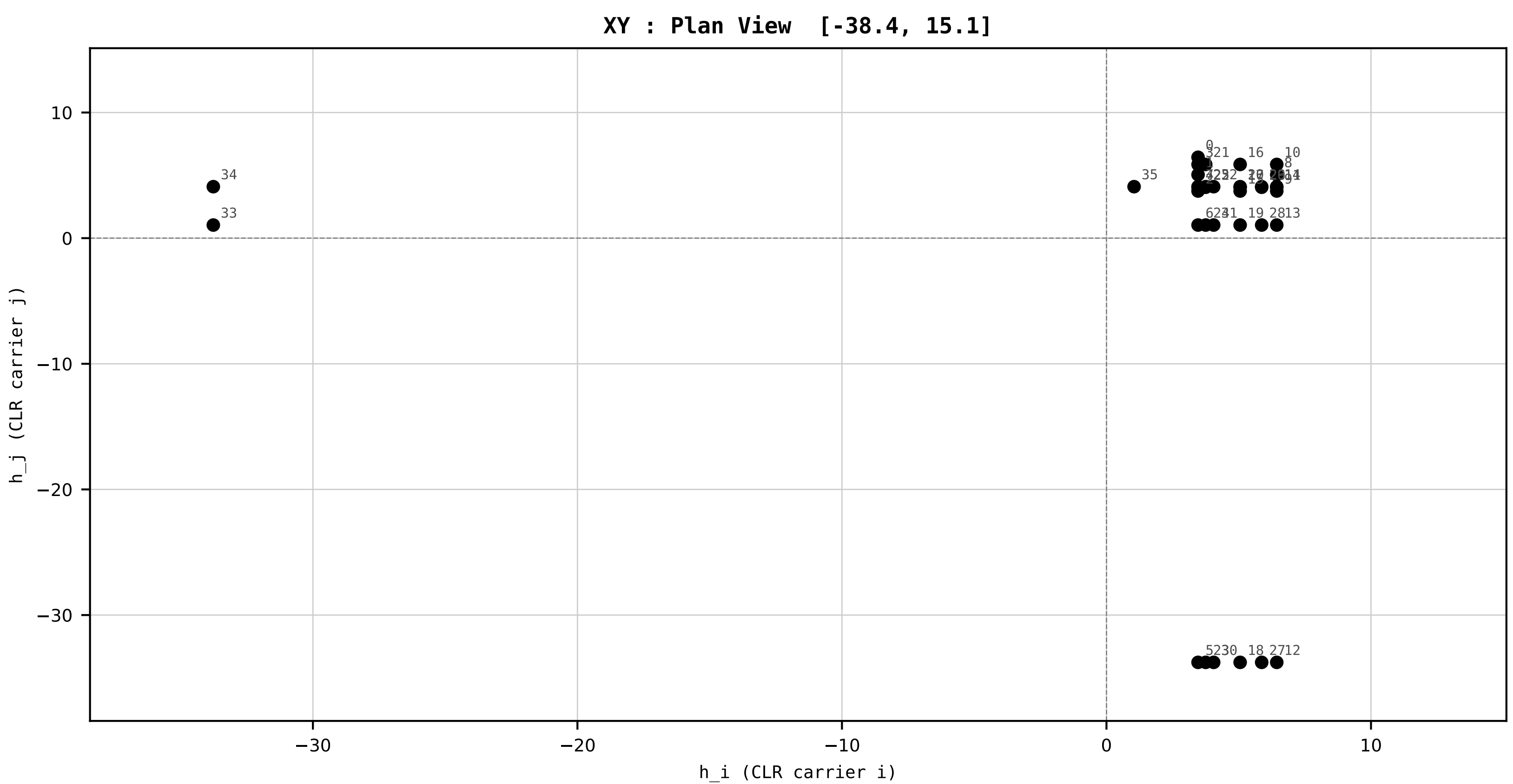


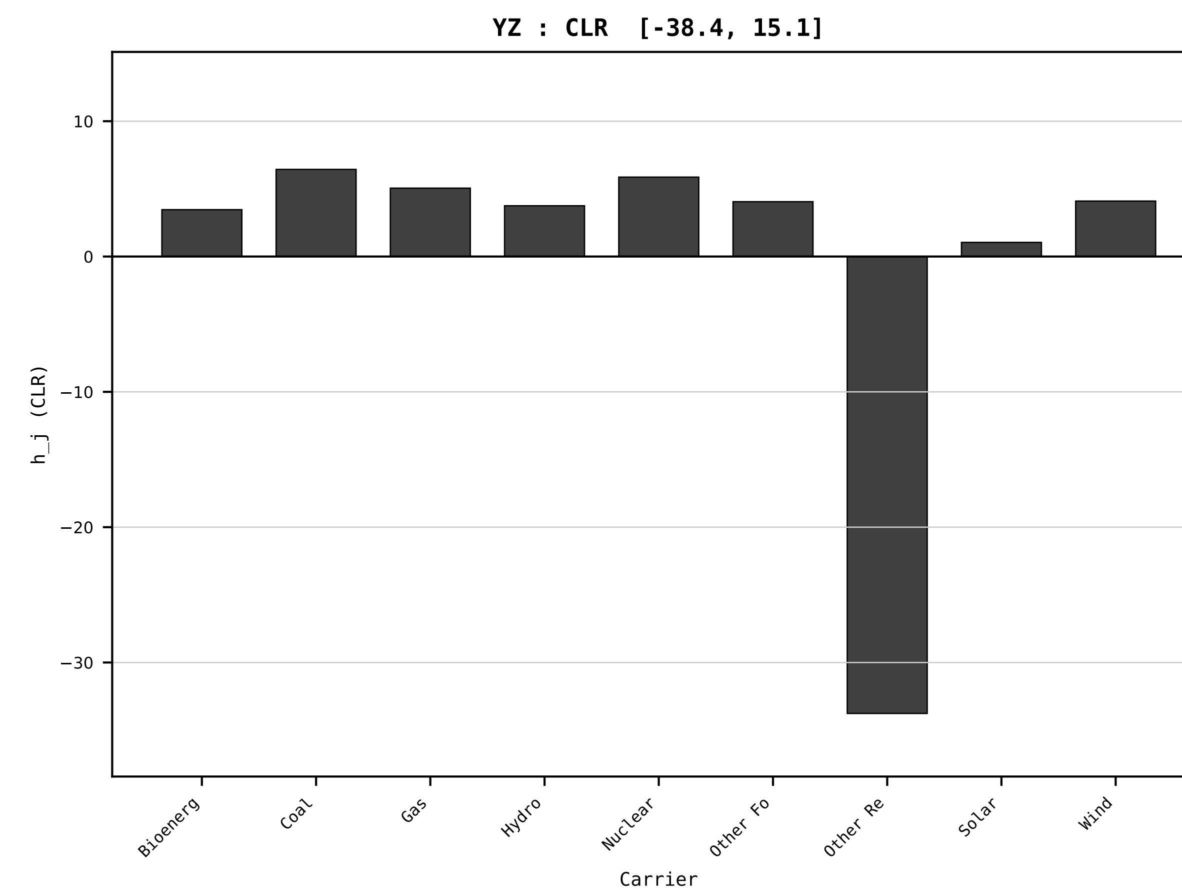
SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2005
t       : 5 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.341114
Ring    = Hs-3
E_metric= 904.9926
kappa_HS= 288139999999999968.00
omega   = 1.2830 deg
d_A     = 0.810406
Helm    = Solar
Helm d  = +0.694037
DR      = 40.202 HLR
DR ratio= 288140000563598304.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```





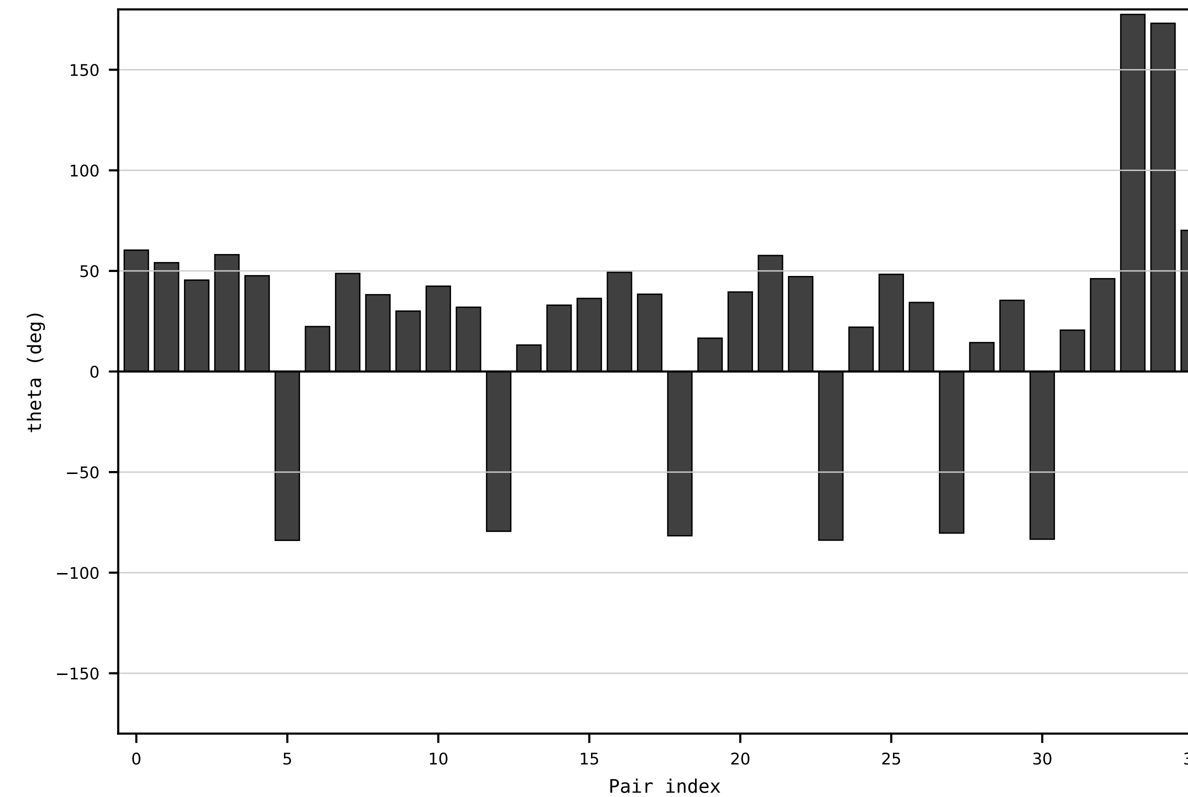


SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

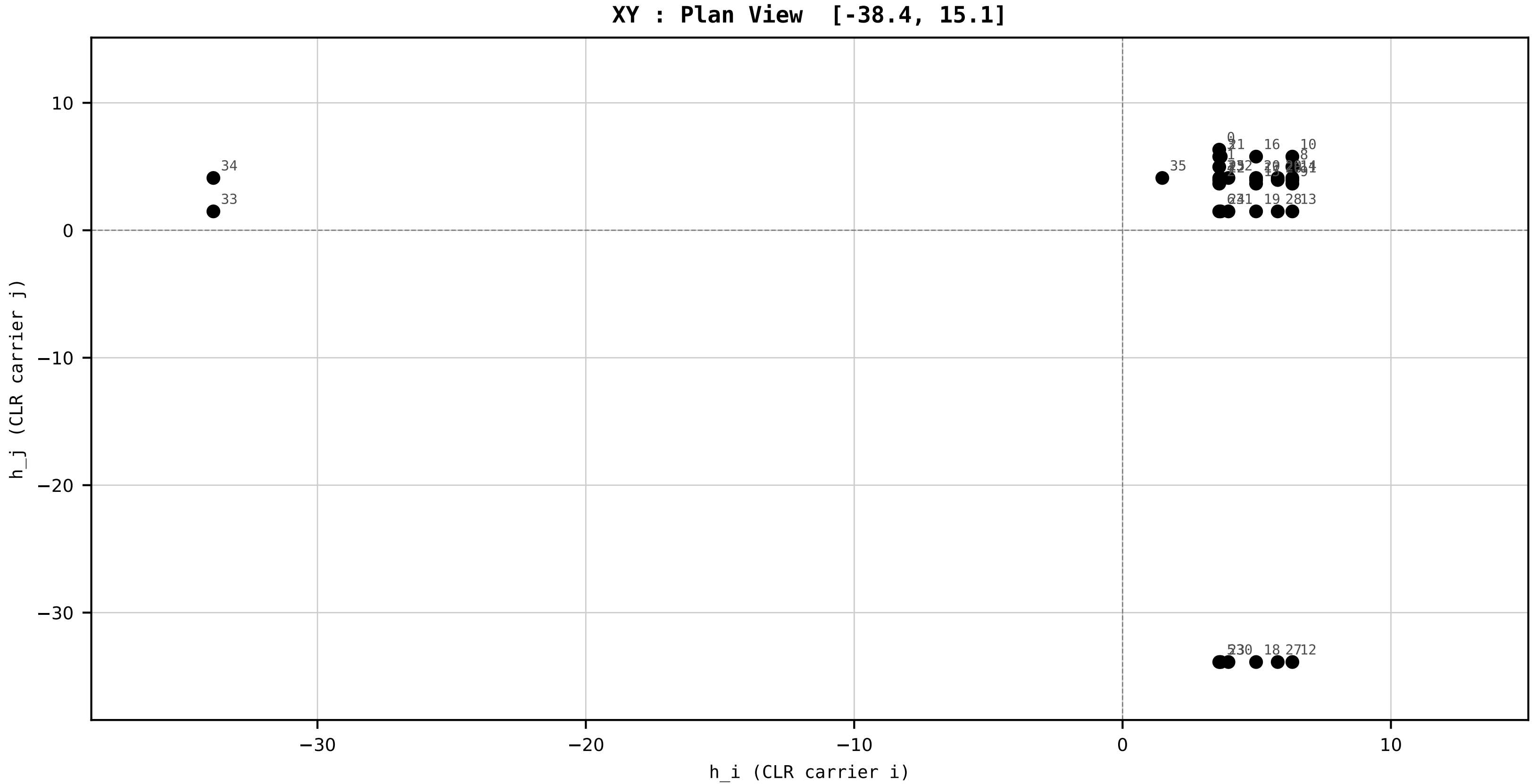
```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2006
t       : 6 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.326053
Ring    = Hs-3
E_metric= 907.0614
kappa_HS= 288930000000000000.00
omega   = 0.8154 deg
d_A     = 0.518705
Helm    = Solar
Helm d  = +0.437297
DR      = 40.205 HLR
DR ratio= 288930000245136960.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal        18=Gas-Other Renewables
 1=Bioenergy-Gas         19=Gas-Solar
 2=Bioenergy-Hydro       20=Gas-Wind
 3=Bioenergy-Nuclear     21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar       24=Hydro-Solar
 7=Bioenergy-Wind        25=Hydro-Wind
 8=Coal-Gas              26=Nuclear-Other Fossil
 9=Coal-Hydro            27=Nuclear-Other Renewables
10=Coal-Nuclear          28=Nuclear-Solar
11=Coal-Other Fossil     29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar            31=Other Fossil-Solar
14=Coal-Wind             32=Other Fossil-Wind
15=Gas-Hydro             33=Other Renewables-Solar
16=Gas-Nuclear           34=Other Renewables-Wind
17=Gas-Other Fossil      35=Solar-Wind
```

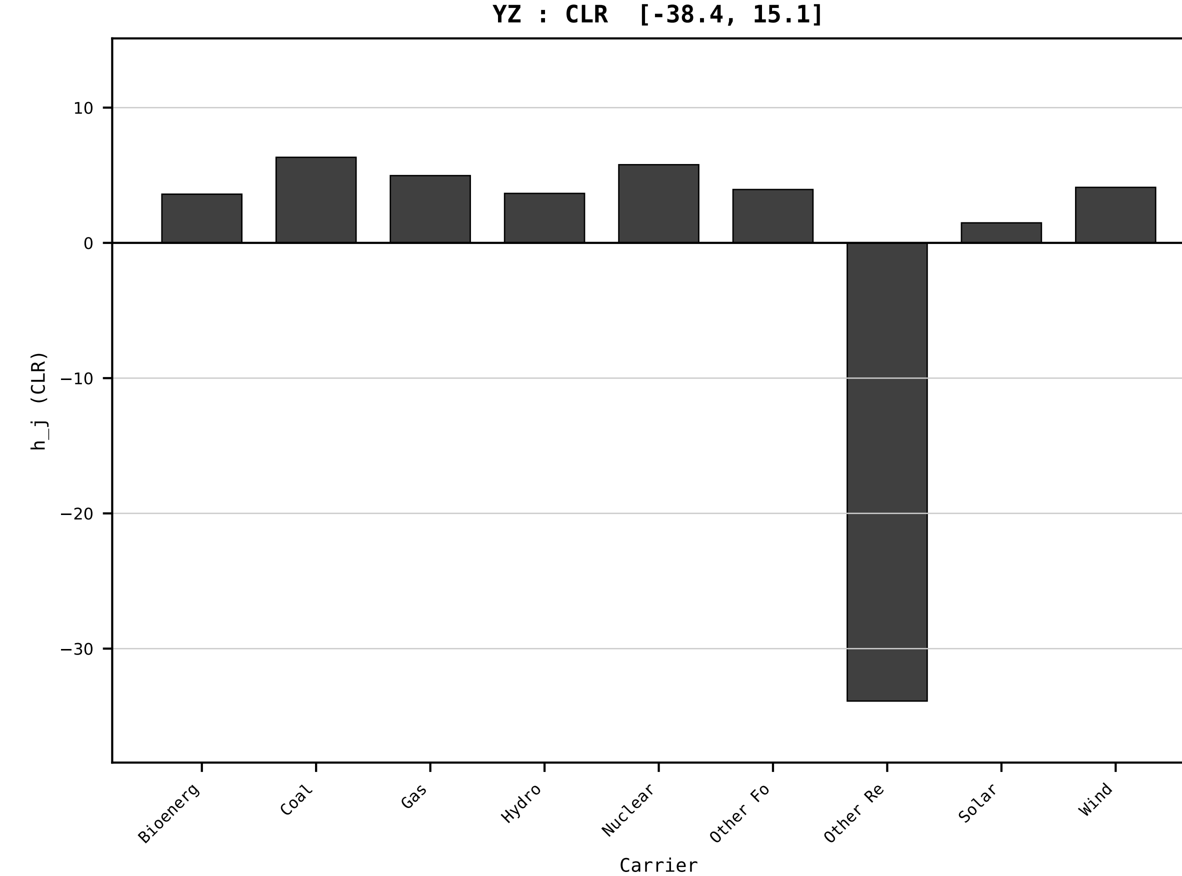
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-38.4, 15.1]



YZ : CLR [-38.4, 15.1]

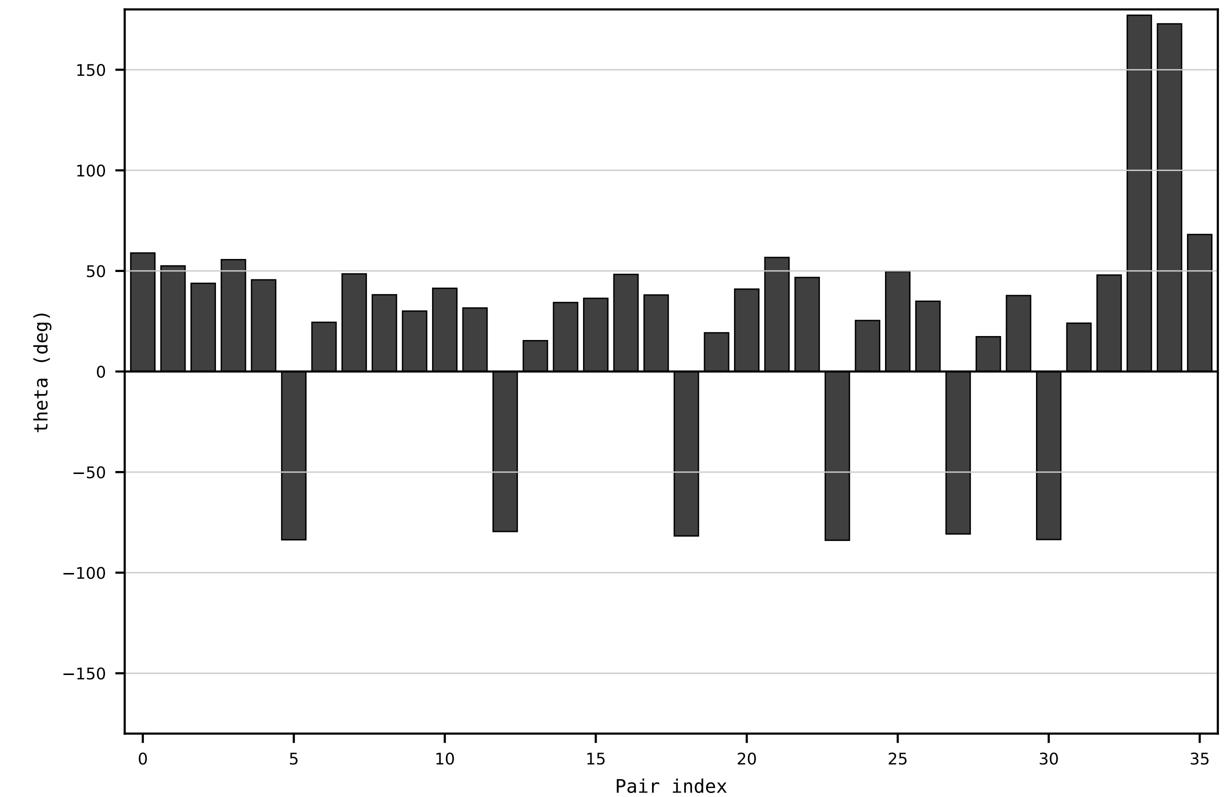


SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

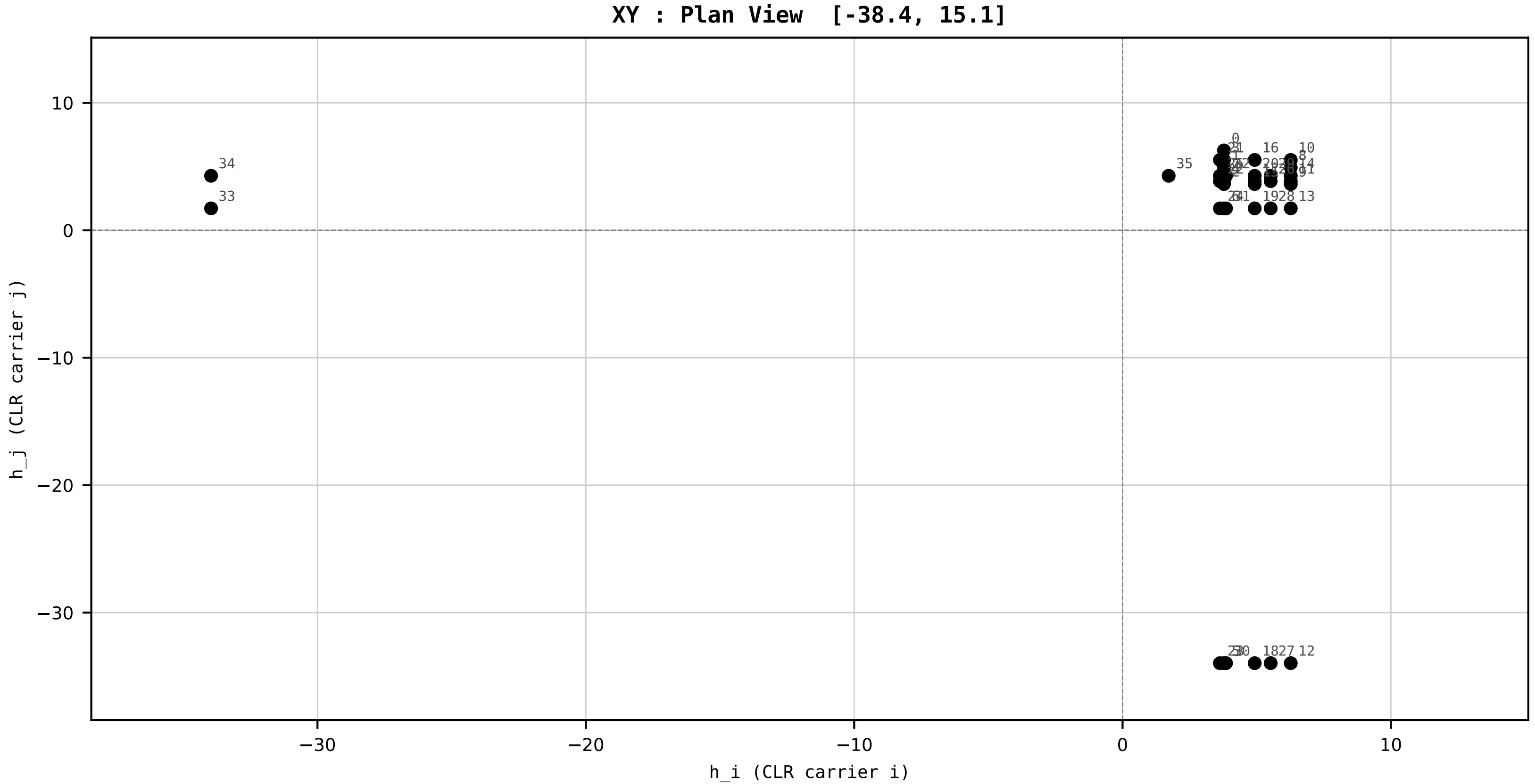
```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2007
t       : 7 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.306825
Ring    = Hs-3
E_metric= 910.0899
kappa_HS= 297100000000000000.00
omega   = 0.7153 deg
d_A     = 0.455473
Helm    = Nuclear
Helm d  = -0.261293
DR      = 40.233 HLR
DR ratio= 297099997807627008.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```

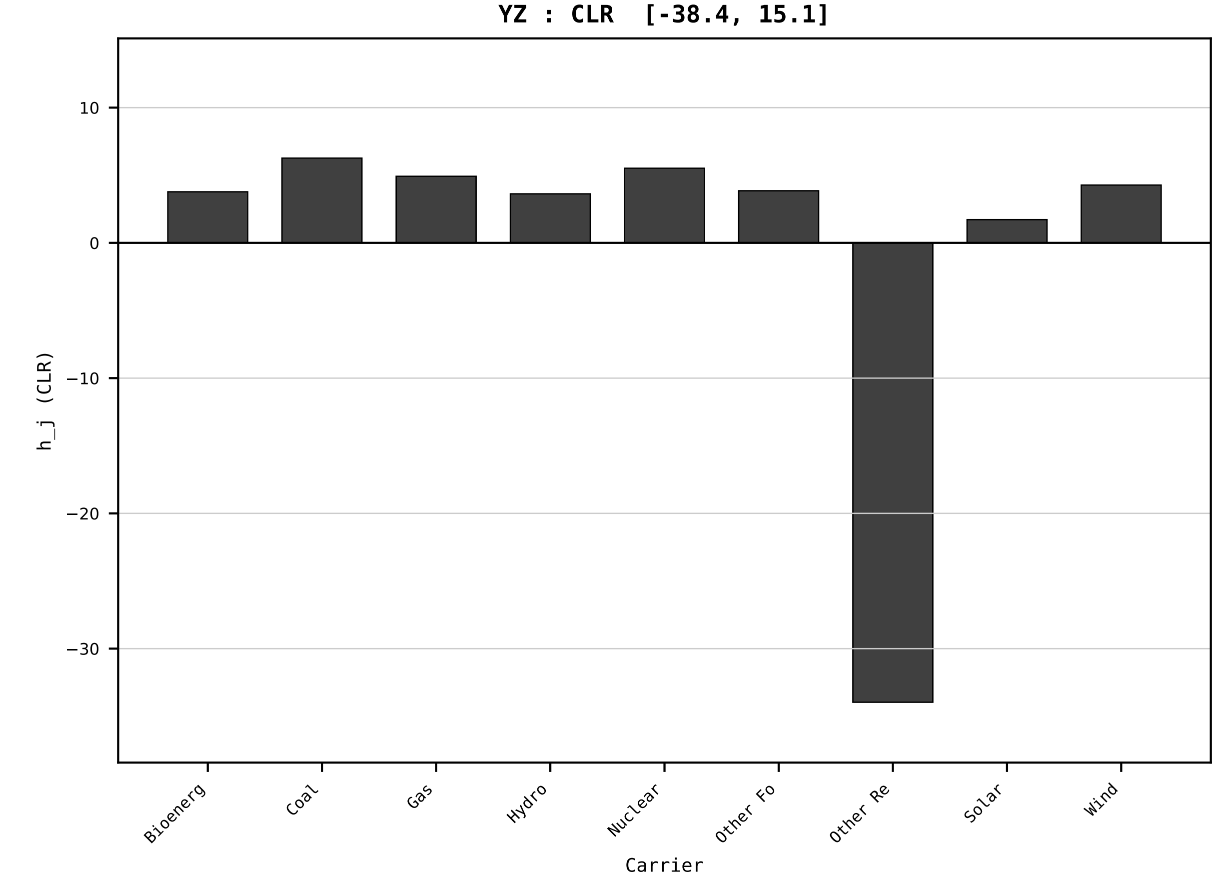
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-38.4, 15.1]



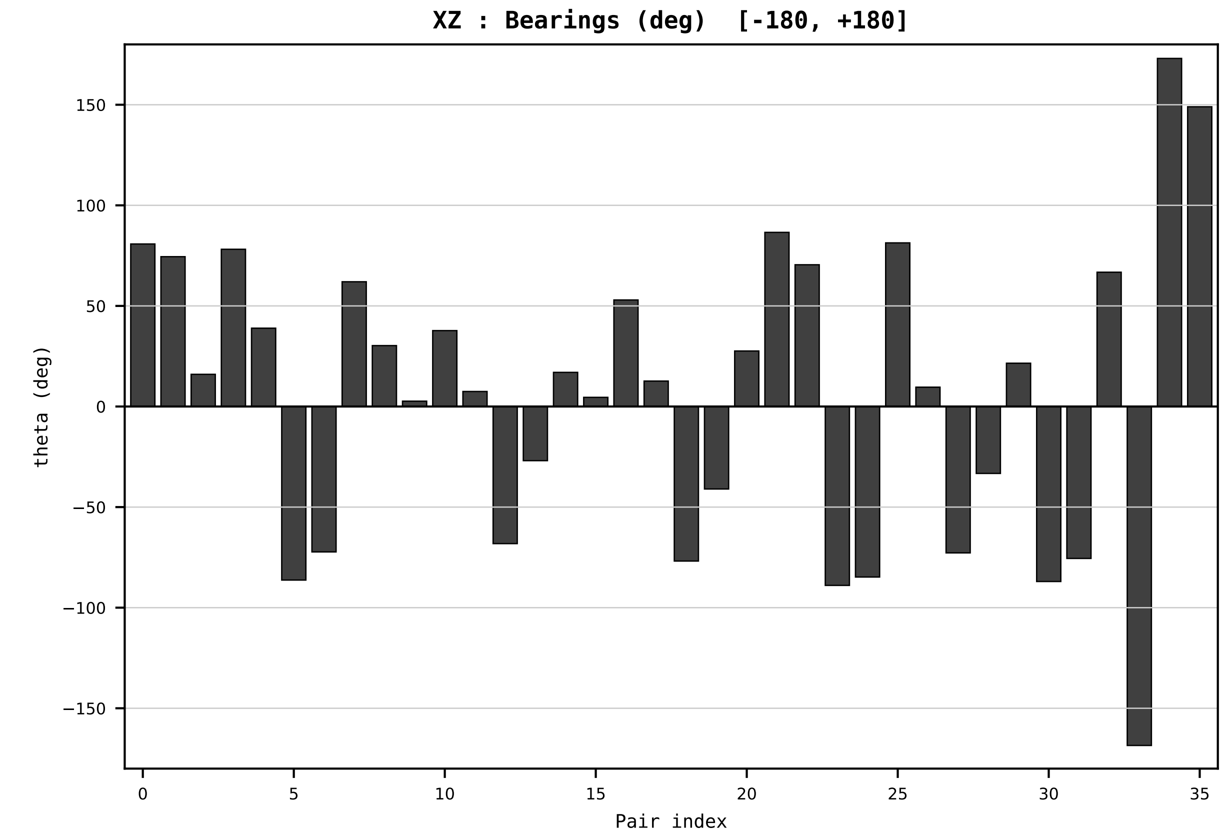
YZ : CLR [-38.4, 15.1]

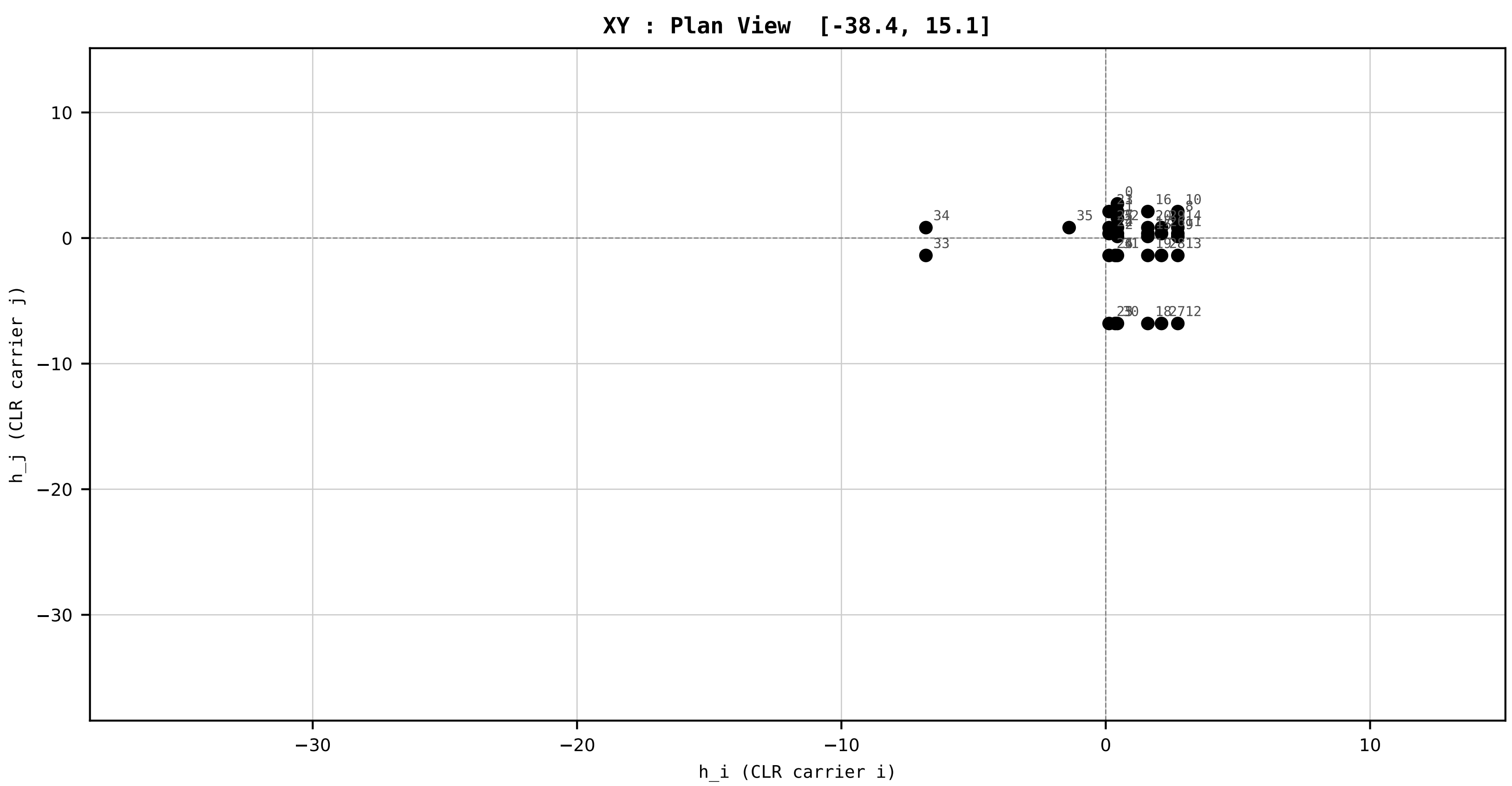


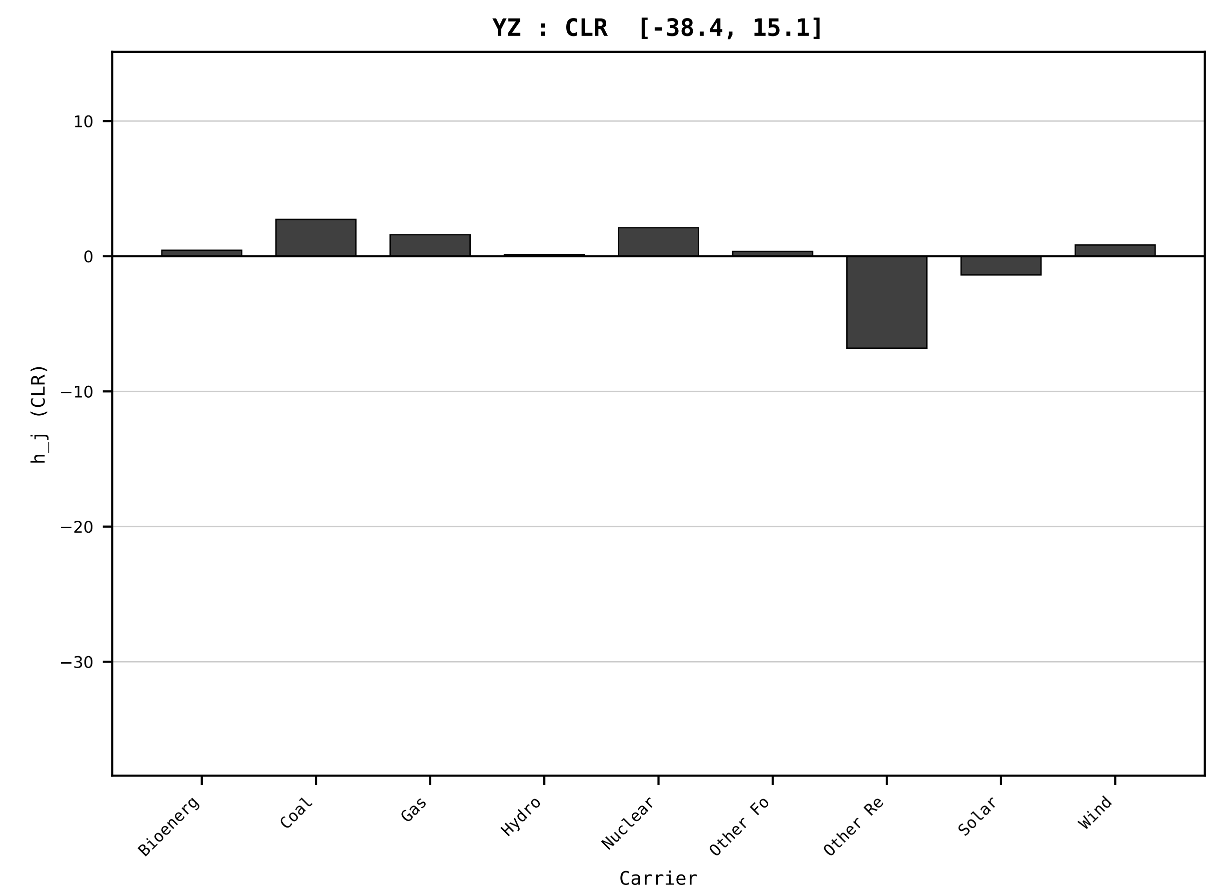
SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2008
t       : 8 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.285057
Ring    = Hs-2
E_metric= 63.6470
kappa_HS= 13760.50
omega   = 19.4937 deg
d_A     = 28.811916
Helm    = Other Renewables
Helm d  = +27.161871
DR      = 9.530 HLR
DR ratio= 13760.5x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```





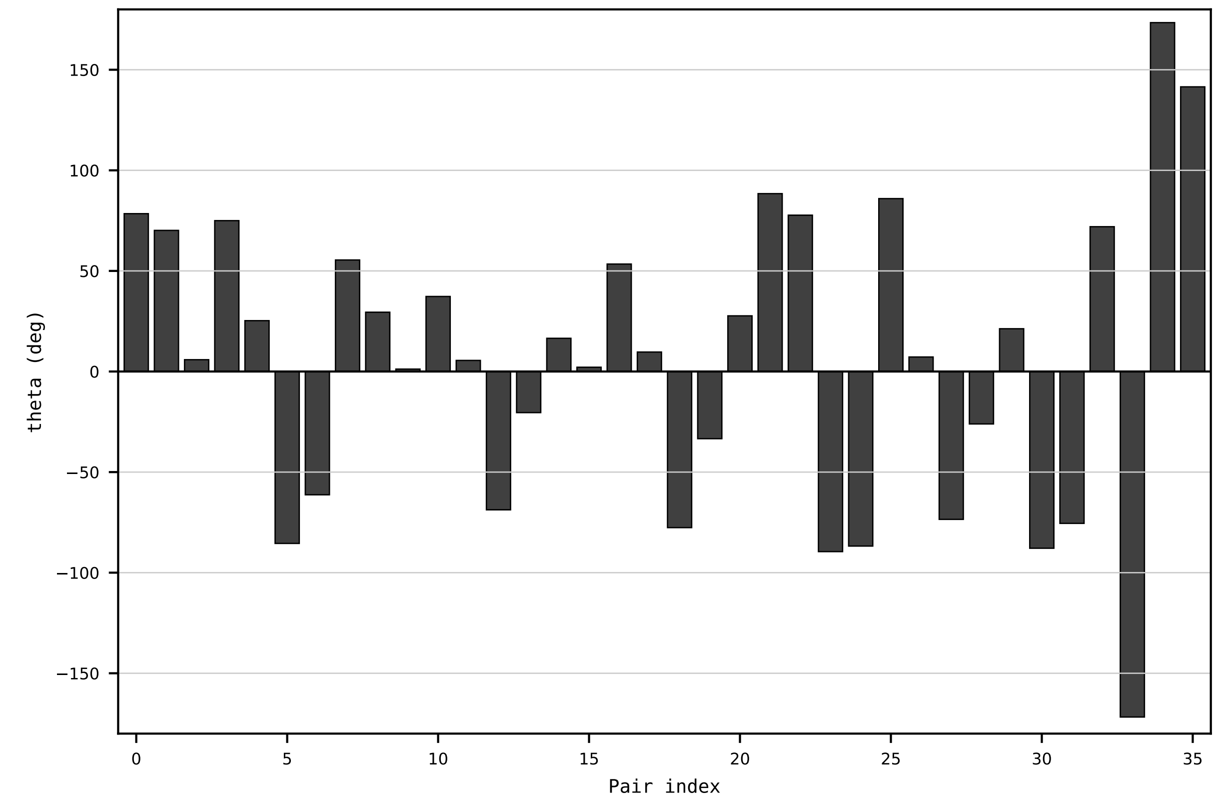


SECTION PLATE t=9 | Year 2009 | D=9 N=26 pairs=36

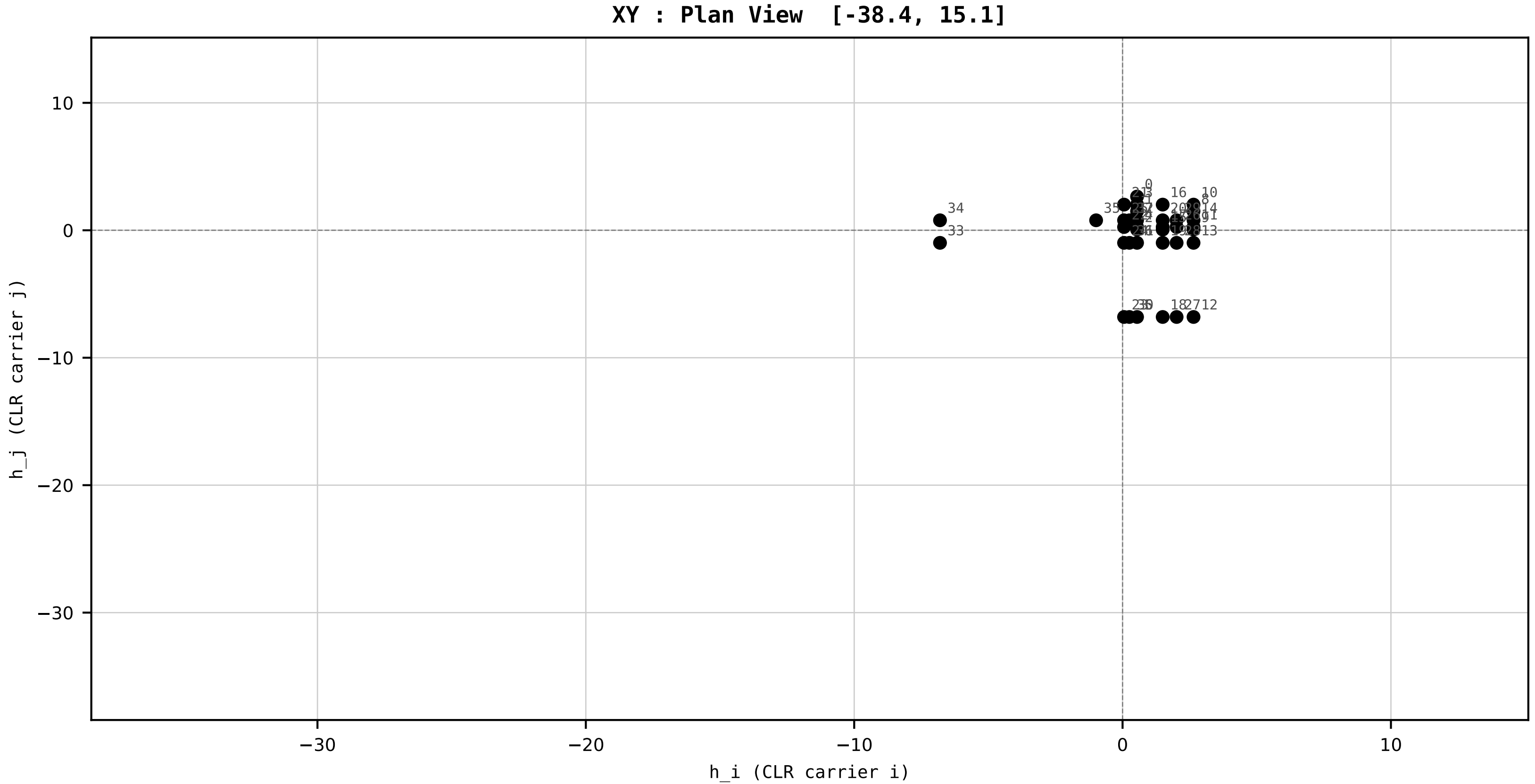
```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2009
t       : 9 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.272489
Ring    = Hs-2
E_metric= 61.5055
kappa_HS= 12672.50
omega   = 3.1880 deg
d_A     = 0.460403
Helm    = Solar
Helm d  = +0.398992
DR      = 9.447 HLR
DR ratio= 12672.5x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

PAIR INDEX:
0=Bioenergy-Coal
1=Bioenergy-Gas
2=Bioenergy-Hydro
3=Bioenergy-Nuclear
4=Bioenergy-Other Fossil
5=Bioenergy-Other Renewables
6=Bioenergy-Solar
7=Bioenergy-Wind
8=Coal-Gas
9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind

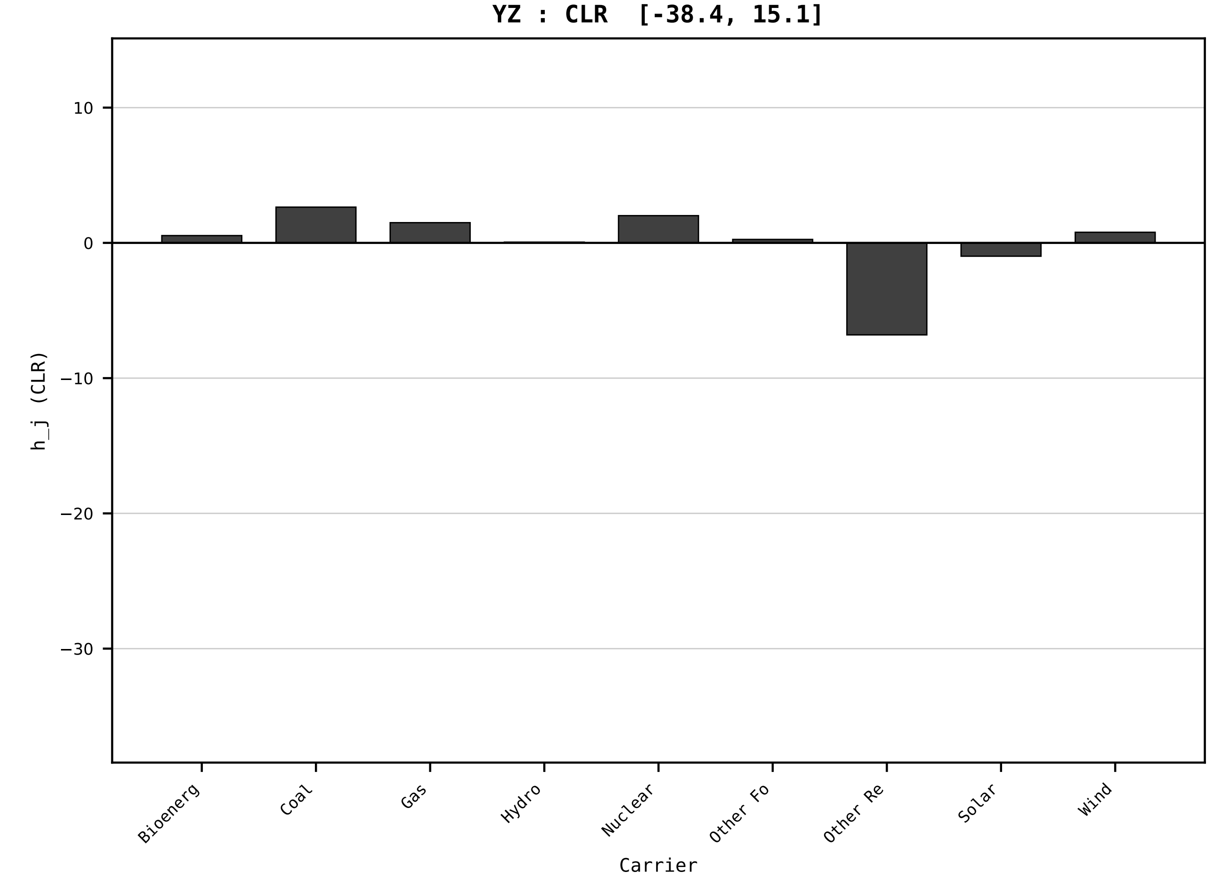
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-38.4, 15.1]



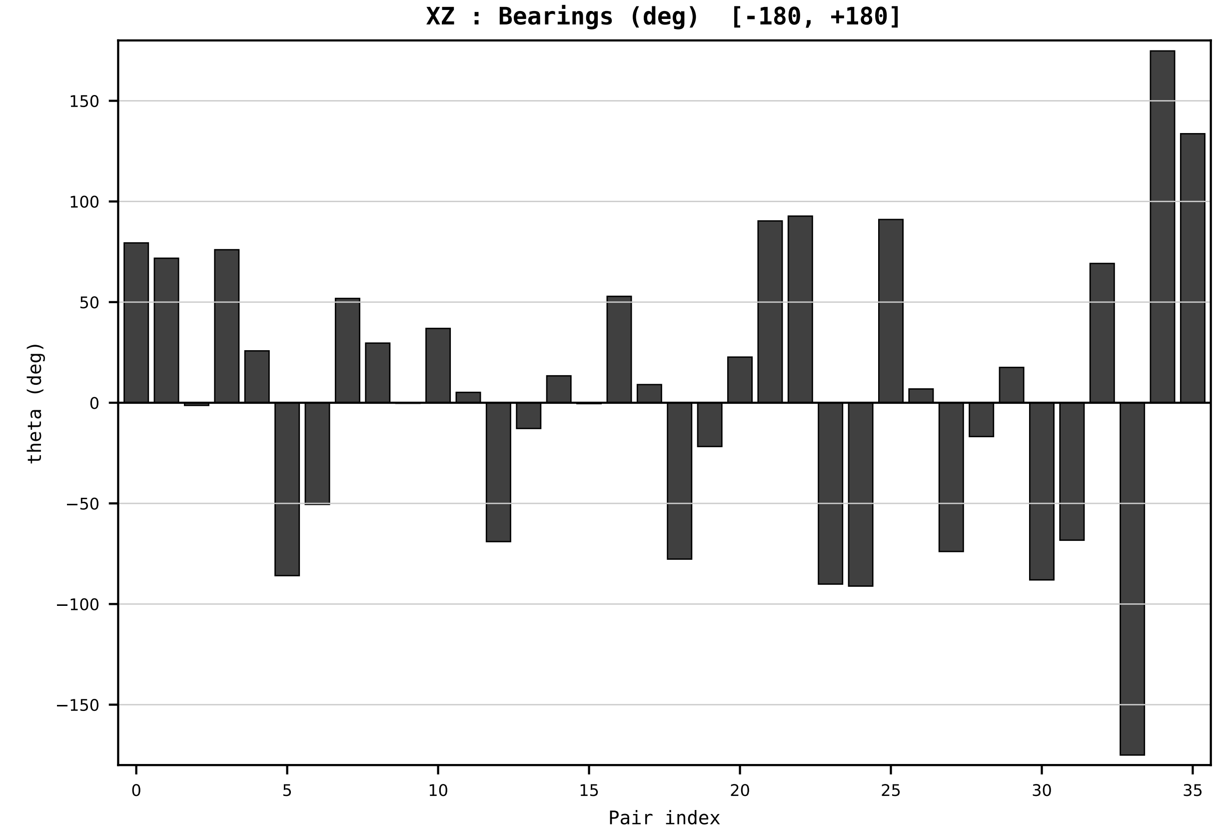
YZ : CLR [-38.4, 15.1]

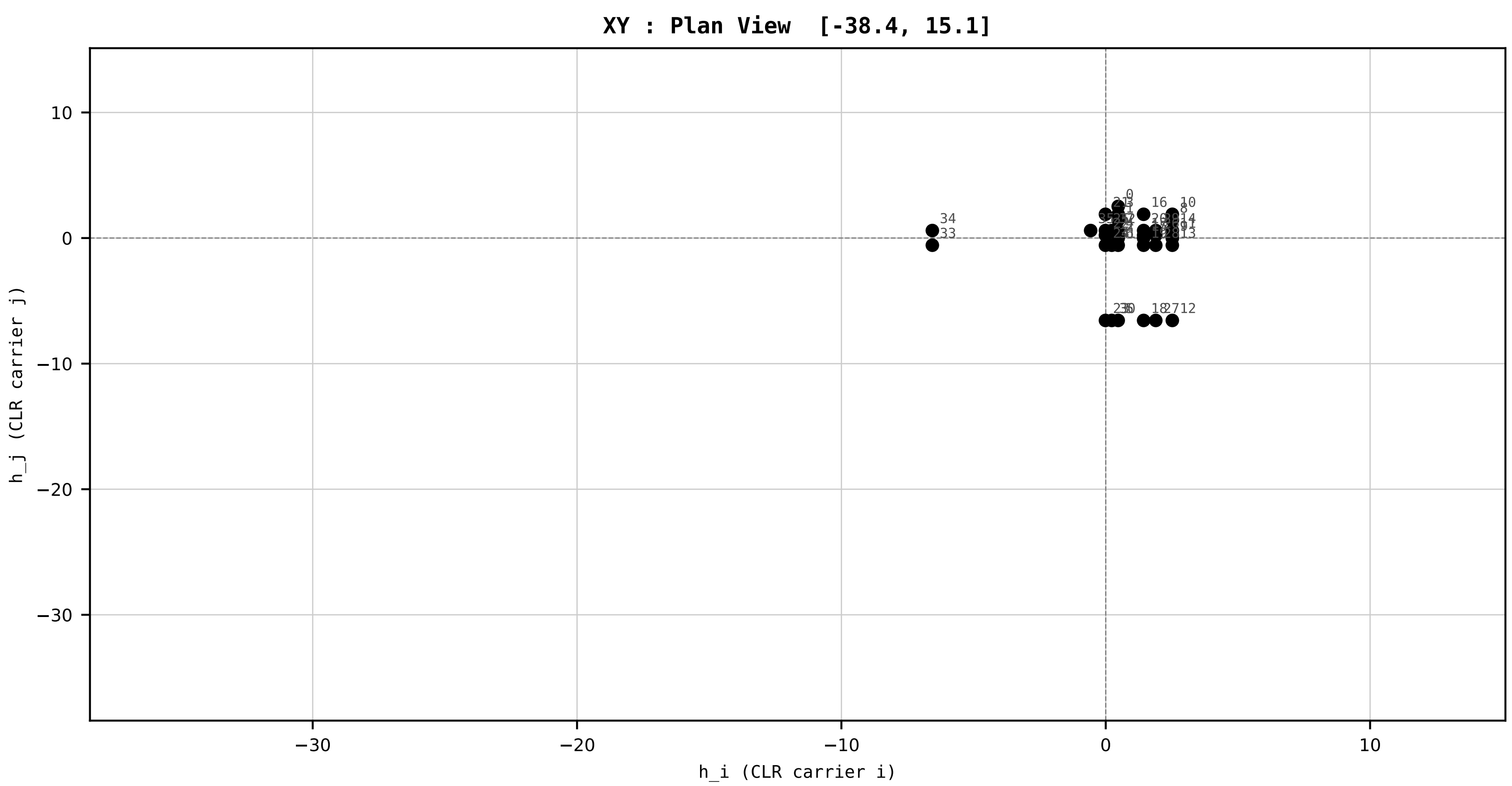


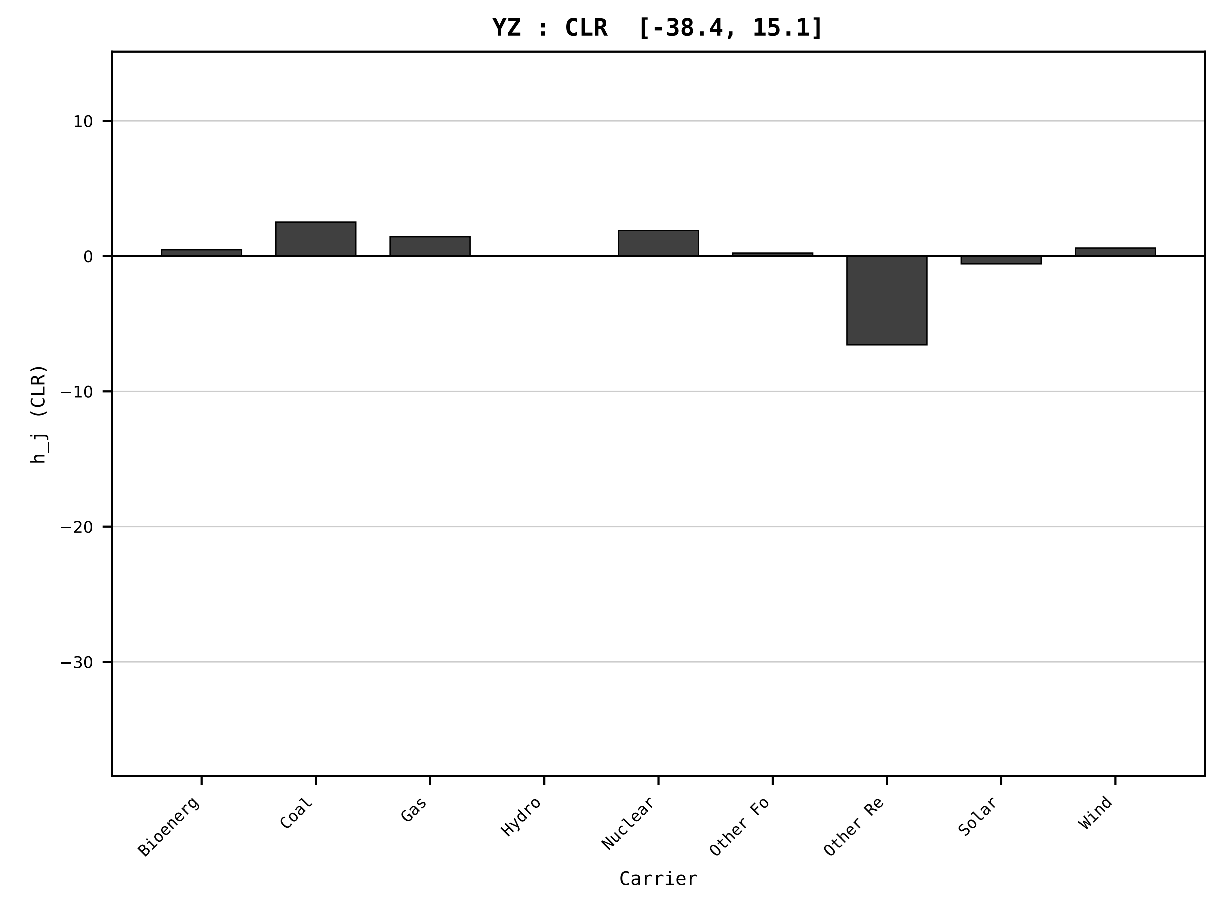
SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2010
t       : 10 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.257193
Ring    = Hs-2
E_metric= 55.9662
kappa_HS= 8763.00
omega   = 3.1589 deg
d_A     = 0.555847
Helm    = Solar
Helm d  = +0.413991
DR      = 9.078 HLR
DR ratio= 8763.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



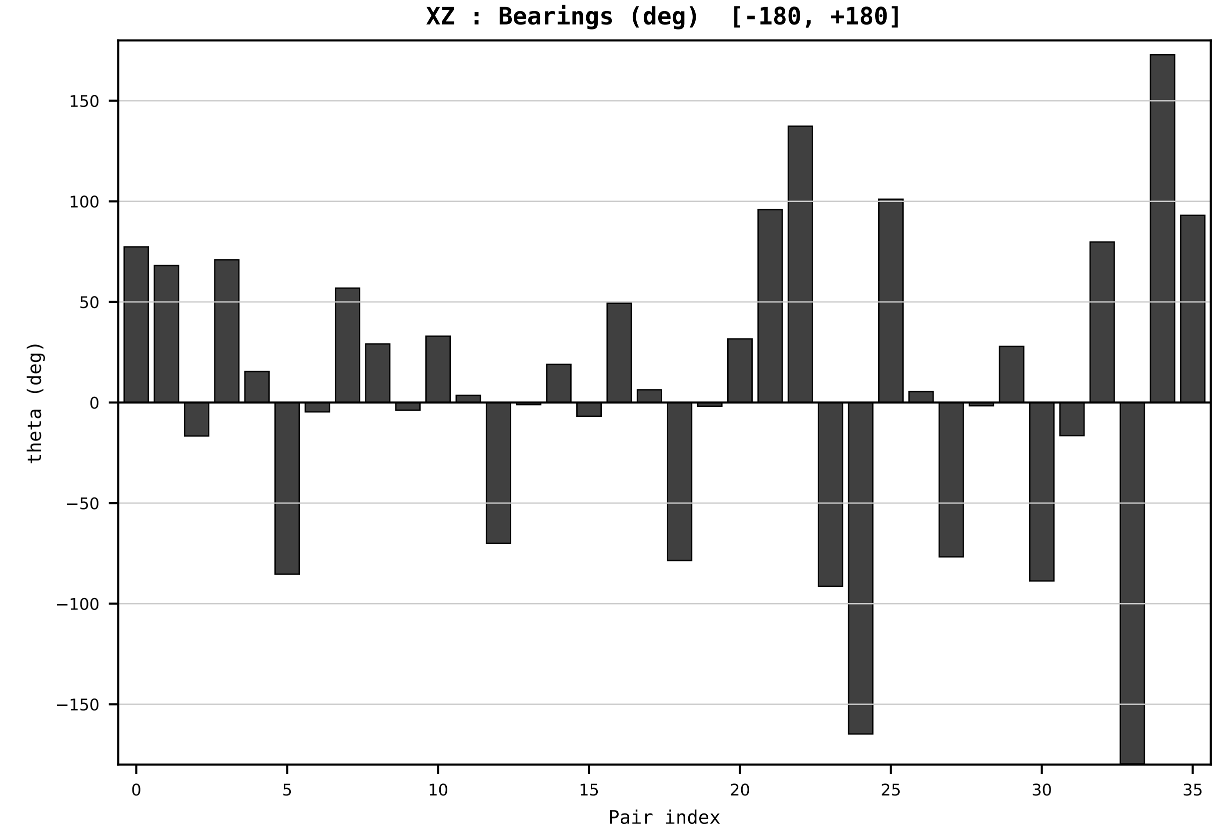


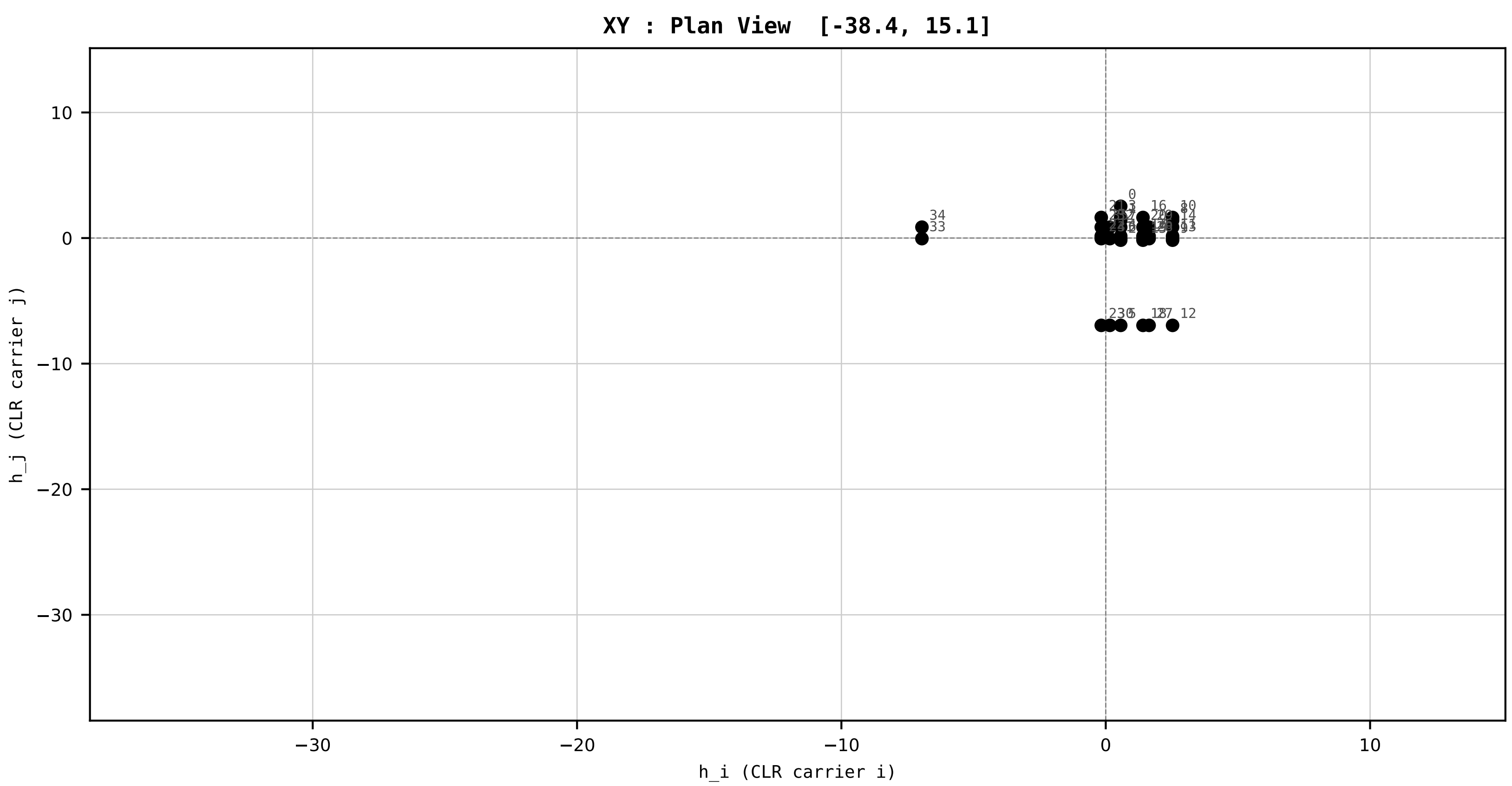


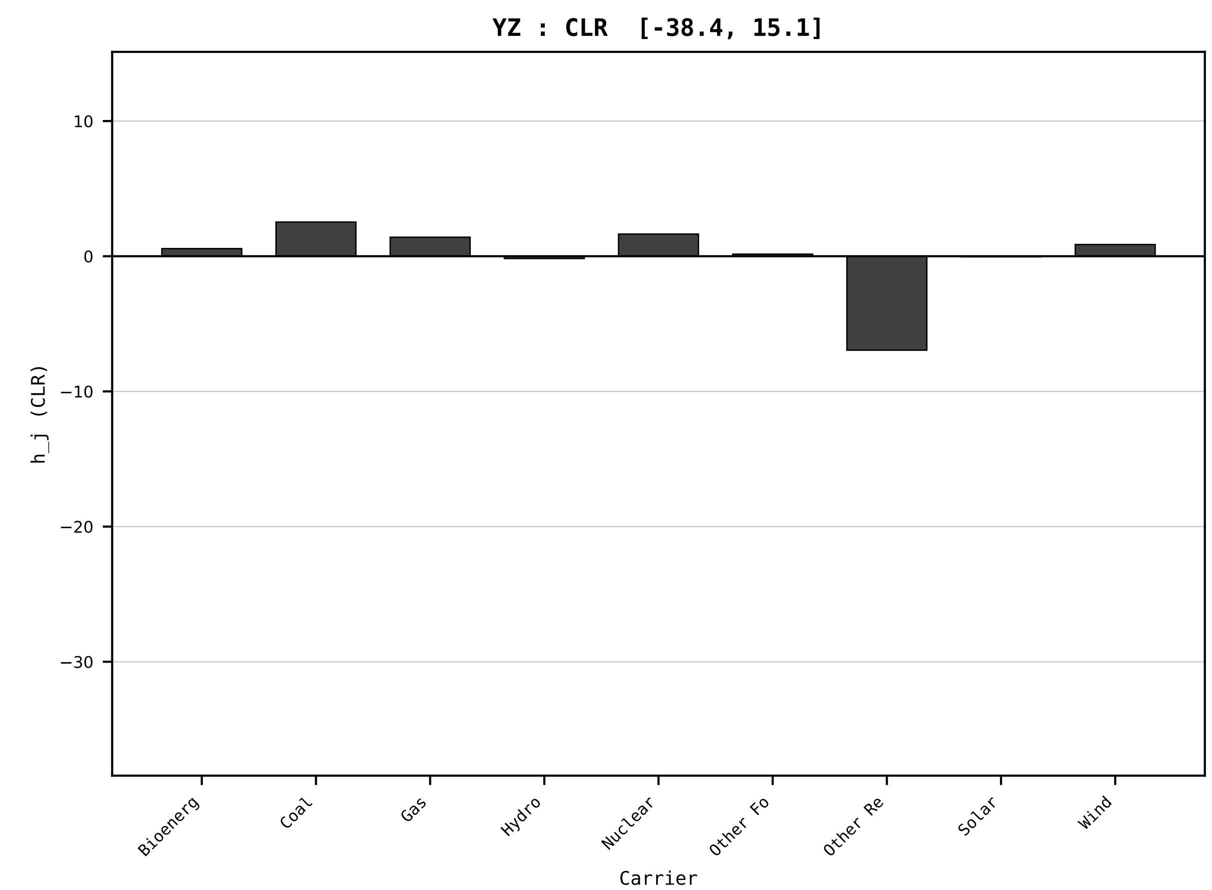
SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2011
t       : 11 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.240465
Ring    = Hs-2
E_metric= 60.5504
kappa_HS= 13123.00
omega   = 5.4004 deg
d_A     = 0.779100
Helm    = Solar
Helm d  = +0.525310
DR      = 9.482 HLR
DR ratio= 13123.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



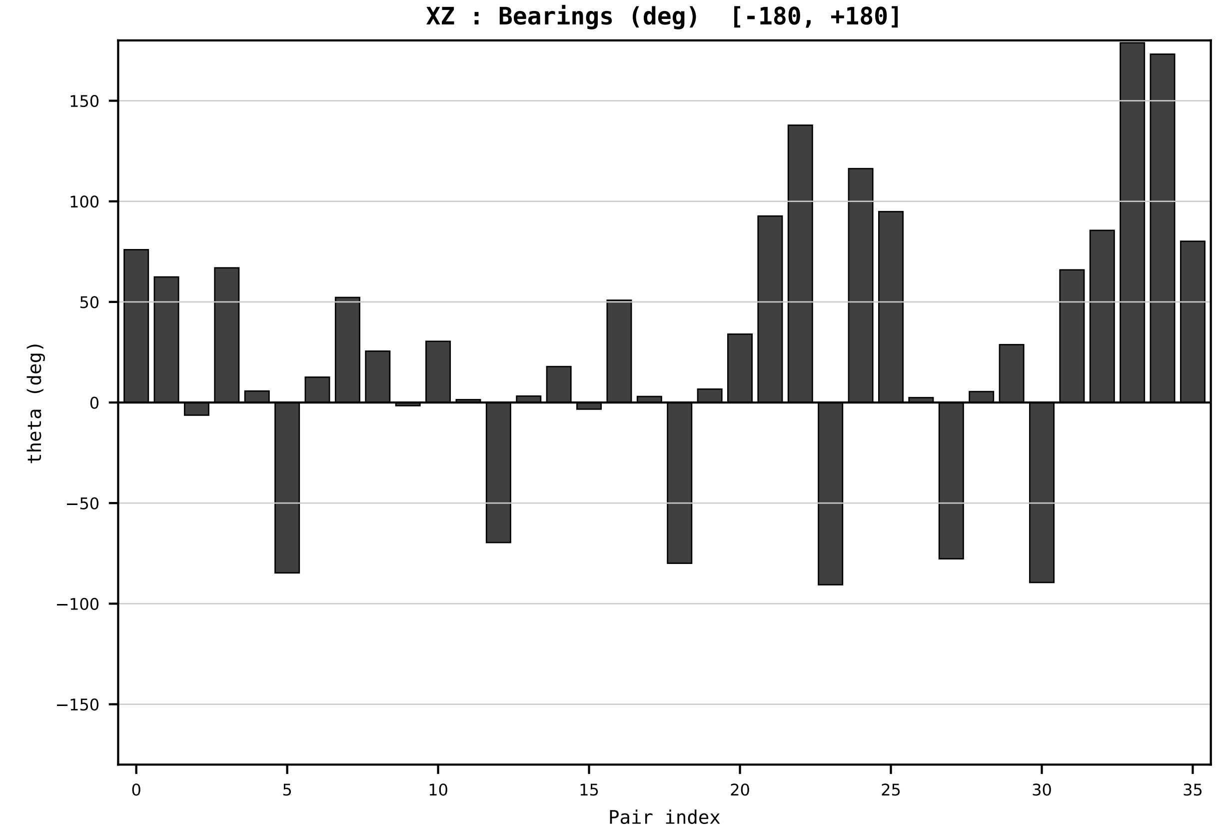


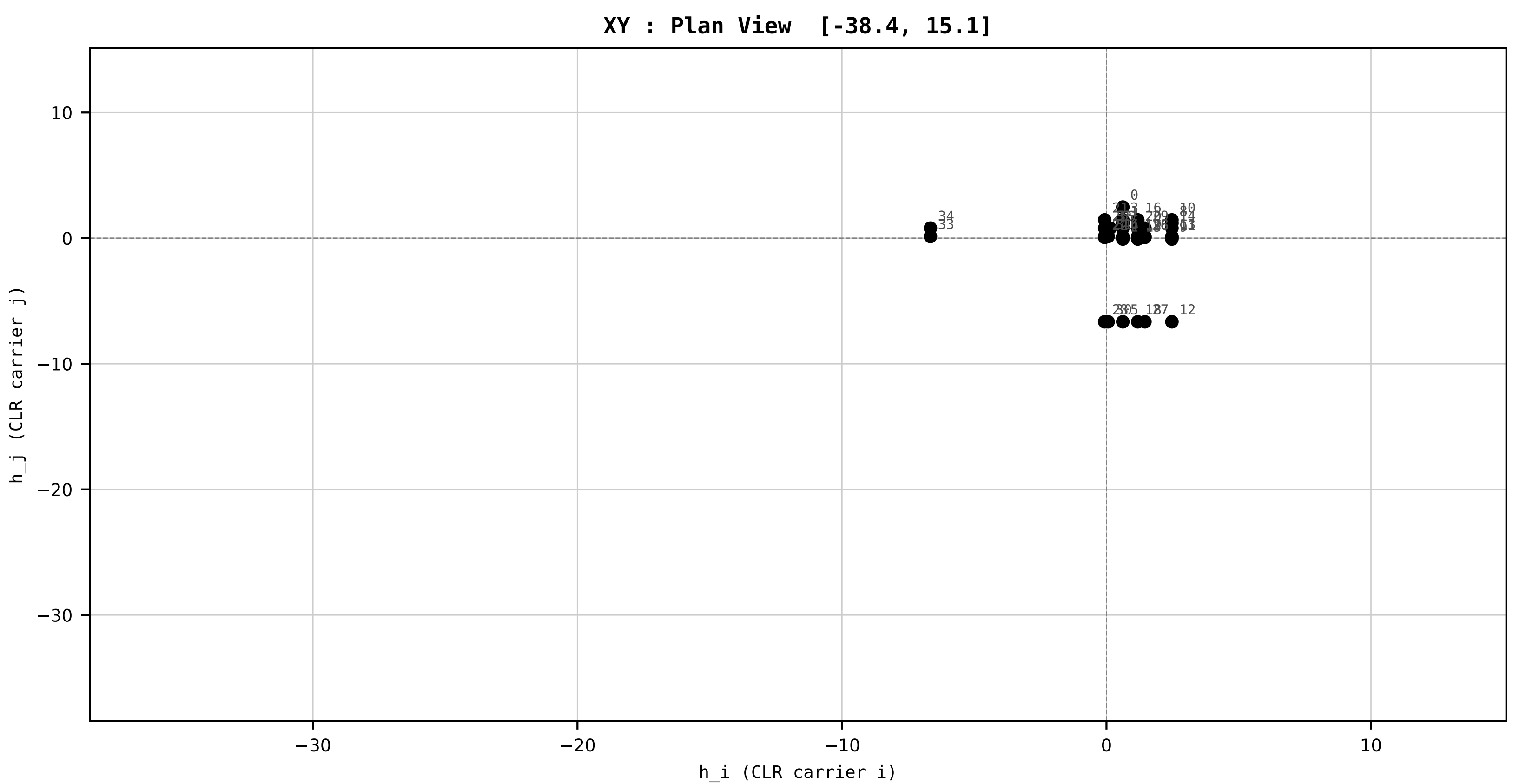


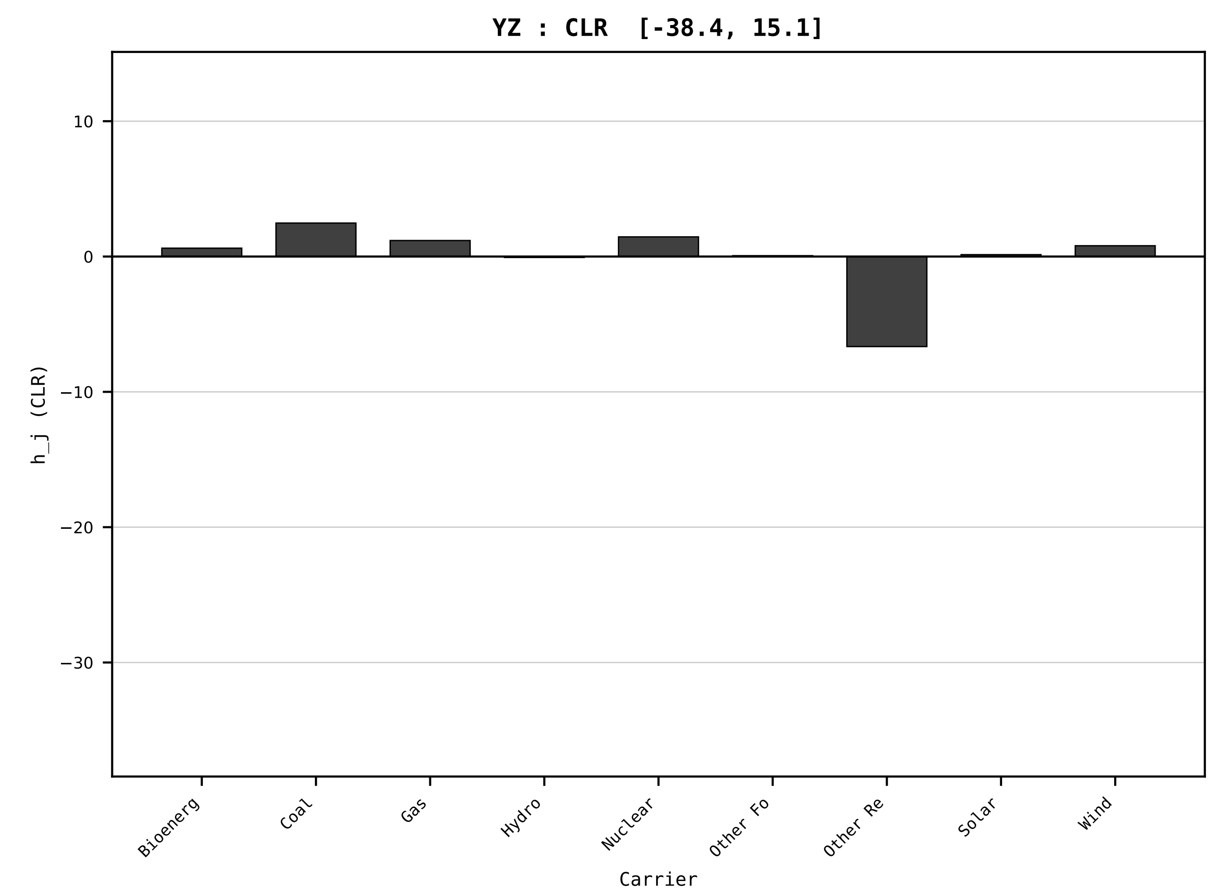
SECTION PLATE t=12 | Year 2012 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2012
t       : 12 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.232607
Ring    = Hs-2
E_metric= 54.9495
kappa_HS= 9207.67
omega   = 2.4380 deg
d_A     = 0.490208
Helm    = Other Renewables
Helm d  = +0.298944
DR      = 9.128 HLR
DR ratio= 9207.7x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```







SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

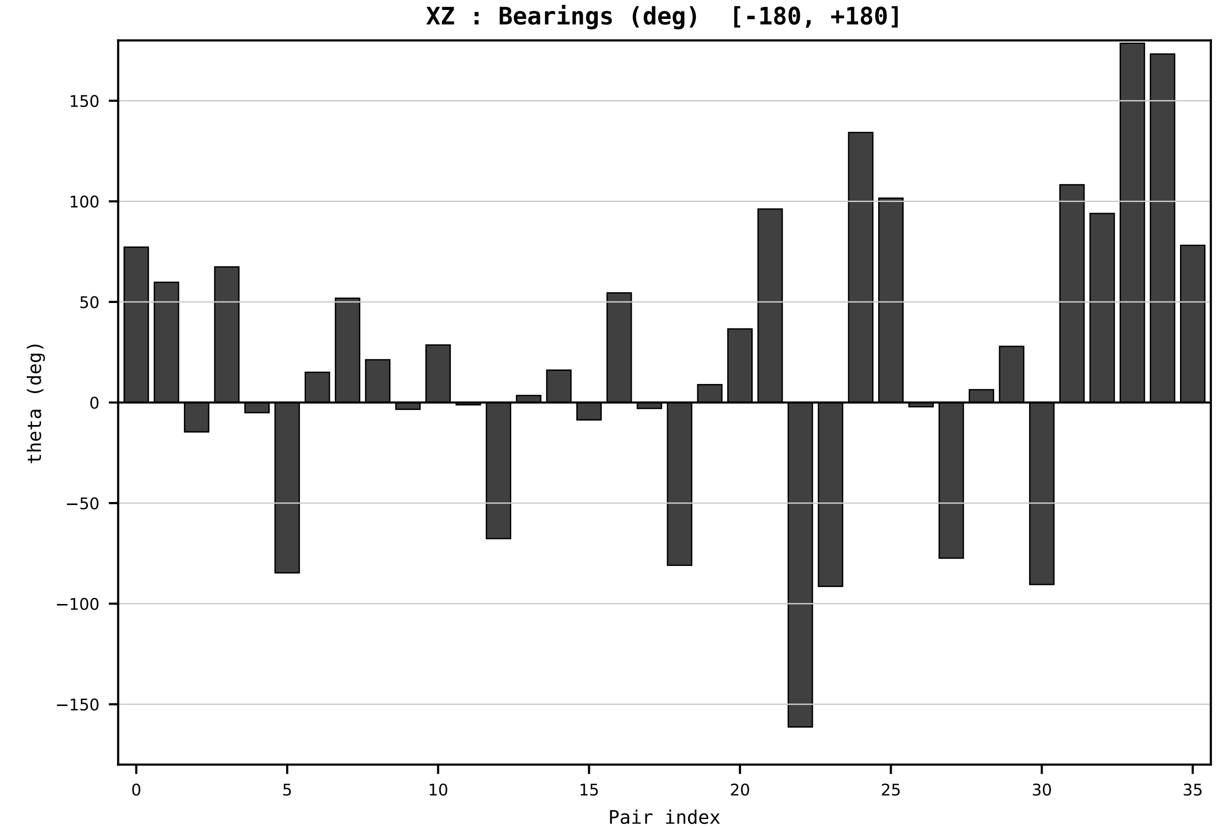
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

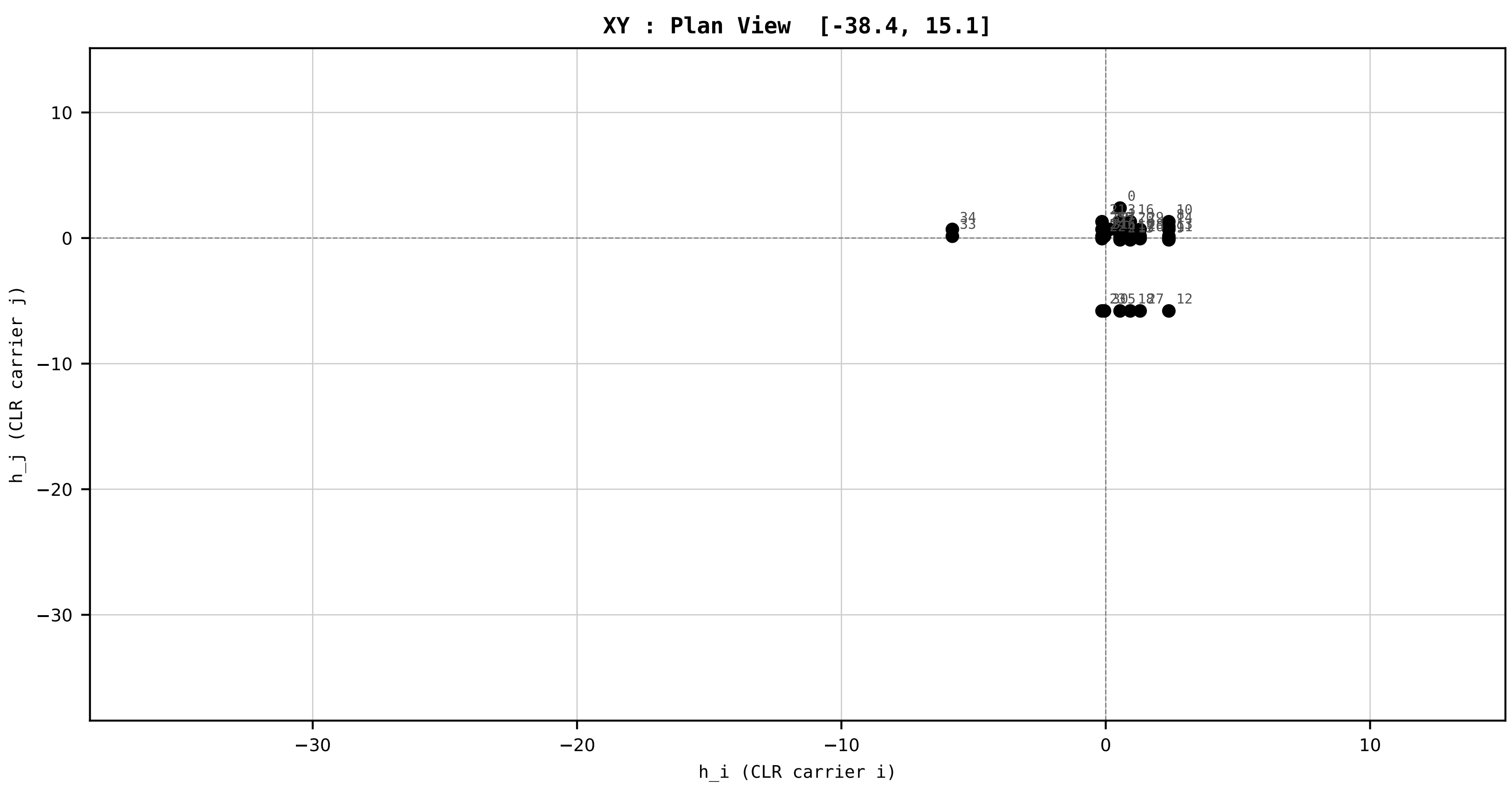
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

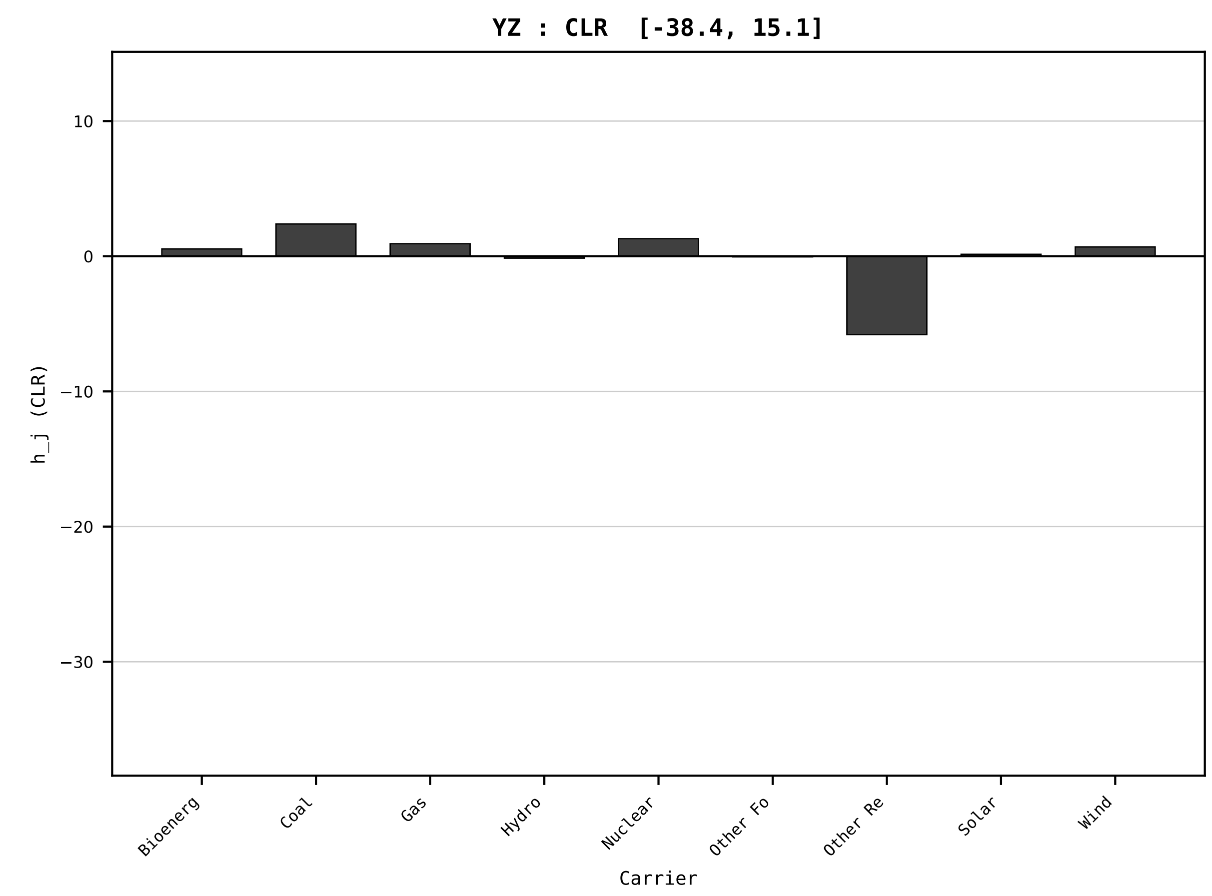
Hs = 0.235275
Ring = Hs-2
E_metric= 42.7299
kappa_HS= 3602.50
omega = 2.4484 deg
d_A = 0.925102
Helm = Other Renewables
Helm d = +0.852077
DR = 8.189 HLR
DR ratio= 3602.5x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:
0=Bioenergy-Coal 18=Gas-Other Renewables
1=Bioenergy-Gas 19=Gas-Solar
2=Bioenergy-Hydro 20=Gas-Wind
3=Bioenergy-Nuclear 21=Hydro-Nuclear
4=Bioenergy-Other Fossil22=Hydro-Other Fossil
5=Bioenergy-Other Renewables23=Hydro-Other Renewables
6=Bioenergy-Solar 24=Hydro-Solar
7=Bioenergy-Wind 25=Hydro-Wind
8=Coal-Gas 26=Nuclear-Other Fossil
9=Coal-Hydro 27=Nuclear-Other Renewables
10=Coal-Nuclear 28=Nuclear-Solar
11=Coal-Other Fossil 29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar 31=Other Fossil-Solar
14=Coal-Wind 32=Other Fossil-Wind
15=Gas-Hydro 33=Other Renewables-Solar
16=Gas-Nuclear 34=Other Renewables-Wind
17=Gas-Other Fossil 35=Solar-Wind





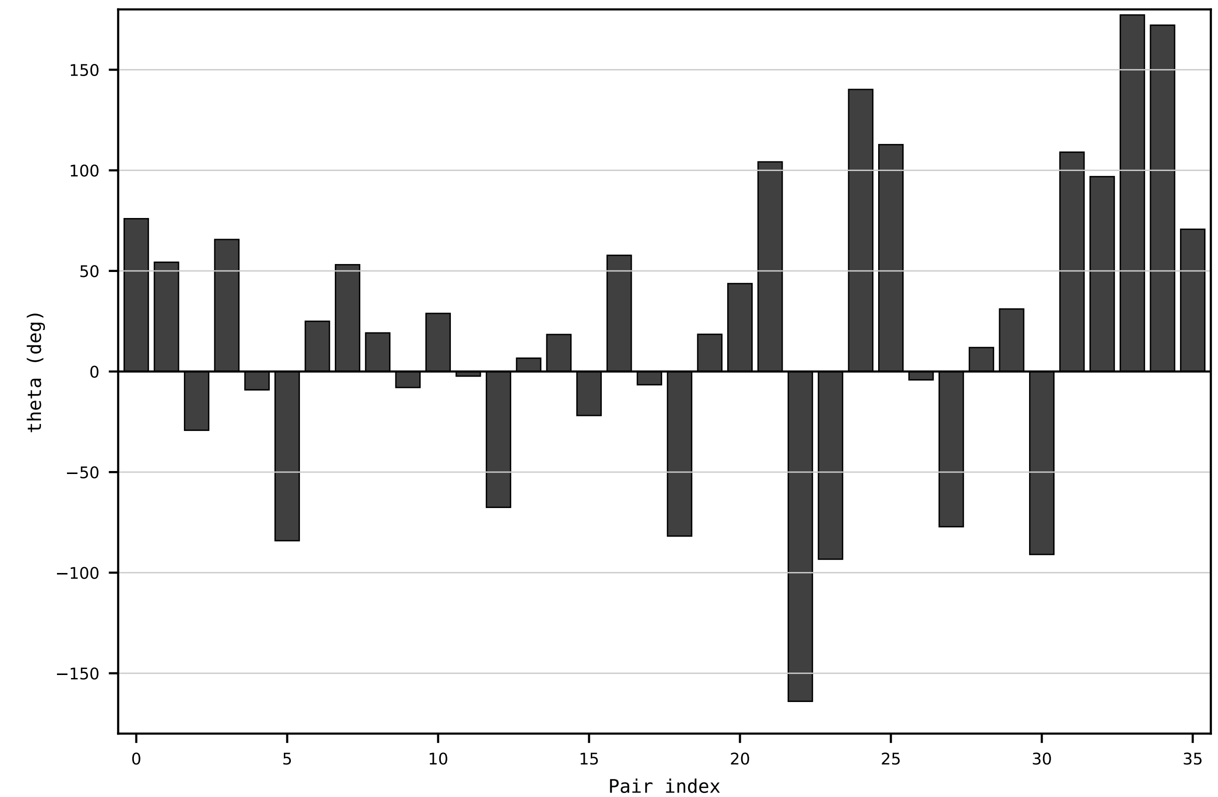


SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

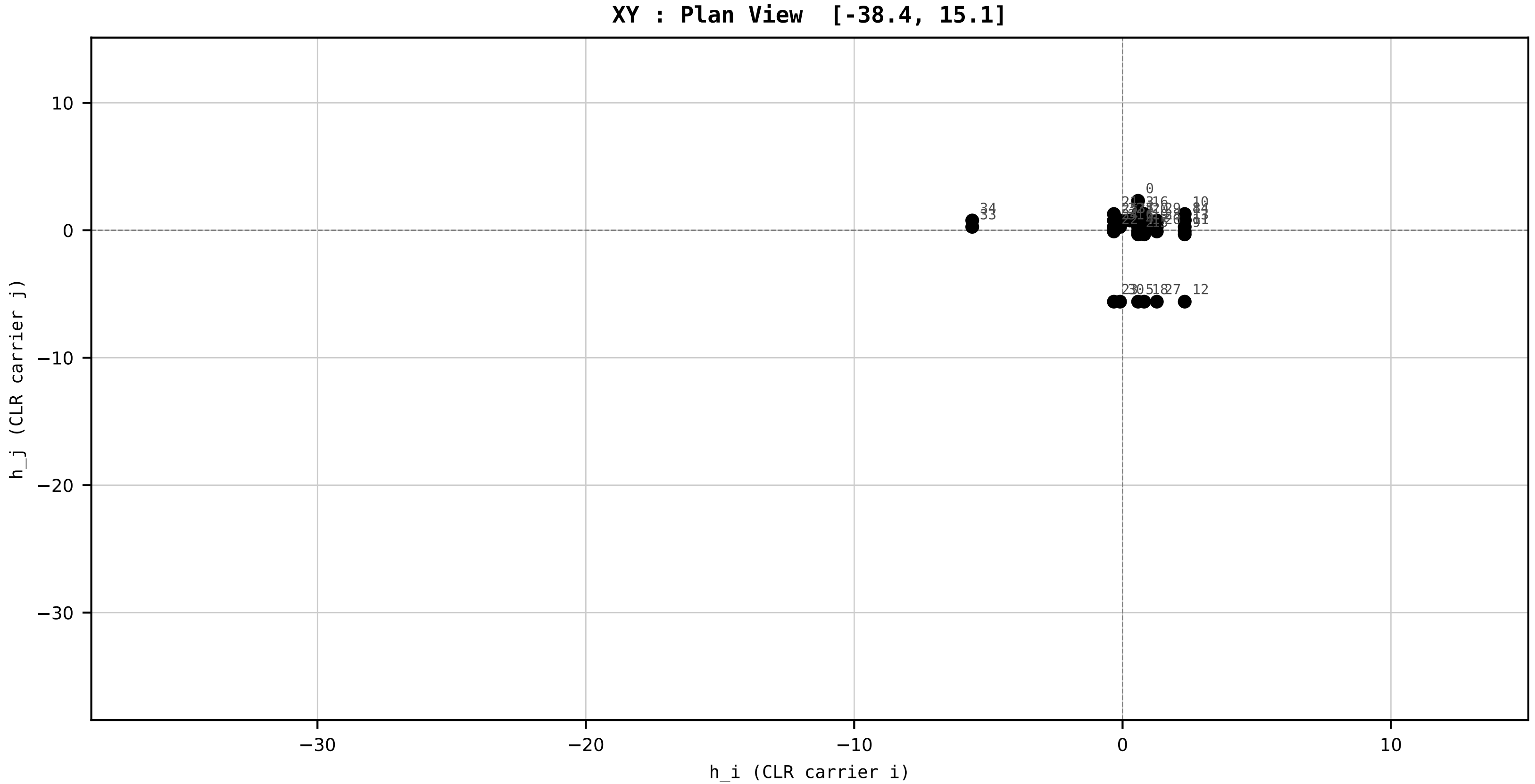
```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2014
t       : 14 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.224515
Ring    = Hs-2
E_metric= 40.1329
kappa_HS= 2744.10
omega   = 2.5057 deg
d_A     = 0.346263
Helm    = Other Renewables
Helm d  = +0.201151
DR      = 7.917 HLR
DR ratio= 2744.1x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```

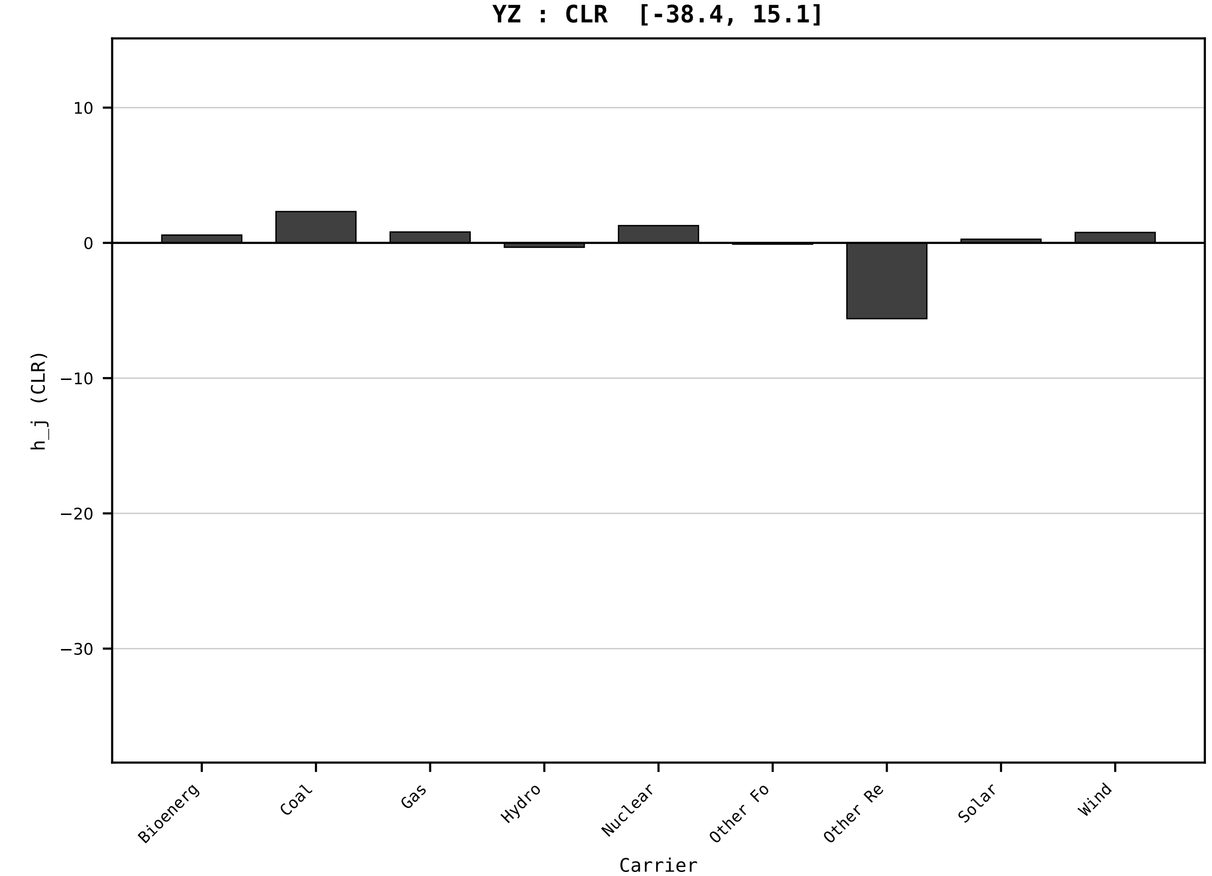
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-38.4, 15.1]



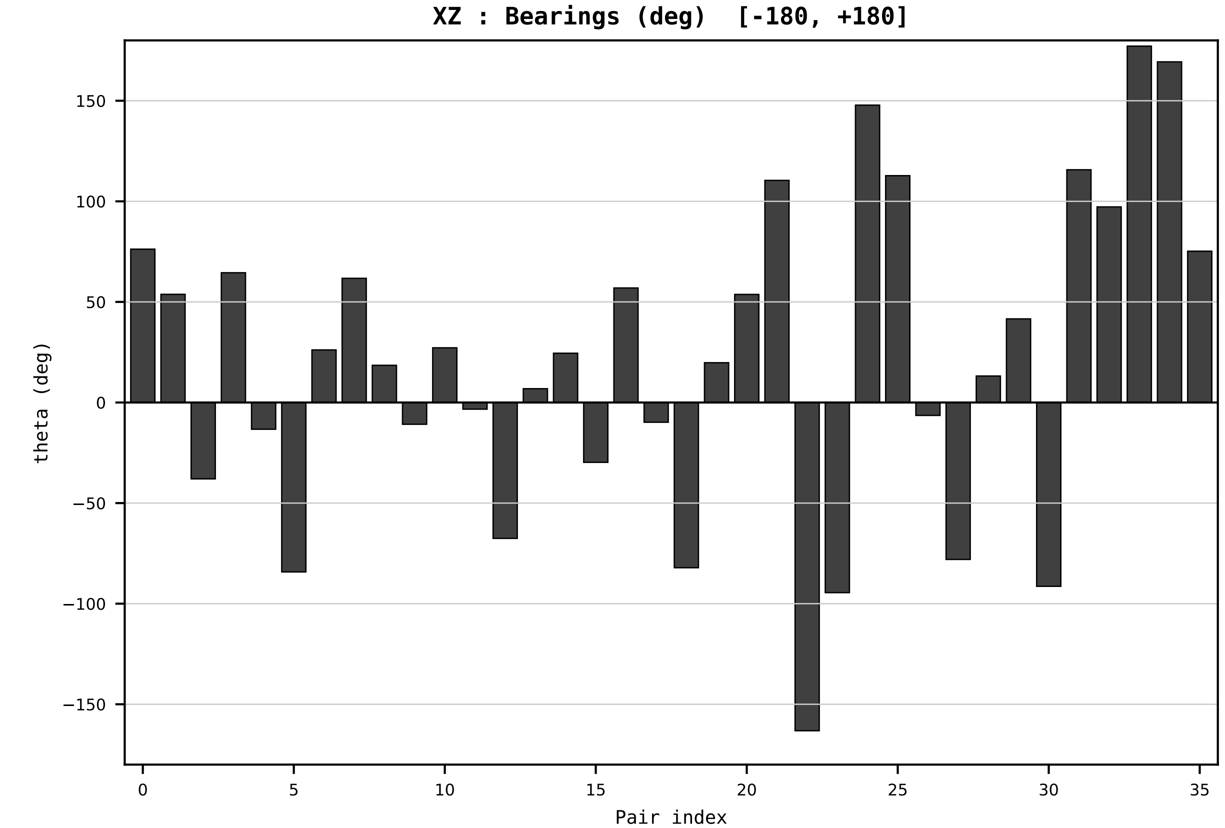
YZ : CLR [-38.4, 15.1]

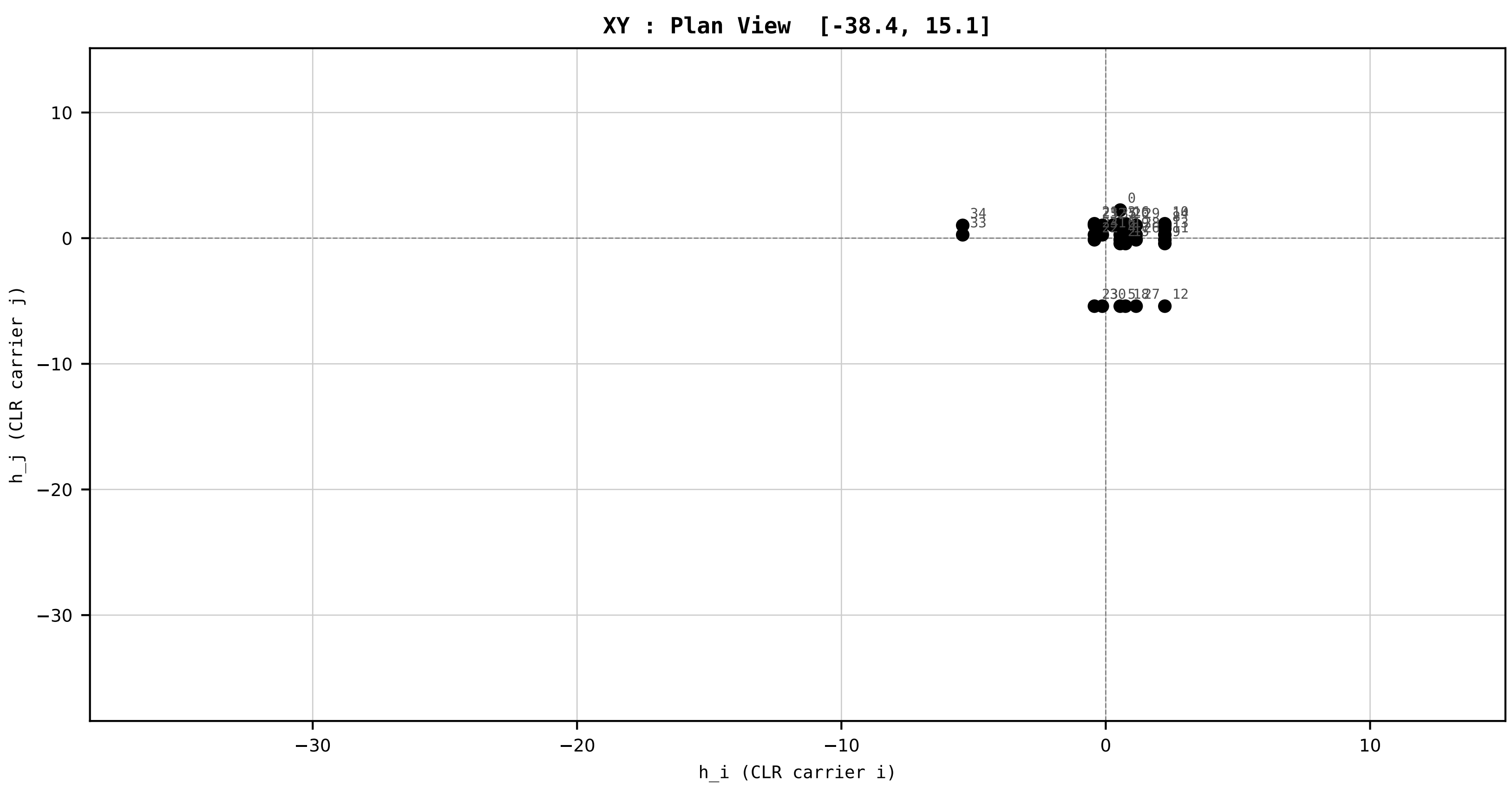


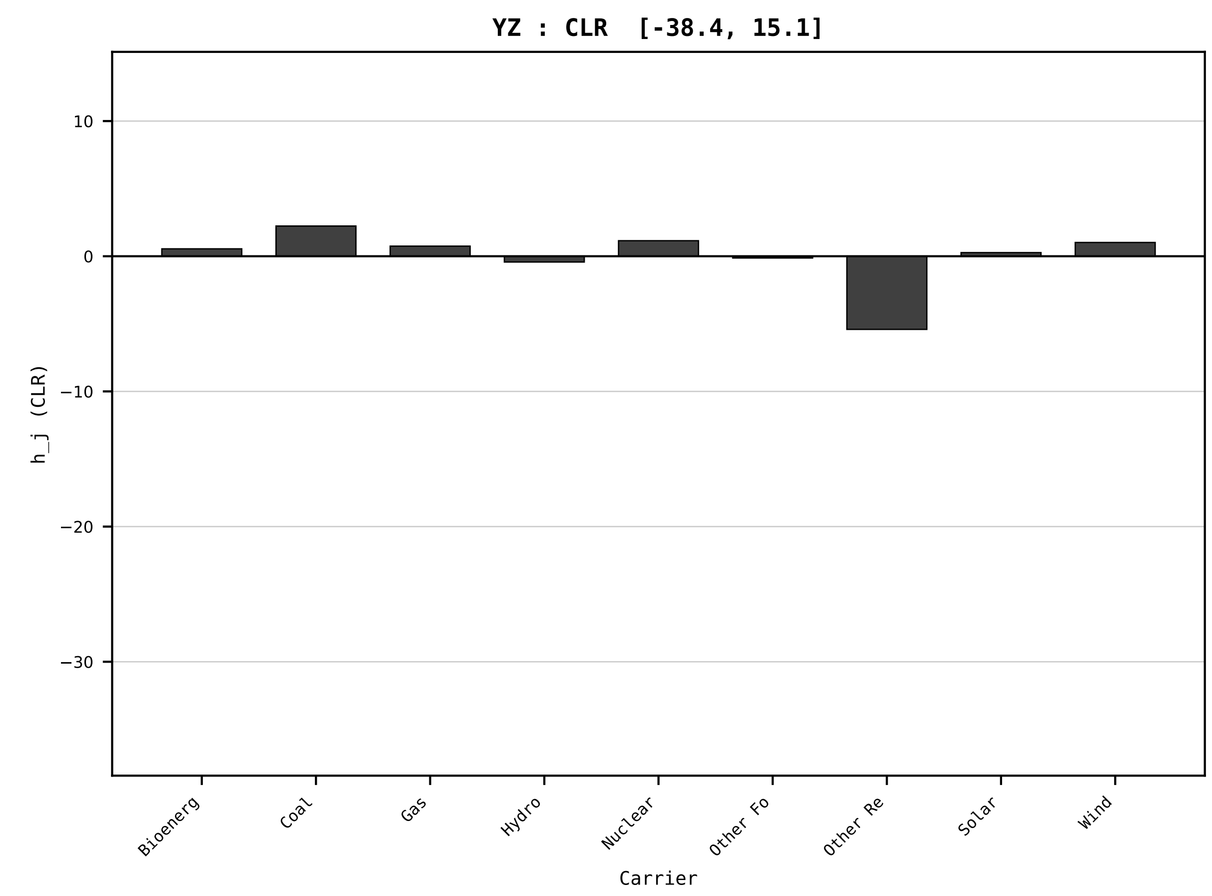
SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2015
t       : 15 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.211885
Ring    = Hs-2
E_metric= 37.7664
kappa_HS= 2093.85
omega   = 2.9247 deg
d_A     = 0.370645
Helm    = Wind
Helm d  = +0.248533
DR      = 7.647 HLR
DR ratio= 2093.8x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```





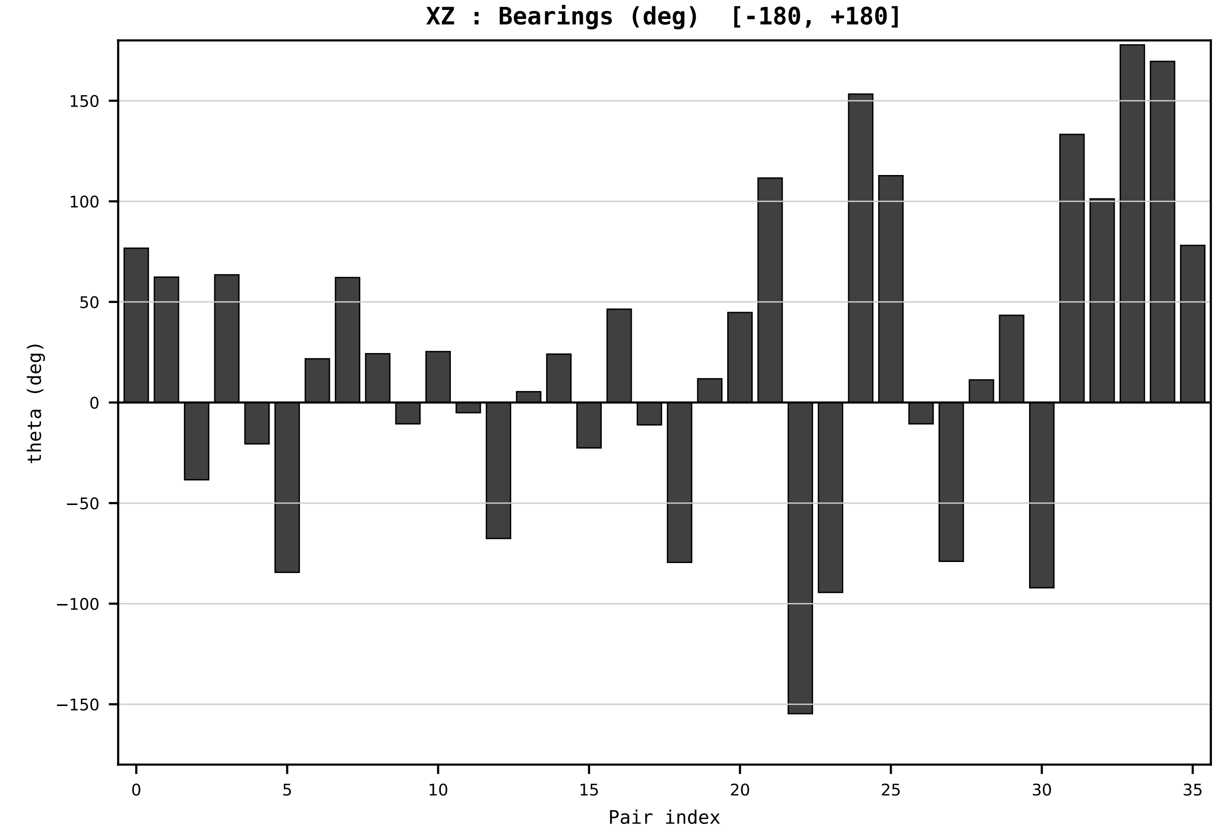


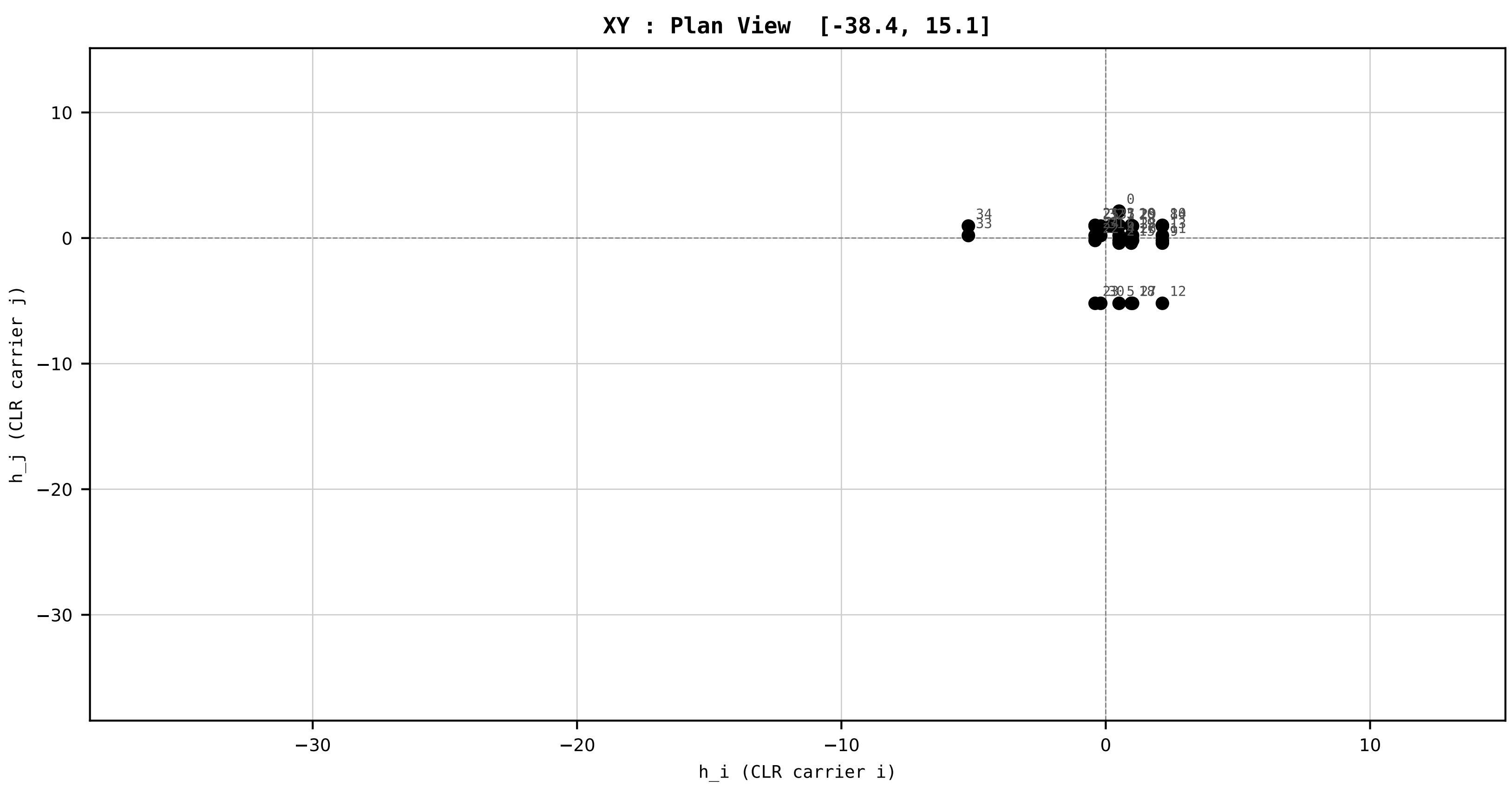
SECTION PLATE t=16 | Year 2016 | D=9 N=26 pairs=36

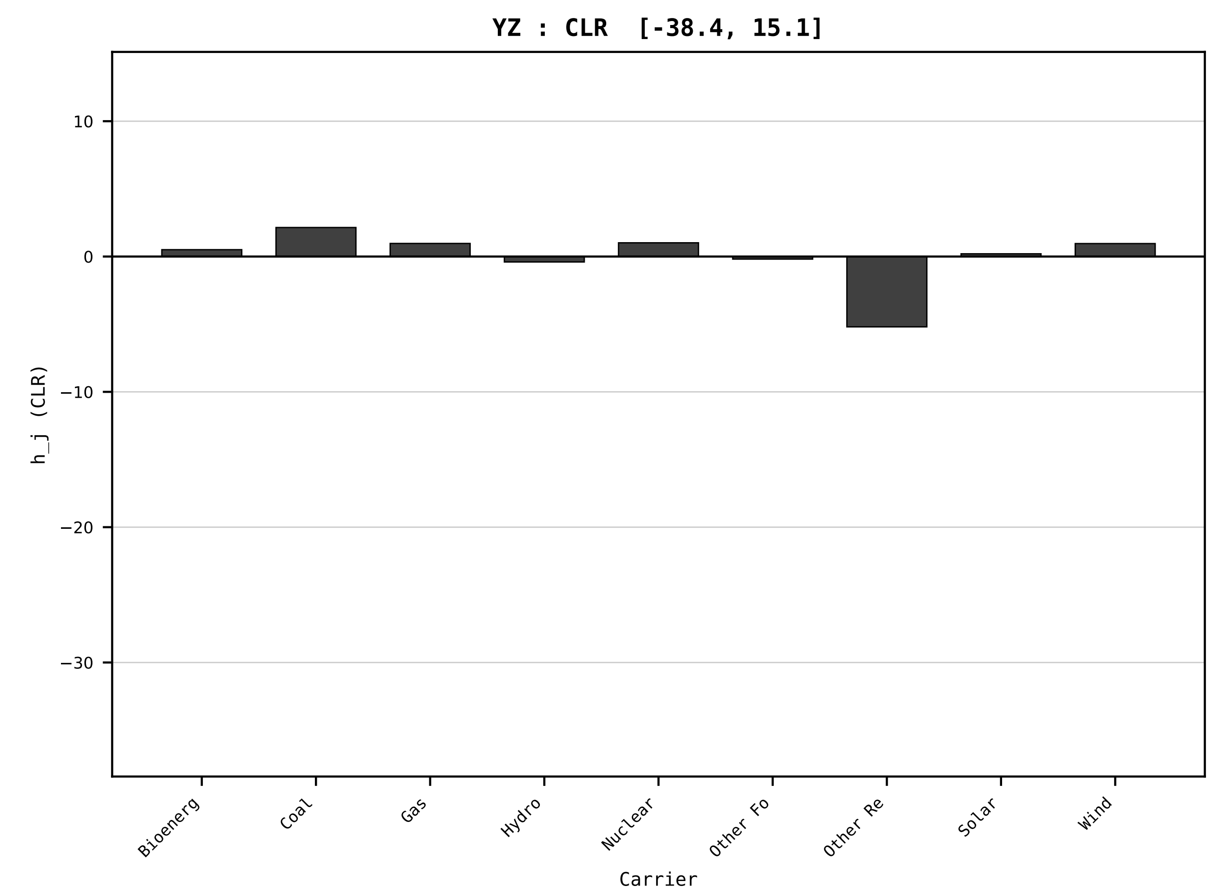
```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2016
t       : 16 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.198712
Ring    = Hs-2
E_metric= 34.9636
kappa_HS= 1539.71
omega   = 2.7002 deg
d_A     = 0.367040
Helm    = Gas
Helm d  = +0.217765
DR      = 7.339 HLR
DR ratio= 1539.7x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind







SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

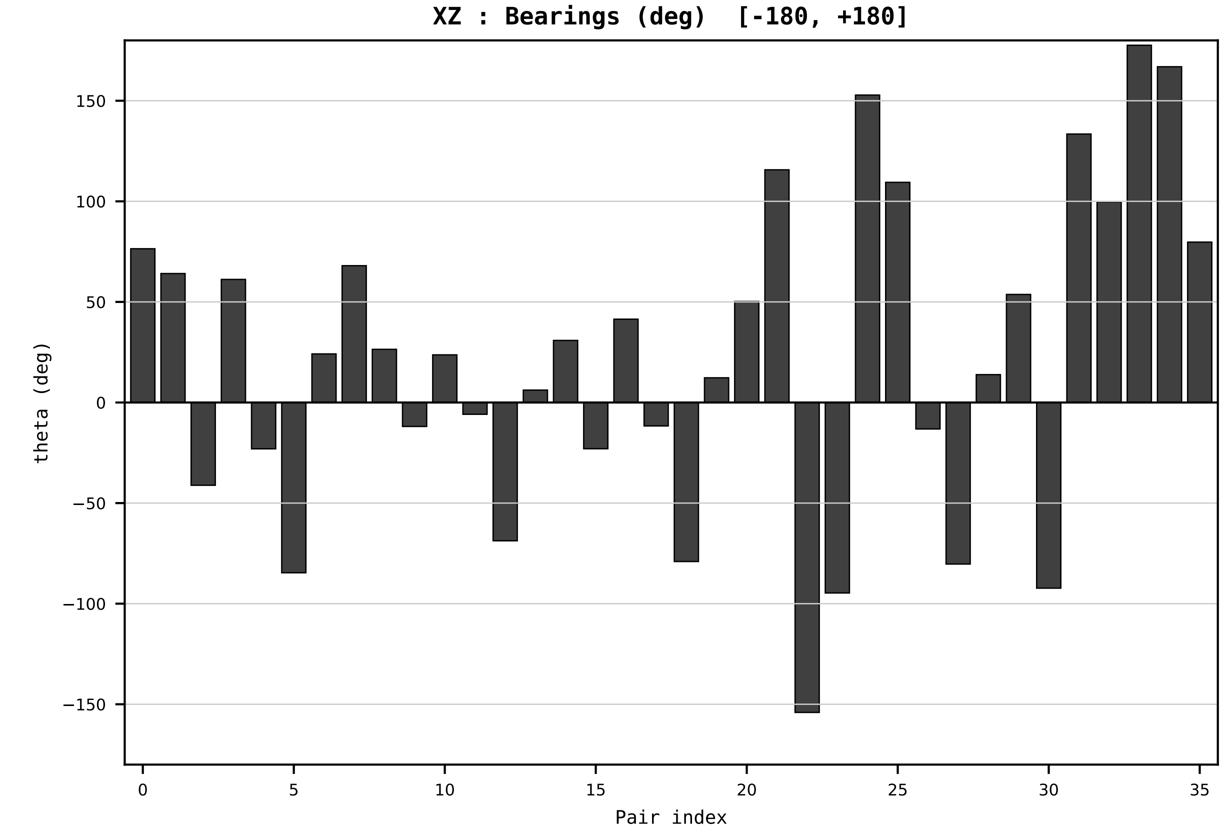
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

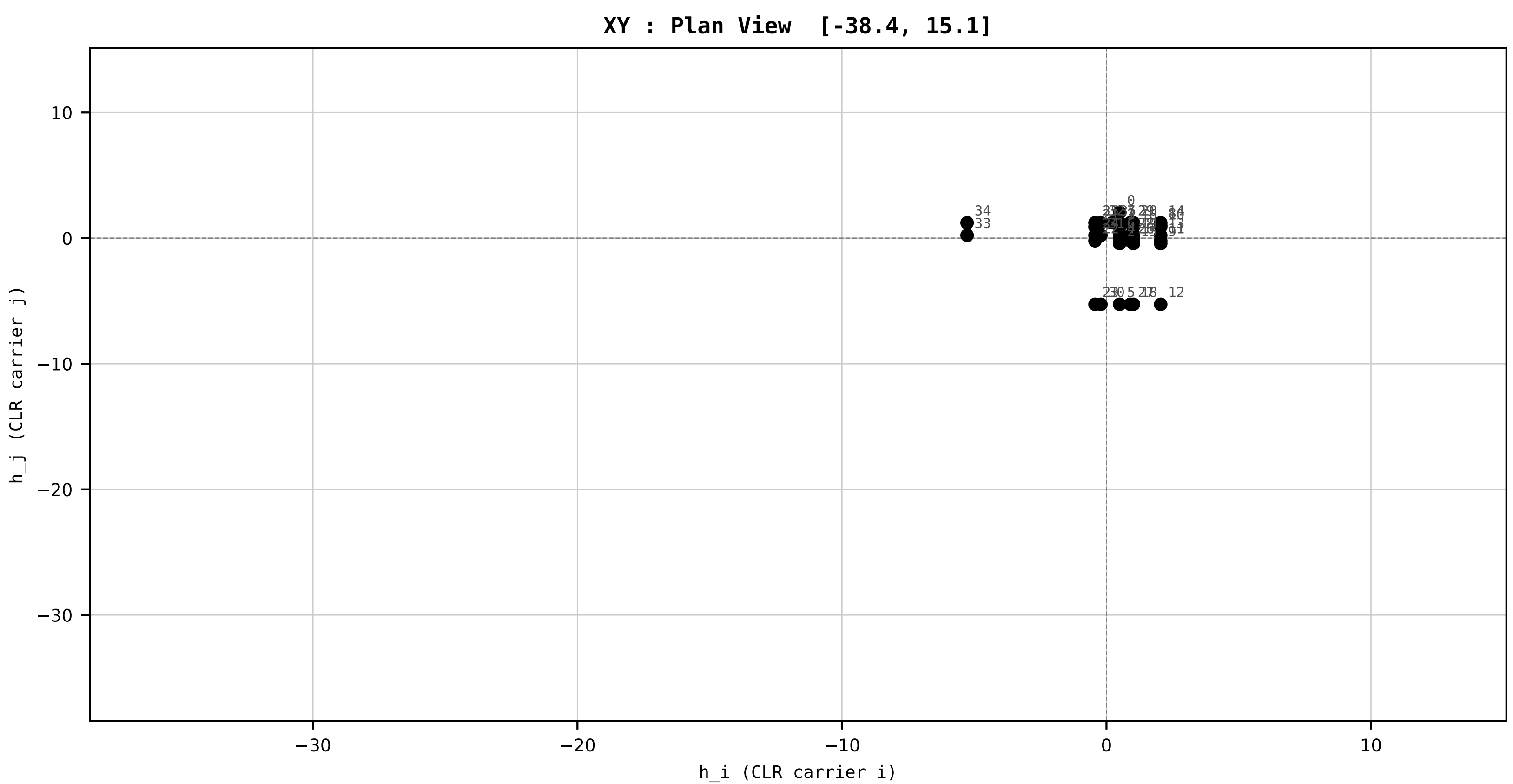
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

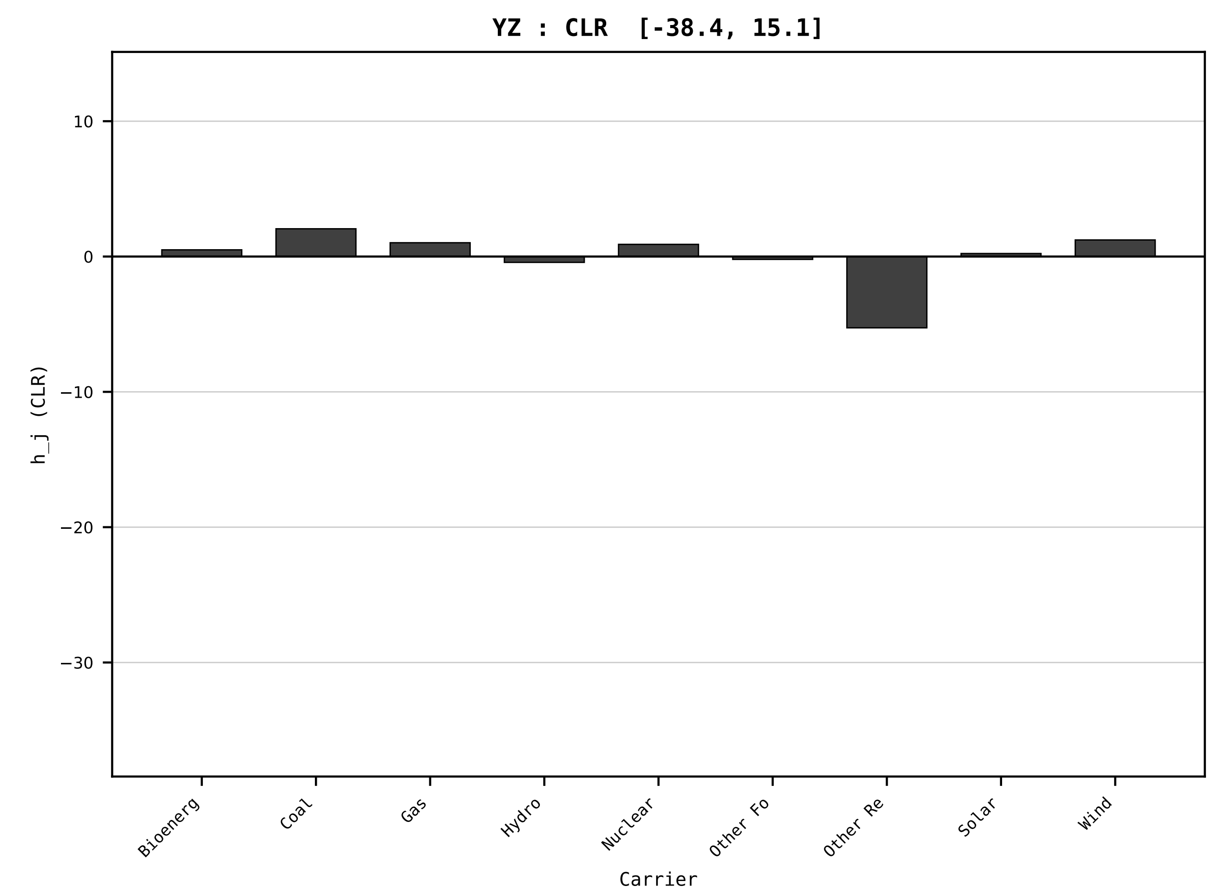
Hs = 0.184230
Ring = Hs-2
E_metric= 35.8305
kappa_HS= 1507.88
omega = 3.0206 deg
d_A = 0.321962
Helm = Wind
Helm d = +0.268201
DR = 7.318 HLR
DR ratio= 1507.9x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:
0=Bioenergy-Coal
1=Bioenergy-Gas
2=Bioenergy-Hydro
3=Bioenergy-Nuclear
4=Bioenergy-Other Fossil
5=Bioenergy-Other Renewables
6=Bioenergy-Solar
7=Bioenergy-Wind
8=Coal-Gas
9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind



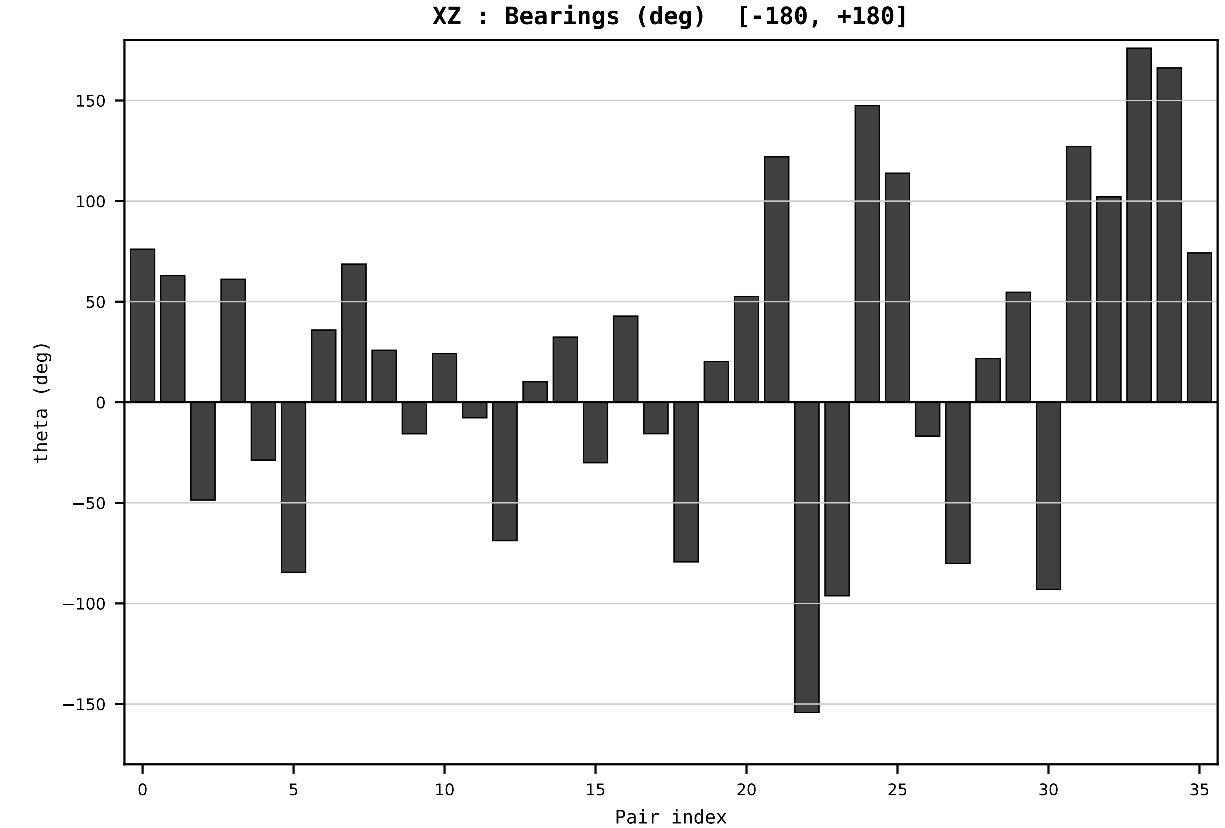


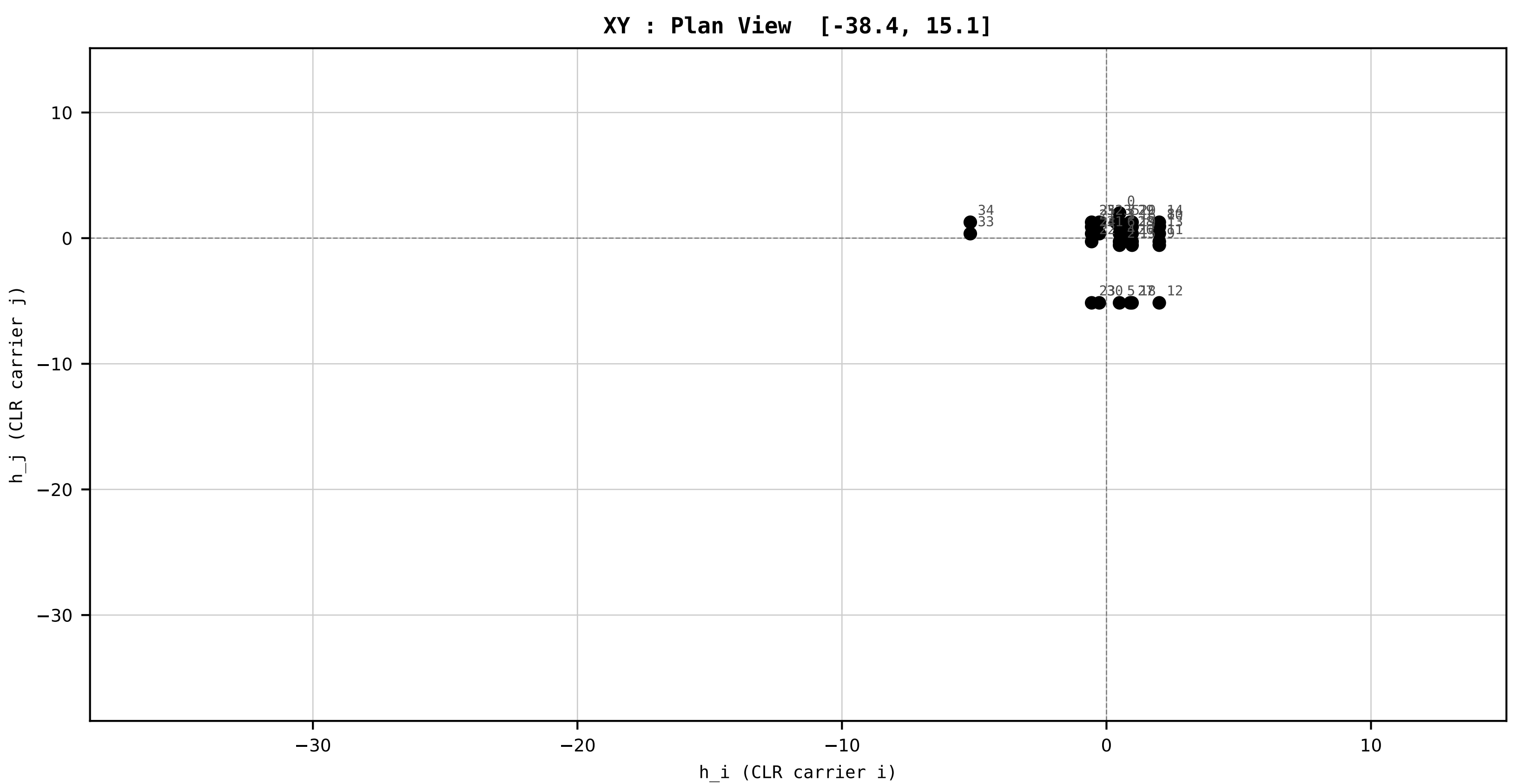


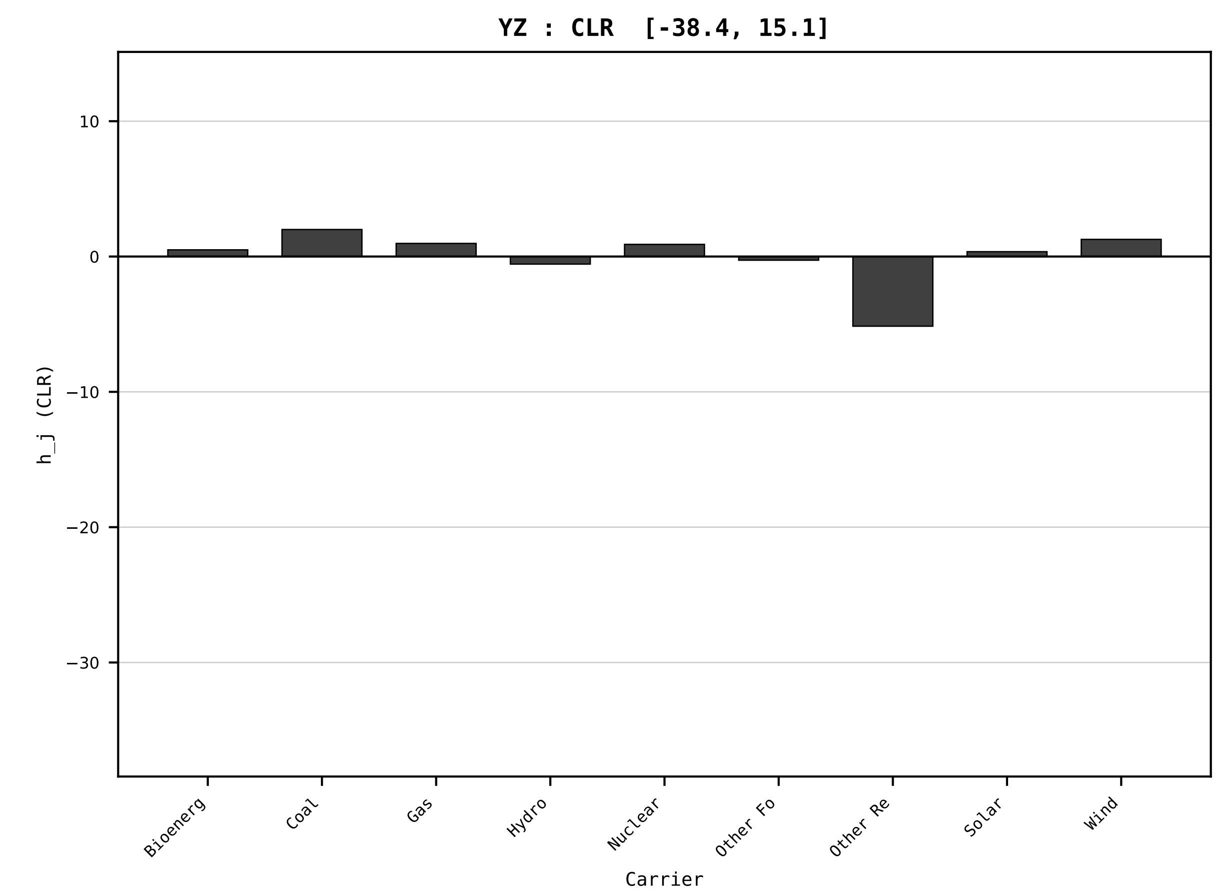
SECTION PLATE t=18 | Year 2018 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2018
t       : 18 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.178543
Ring    = Hs-2
E_metric= 34.5969
kappa_HS= 1267.56
omega   = 2.1475 deg
d_A     = 0.245475
Helm    = Solar
Helm d  = +0.136009
DR      = 7.145 HLR
DR ratio= 1267.6x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



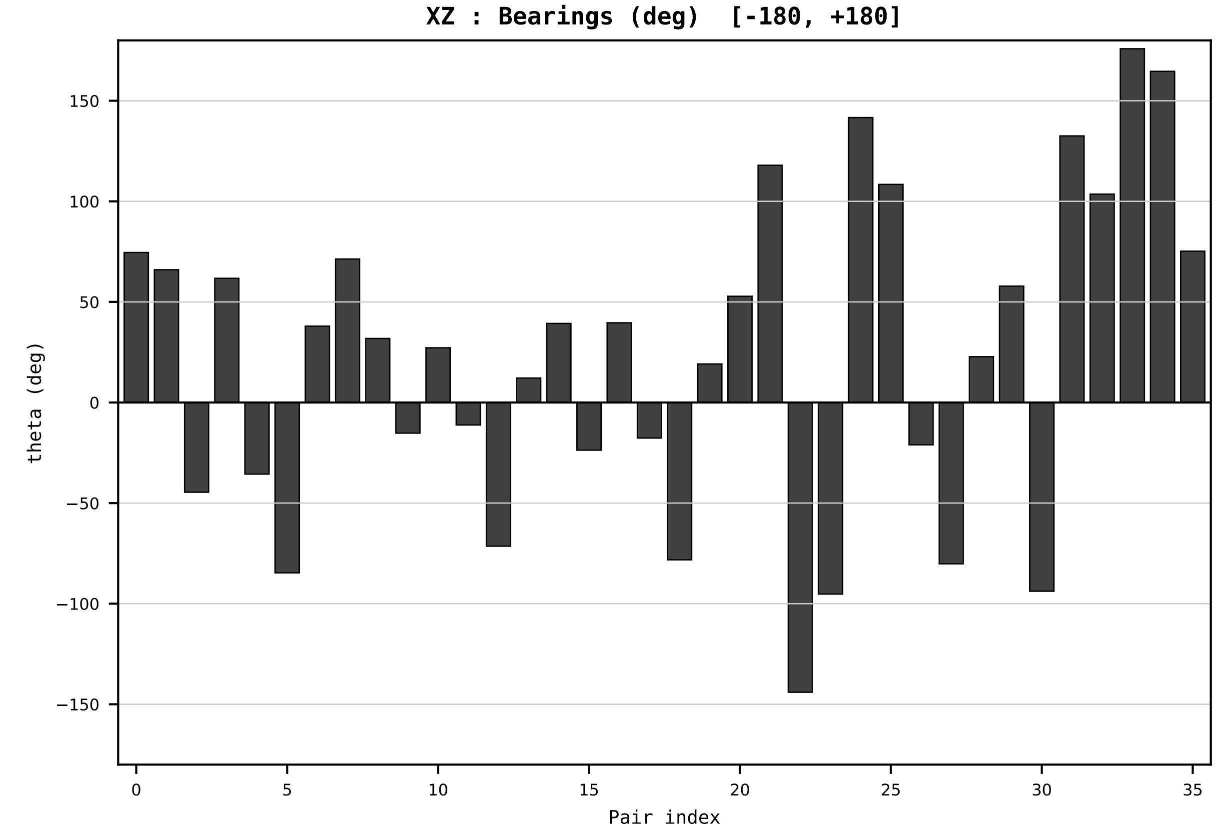


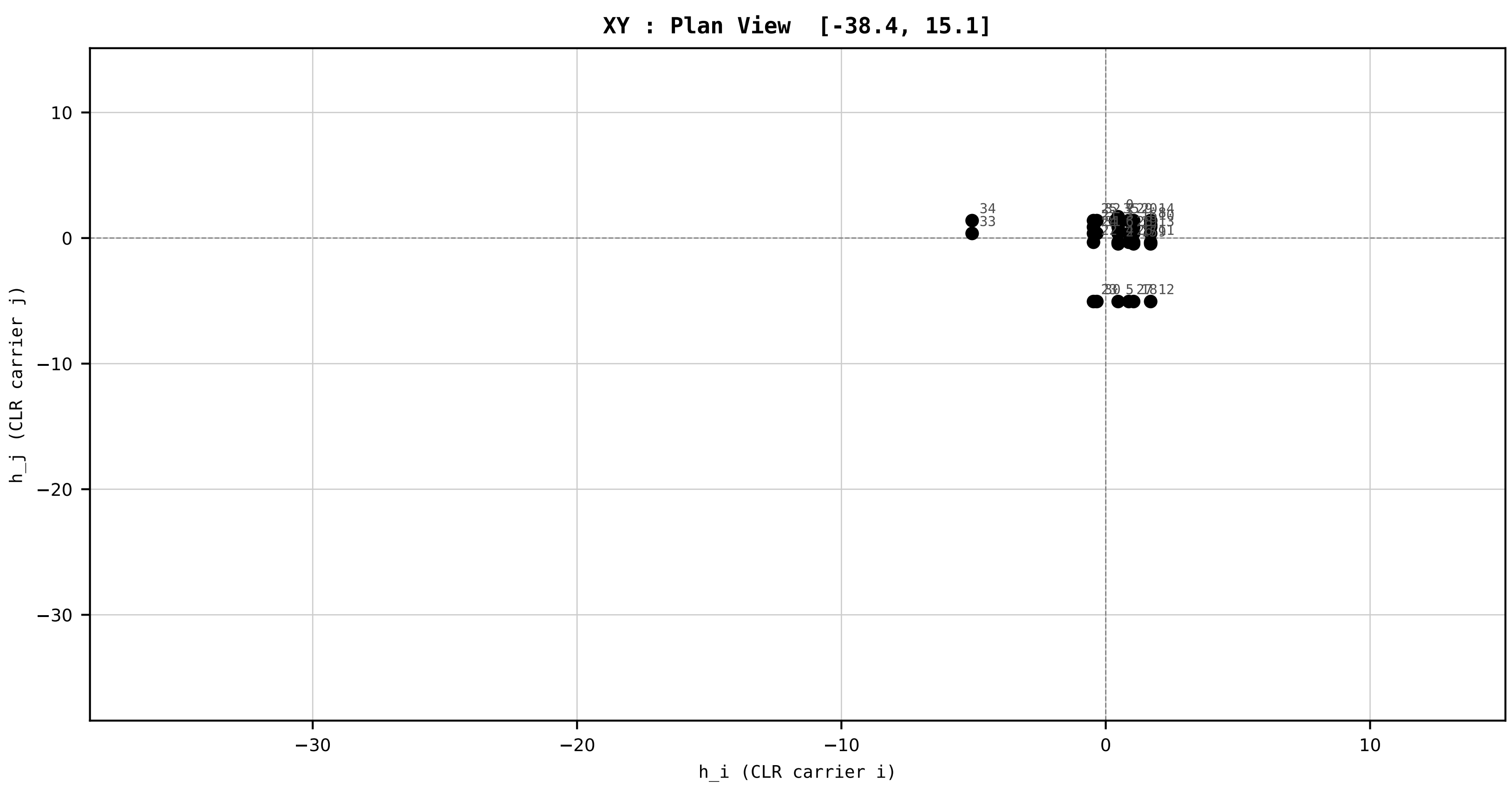


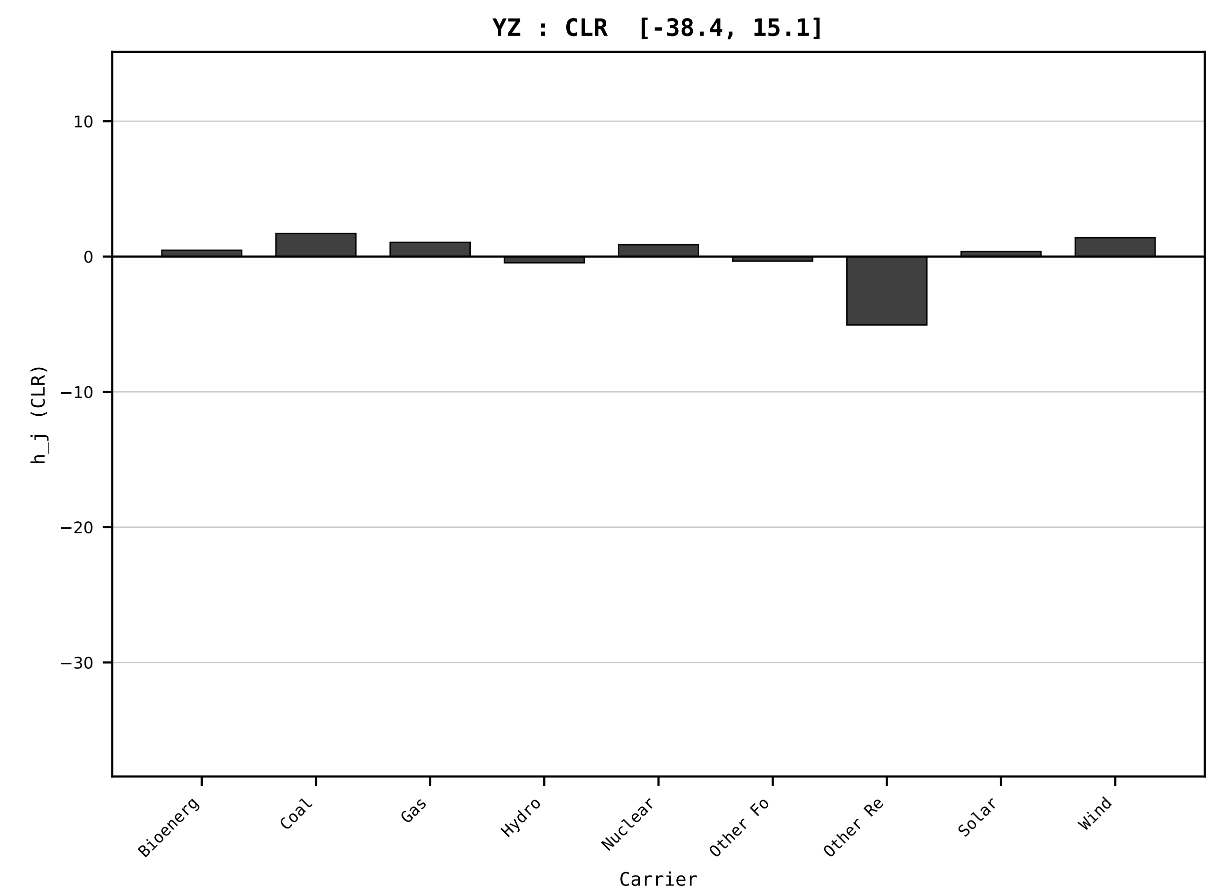
SECTION PLATE t=19 | Year 2019 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2019
t       : 19 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.148916
Ring    = Hs-1
E_metric= 32.9256
kappa_HS= 857.25
omega   = 3.3361 deg
d_A     = 0.367532
Helm    = Coal
Helm d  = -0.297295
DR      = 6.754 HLR
DR ratio= 857.2x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



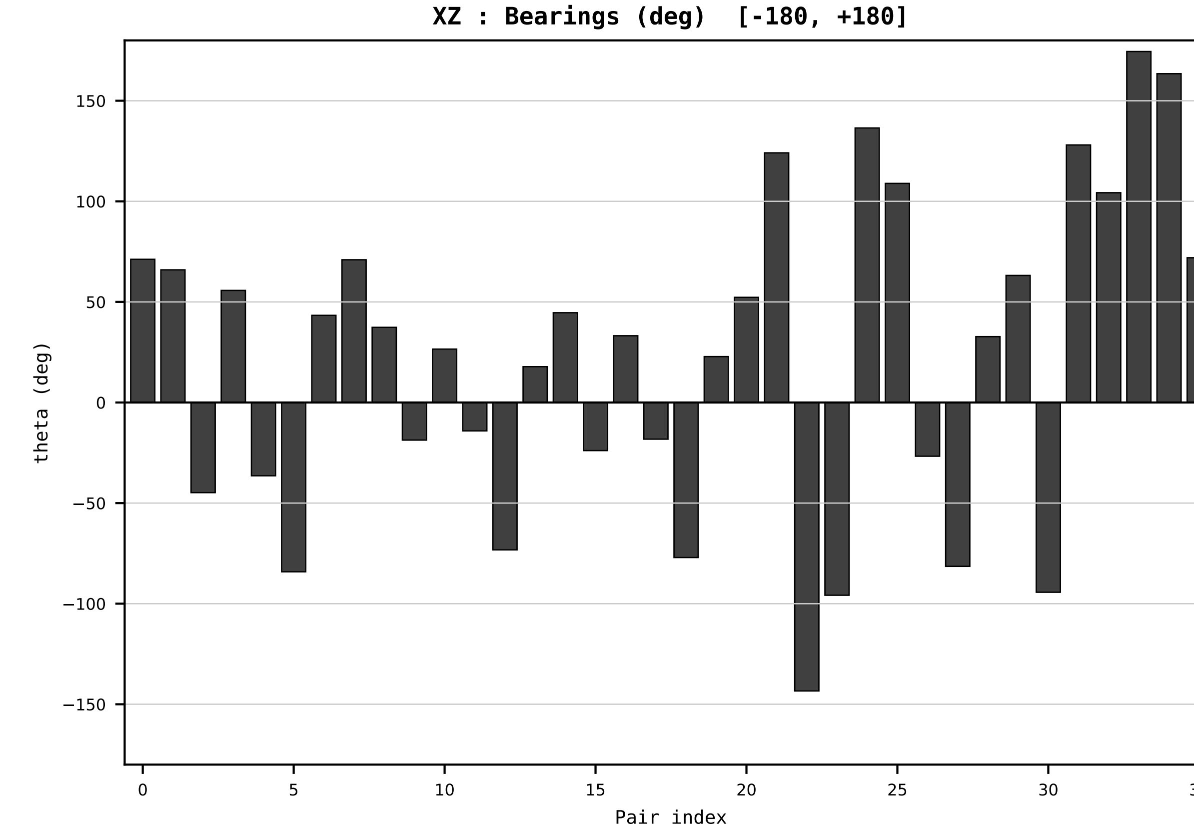


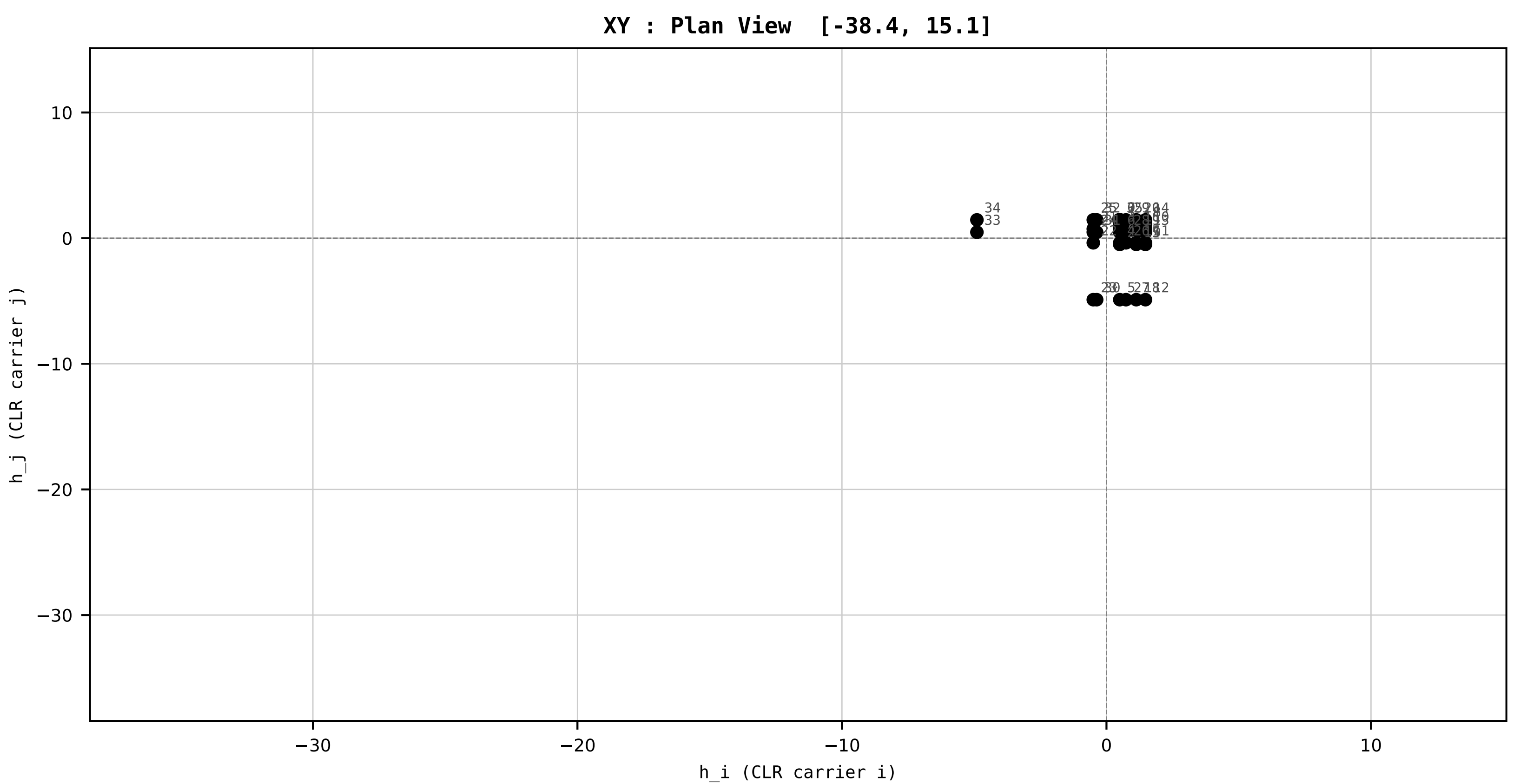


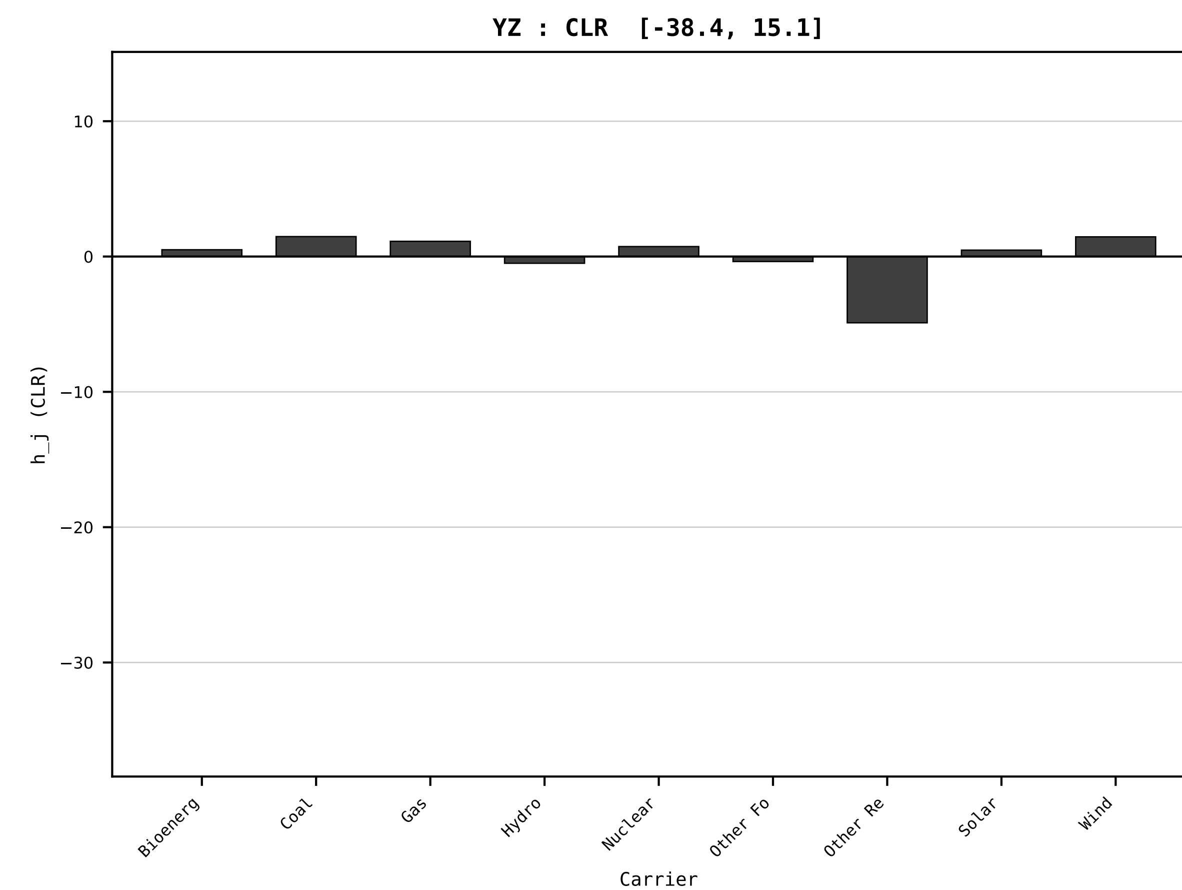
SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2020
t       : 20 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.137161
Ring    = Hs-1
E_metric= 30.9528
kappa_HS= 585.22
omega   = 3.0042 deg
d_A     = 0.343833
Helm    = Coal
Helm d  = -0.225193
DR      = 6.372 HLR
DR ratio= 585.2x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



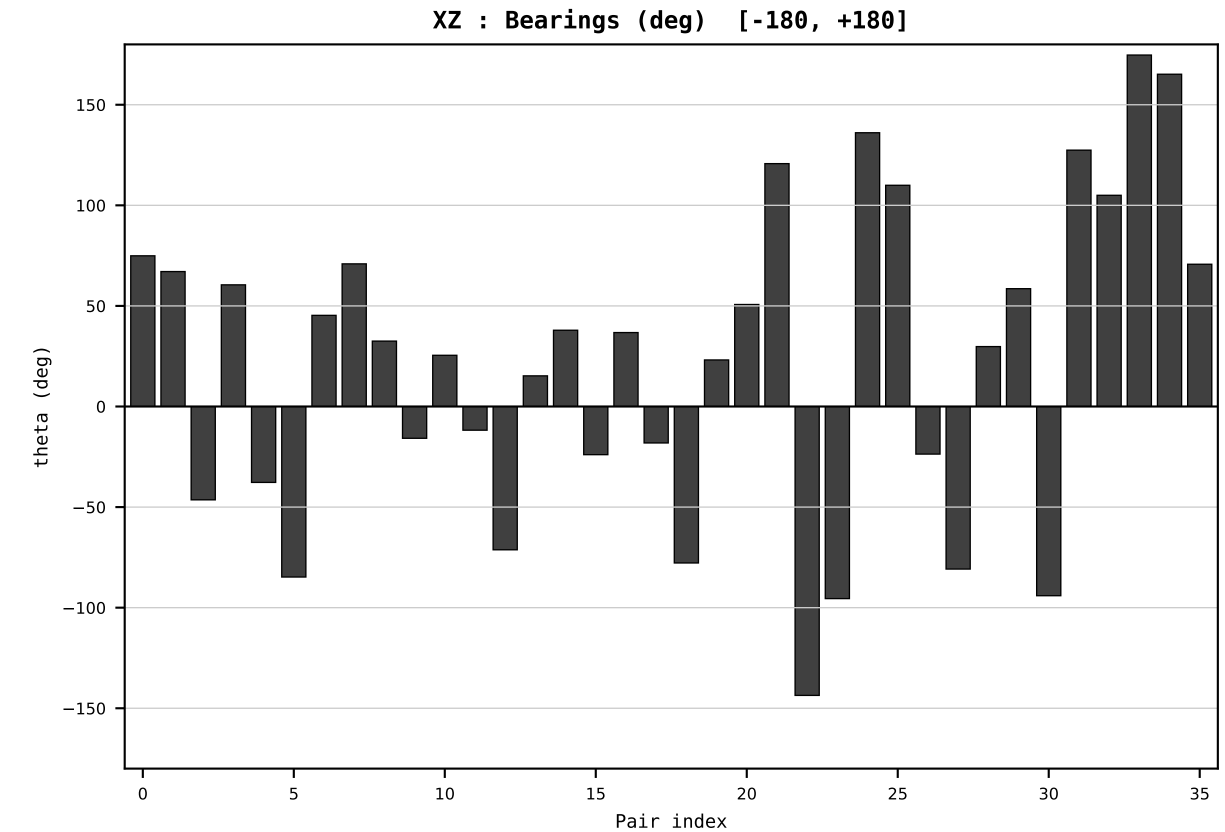


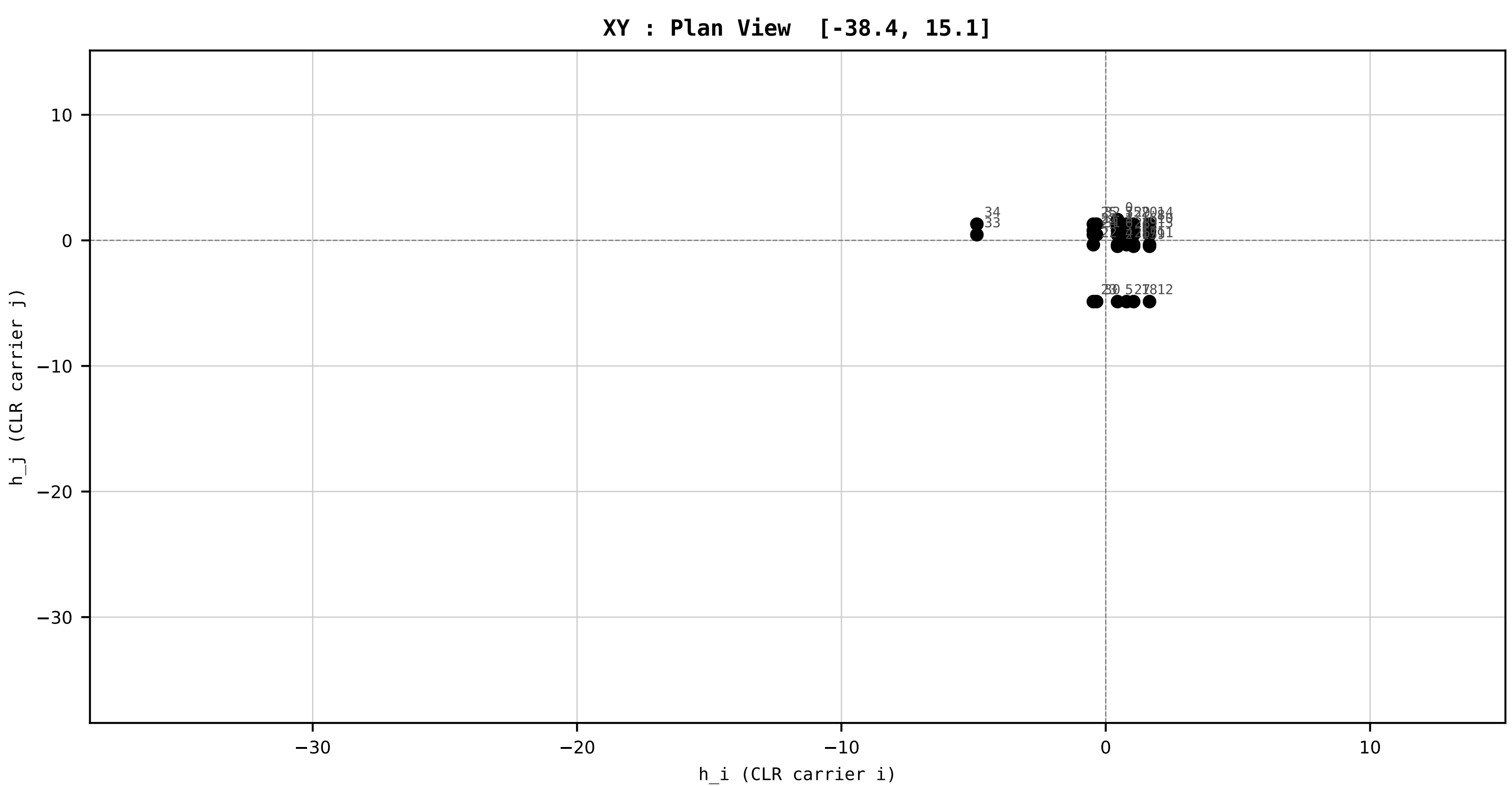


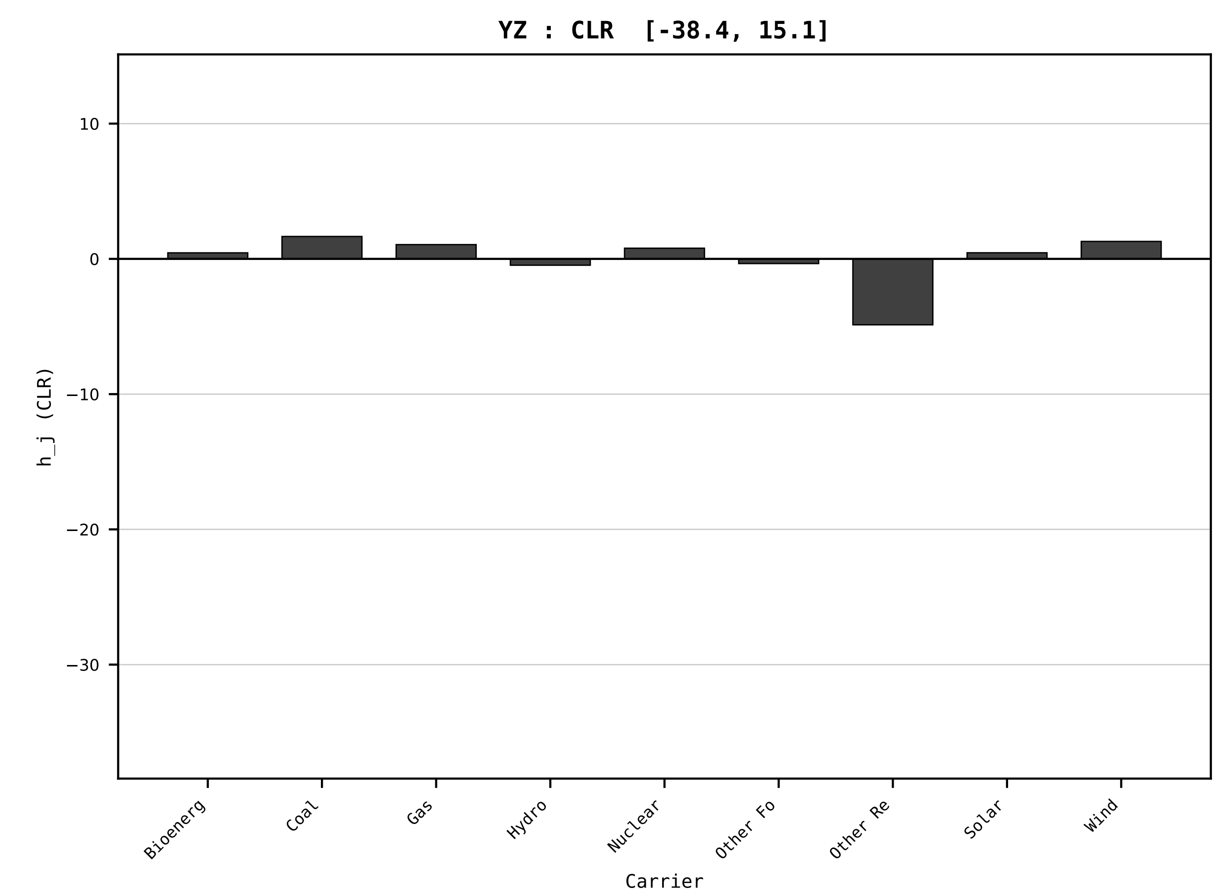
SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2021
t       : 21 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.142585
Ring    = Hs-1
E_metric= 30.6481
kappa_HS= 686.04
omega   = 2.7904 deg
d_A     = 0.271644
Helm    = Coal
Helm d  = +0.182416
DR      = 6.531 HLR
DR ratio= 686.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal
 1=Bioenergy-Gas
 2=Bioenergy-Hydro
 3=Bioenergy-Nuclear
 4=Bioenergy-Other Fossil
 5=Bioenergy-Other Renewables
 6=Bioenergy-Solar
 7=Bioenergy-Wind
 8=Coal-Gas
 9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind
```



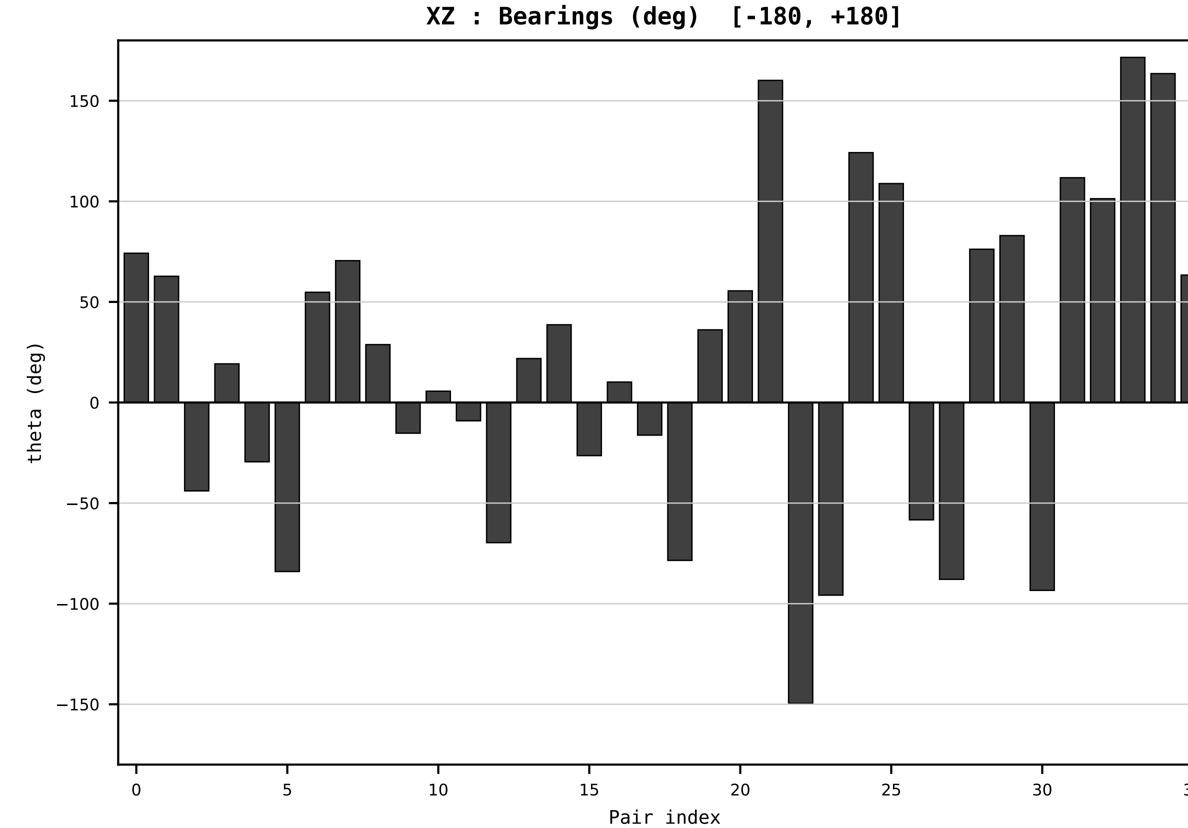


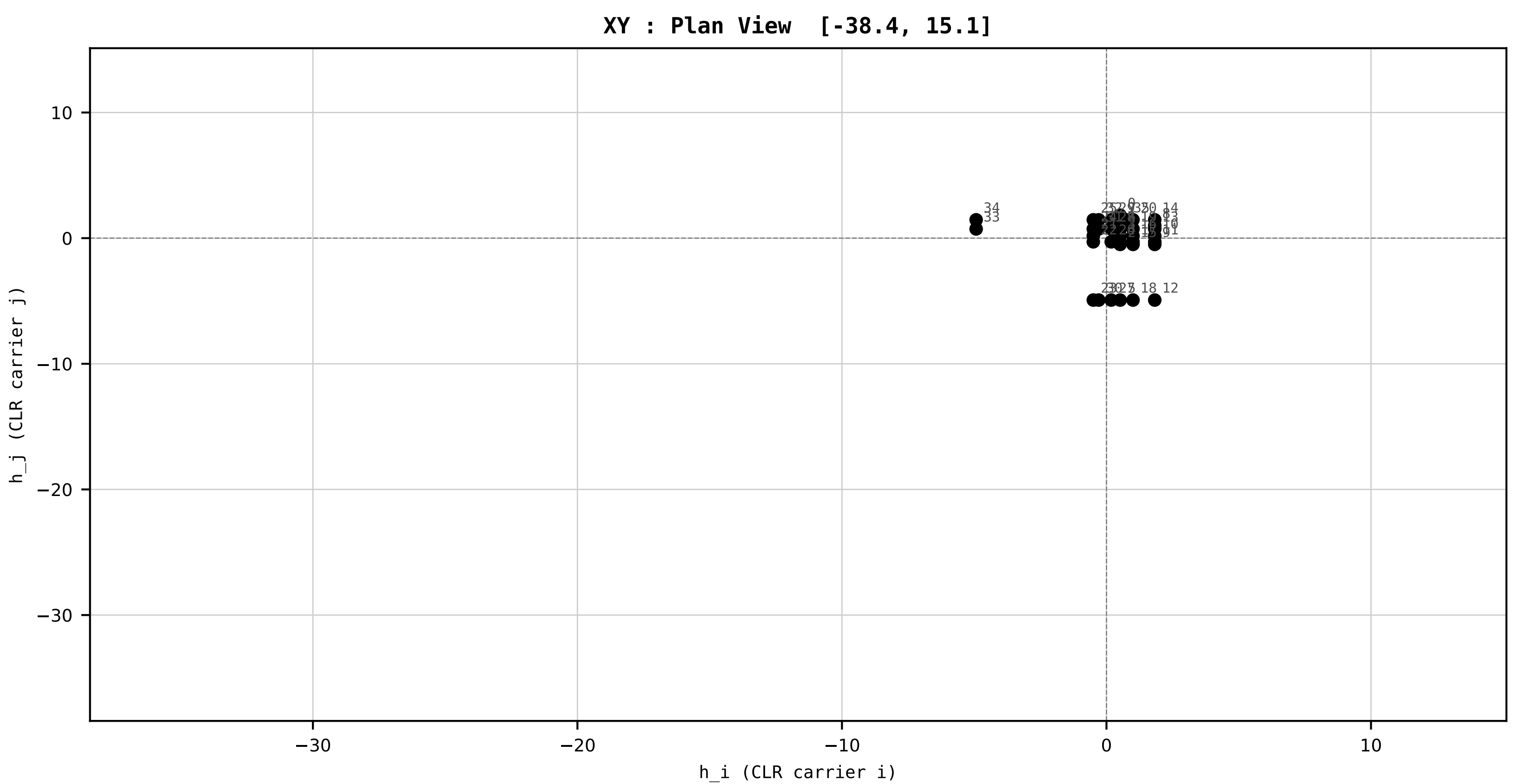


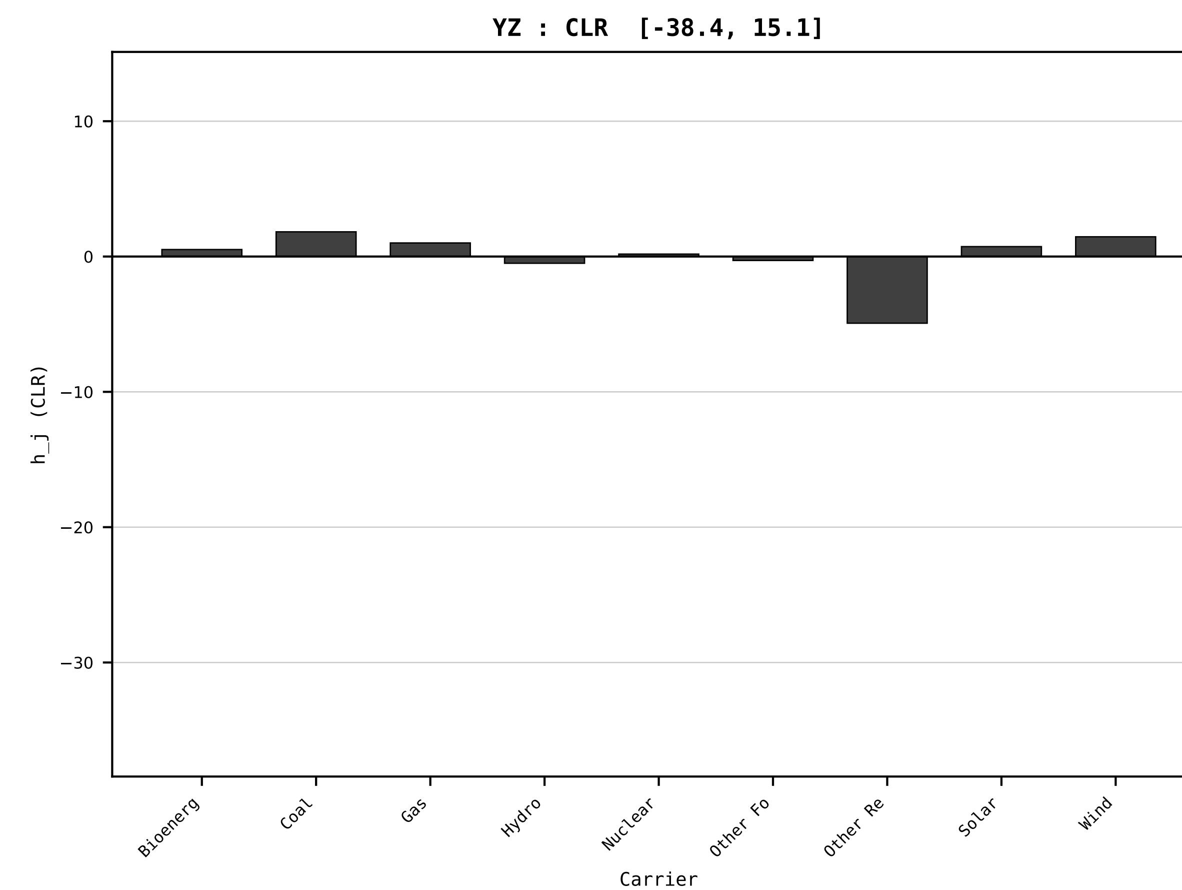
SECTION PLATE t=22 | Year 2022 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2022
t       : 22 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.168193
Ring    = Hs-2
E_metric= 31.9204
kappa_HS= 856.81
omega   = 7.3067 deg
d_A     = 0.721749
Helm    = Nuclear
Helm d  = -0.608539
DR      = 6.753 HLR
DR ratio= 856.8x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal
 1=Bioenergy-Gas
 2=Bioenergy-Hydro
 3=Bioenergy-Nuclear
 4=Bioenergy-Other Fossil
 5=Bioenergy-Other Renewables
 6=Bioenergy-Solar
 7=Bioenergy-Wind
 8=Coal-Gas
 9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind
```







SECTION PLATE t=23 | Year 2023 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

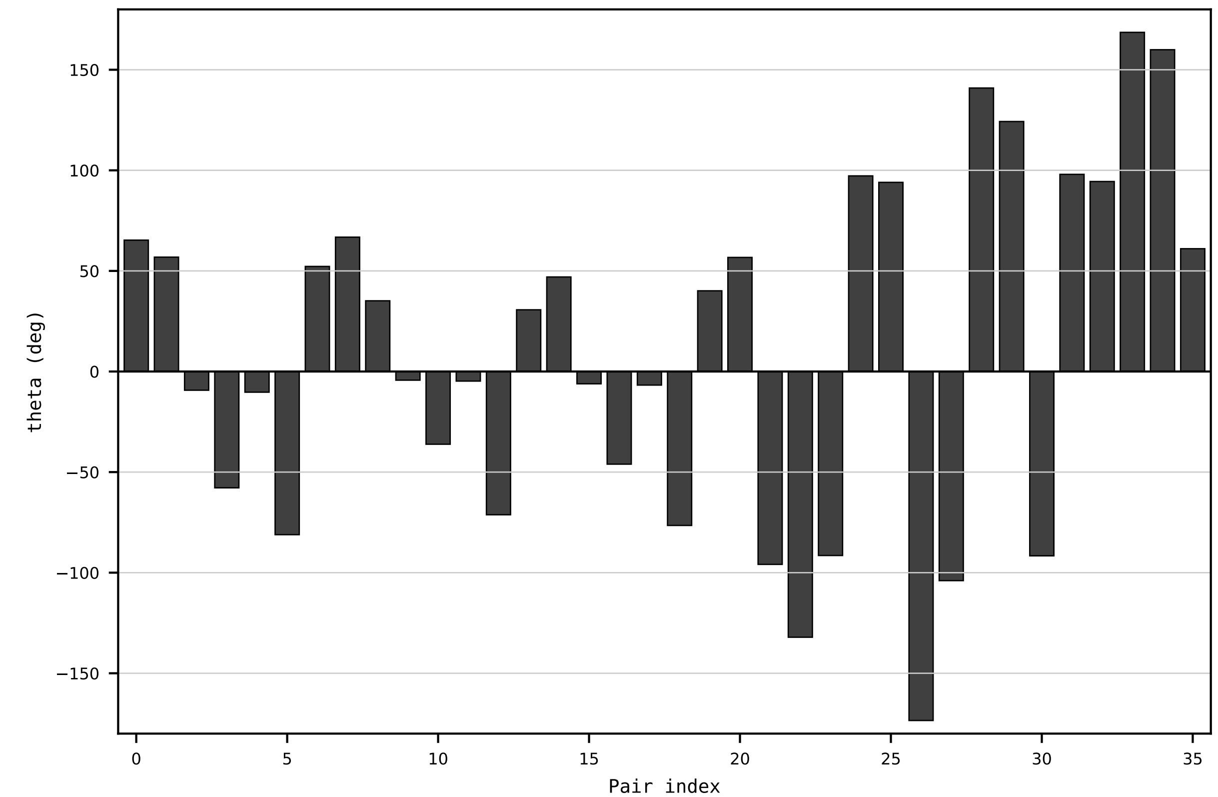
Dataset : ember_DEU_Germany_generation_TWh.csv
Year   : 2023
t   : 23 / 25
D   : 9 carriers
Pairs   : 36 channels

Hs   = 0.177336
Ring   = Hs-2
E_metric= 33.6139
kappa_HS= 739.68
omega   = 15.3266 deg
d_A   = 1.533573
Helm   = Nuclear
Helm d = -1.383172
DR   = 6.606 HLR
DR ratio= 739.7x

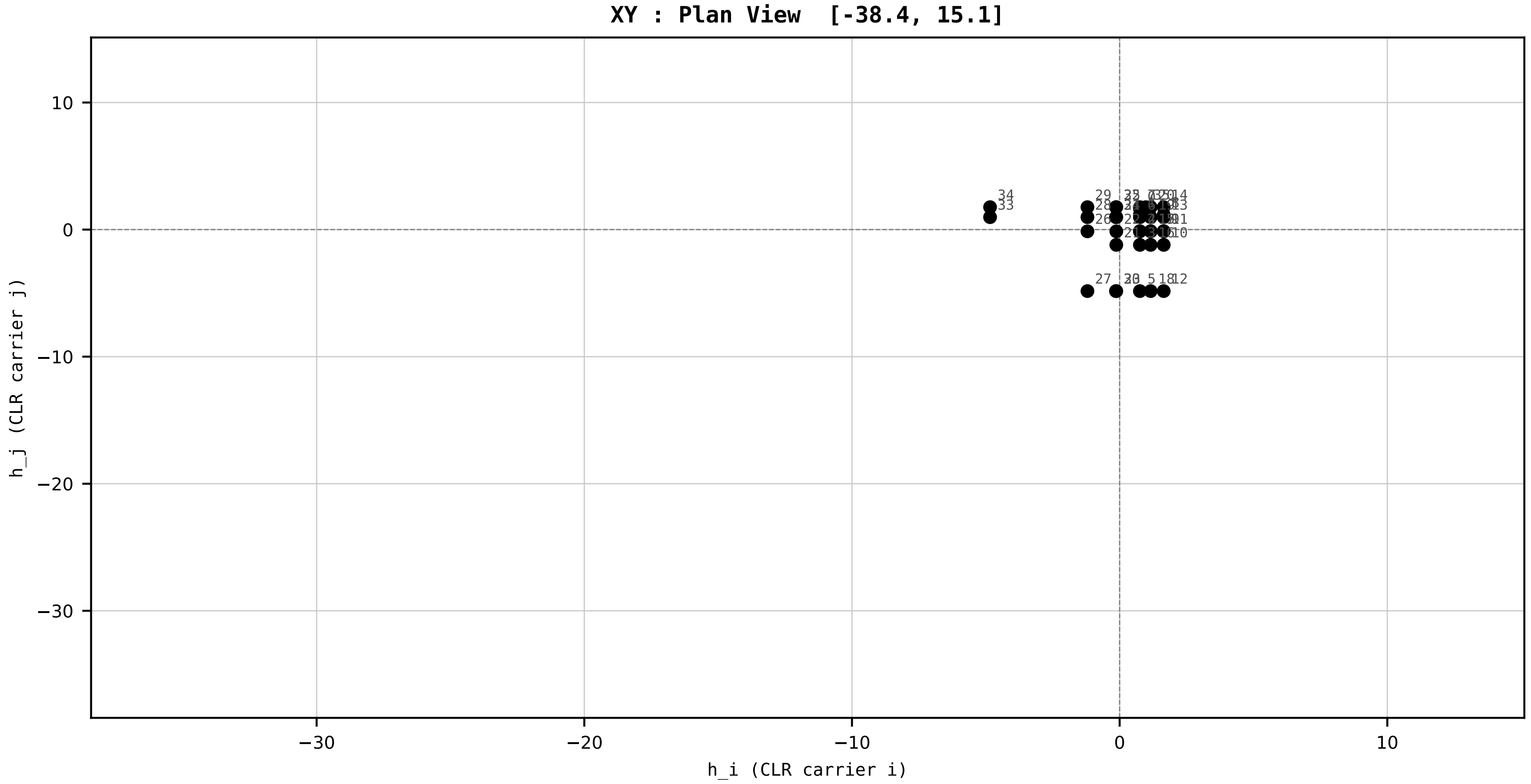
XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:
0=Bioenergy-Coal
1=Bioenergy-Gas
2=Bioenergy-Hydro
3=Bioenergy-Nuclear
4=Bioenergy-Other Fossil
5=Bioenergy-Other Renewables
6=Bioenergy-Solar
7=Bioenergy-Wind
8=Coal-Gas
9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind

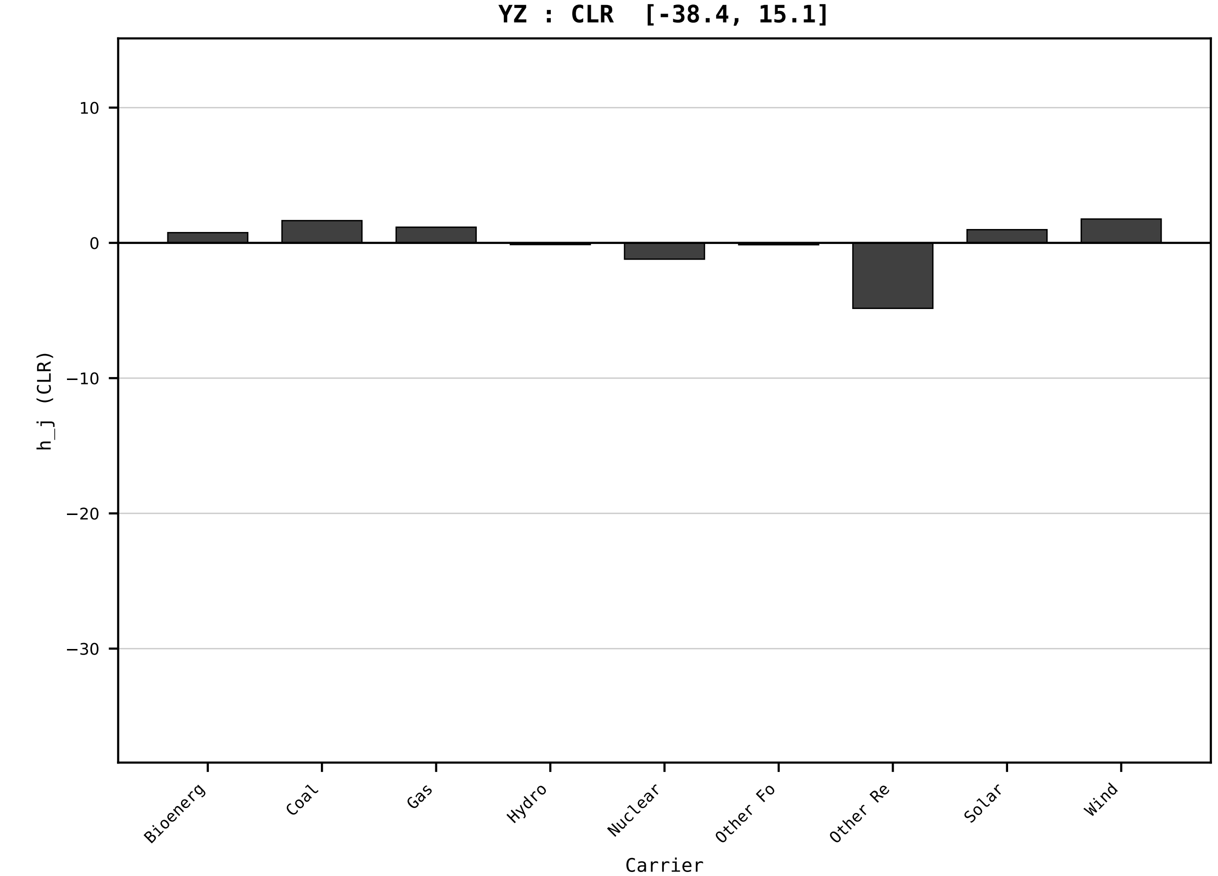
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-38.4, 15.1]



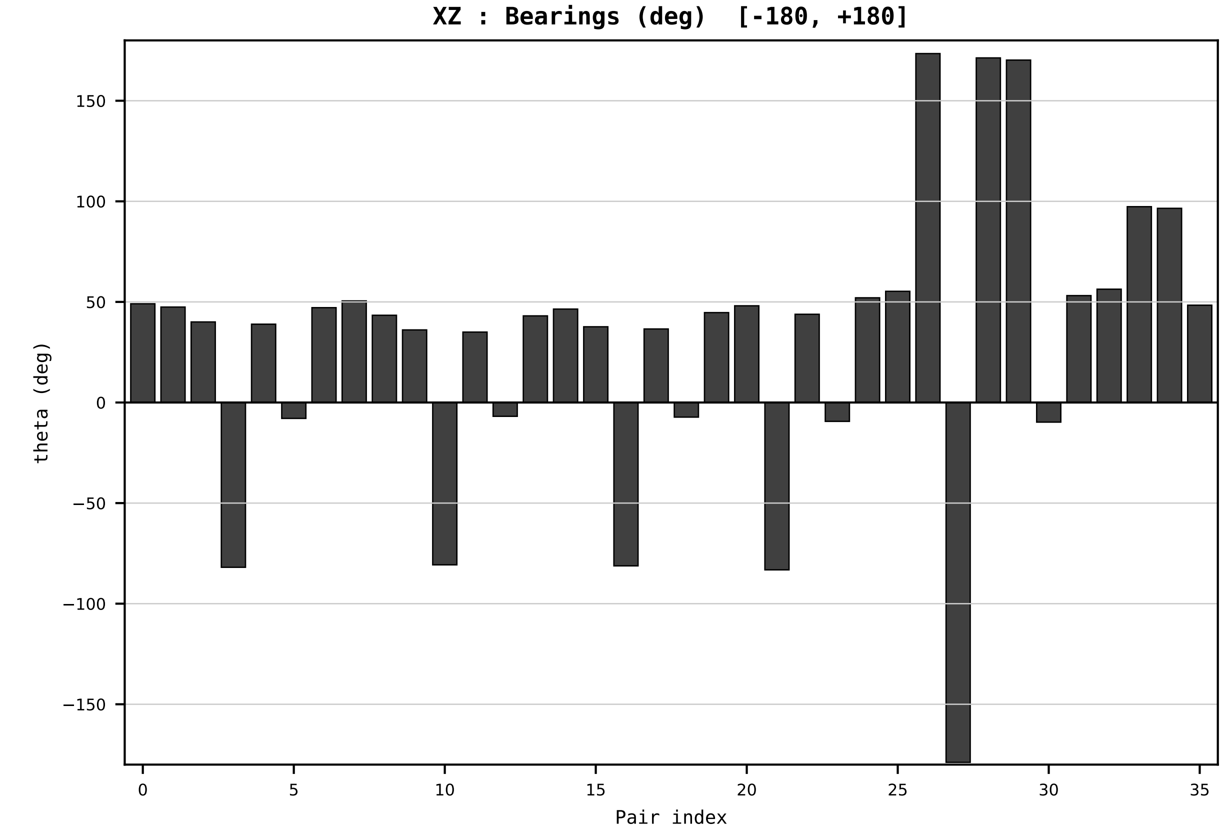
YZ : CLR [-38.4, 15.1]

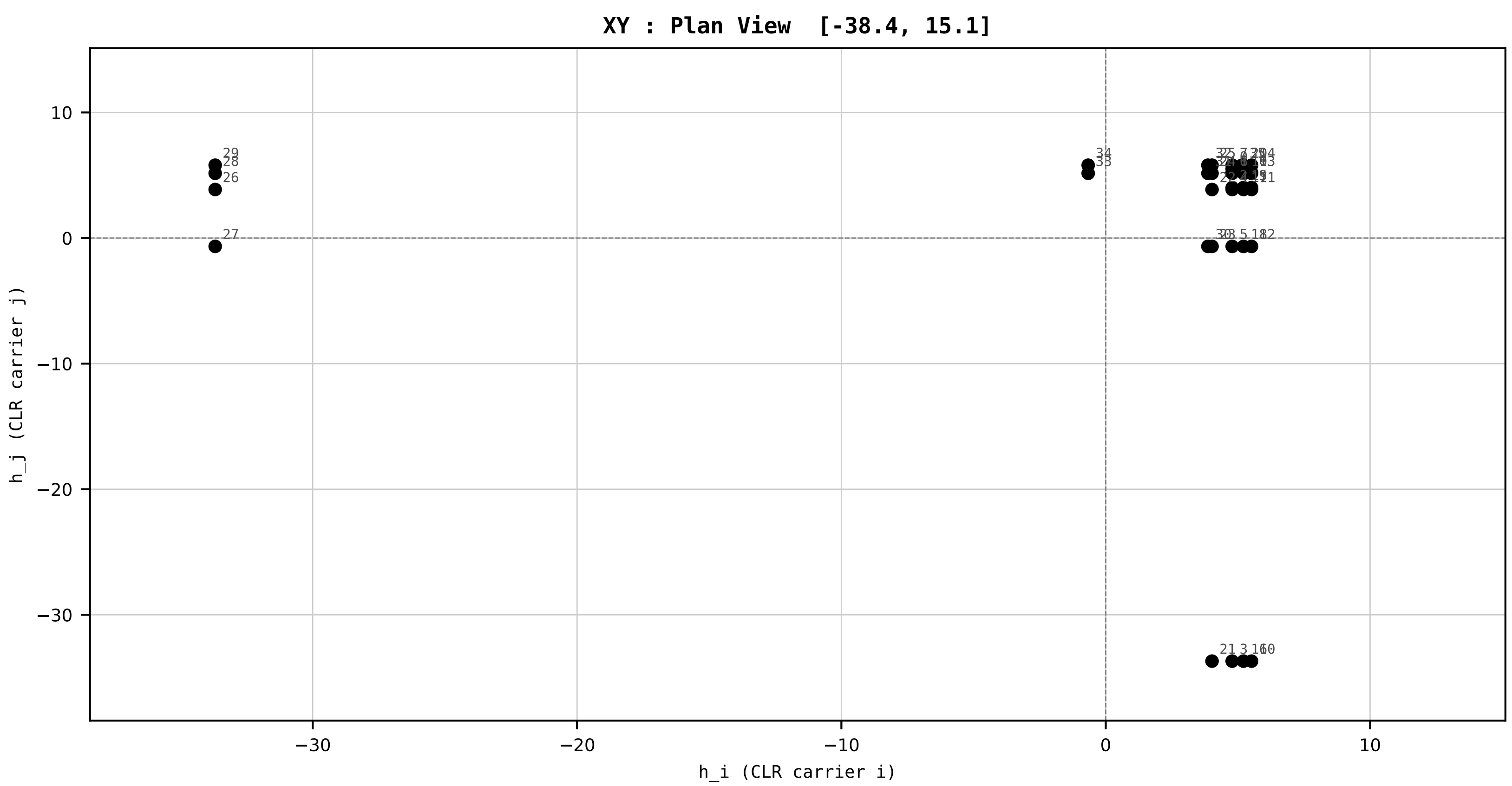


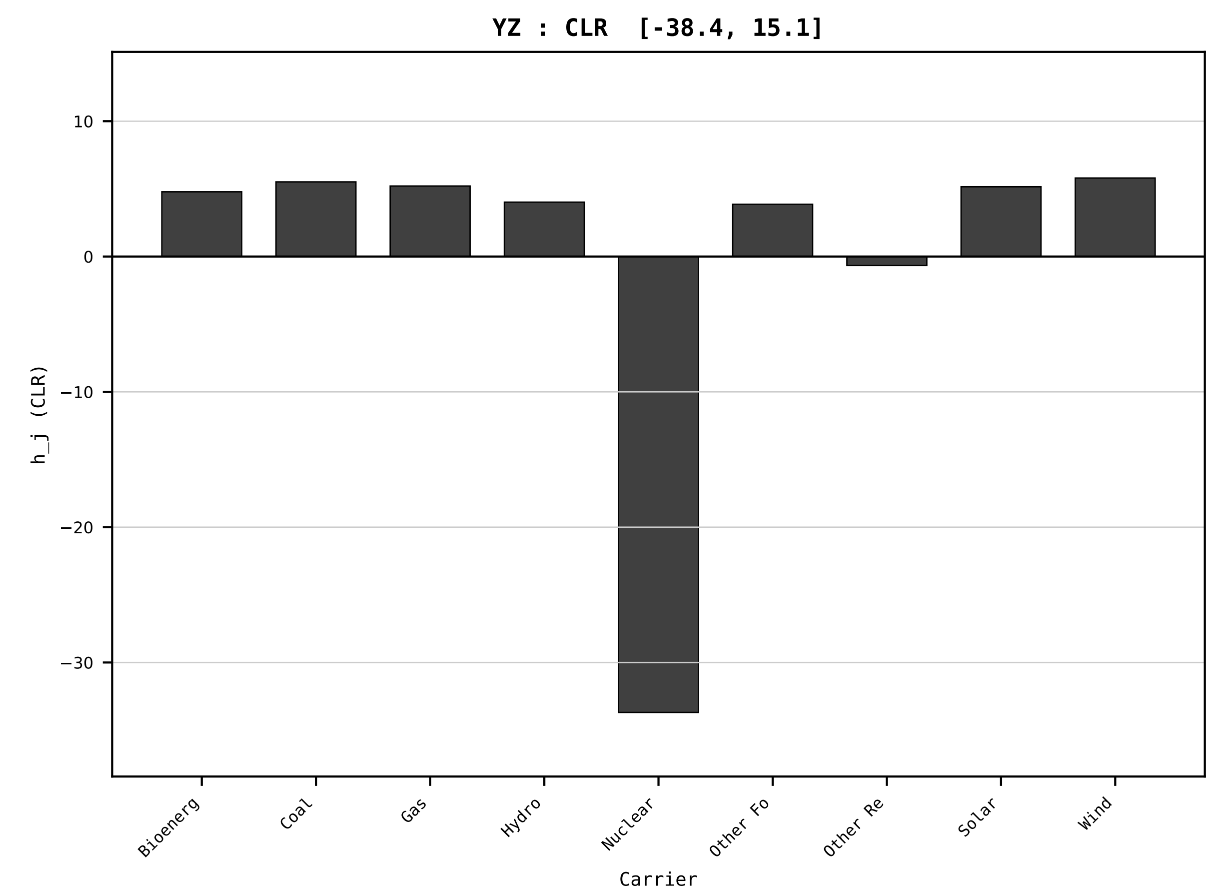
SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2024
t       : 24 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.190666
Ring    = Hs-2
E_metric= 923.2696
kappa_HS= 141599999999999984.00
omega   = 68.5237 deg
d_A     = 34.457790
Helm    = Nuclear
Helm d  = -32.486075
DR      = 39.492 HLR
DR ratio= 141599999126284592.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



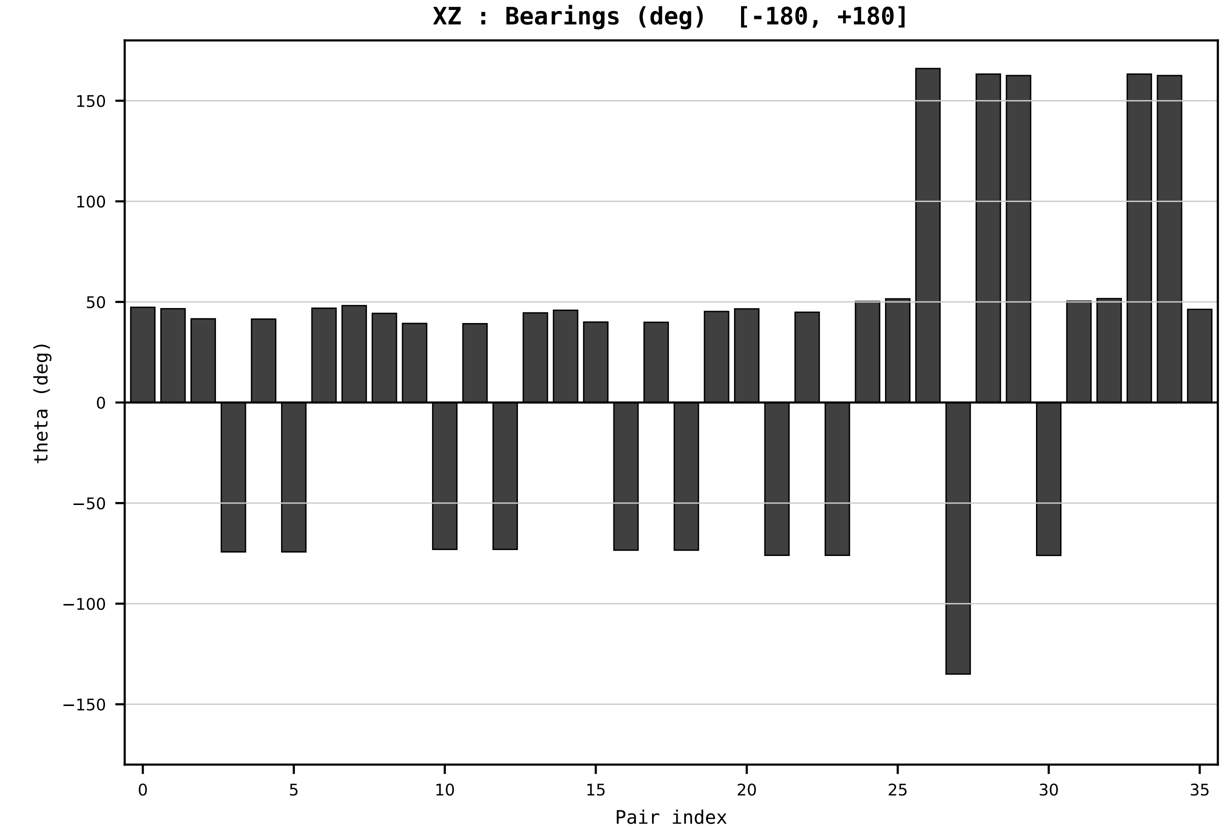


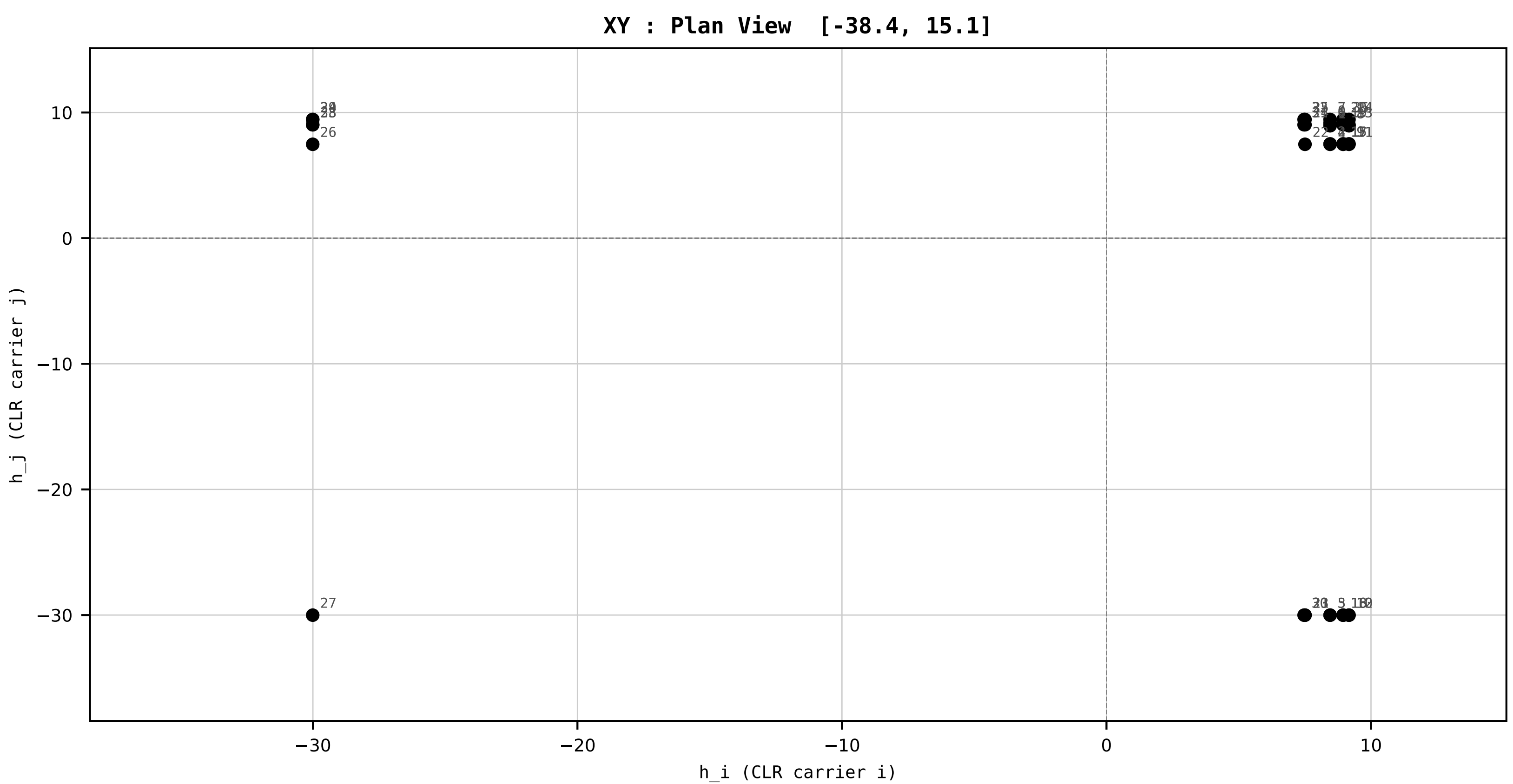


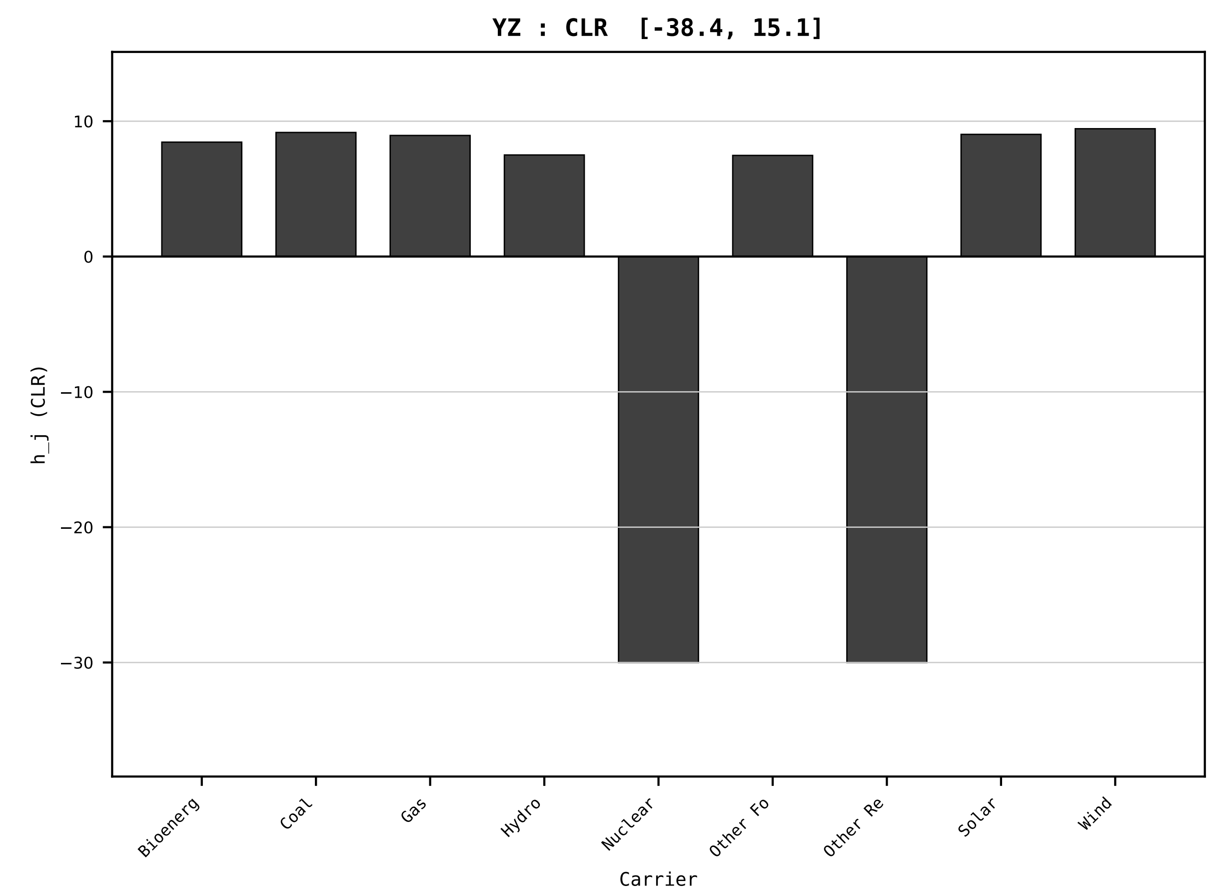
SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year    : 2025
t       : 25 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.195789
Ring    = Hs-2
E_metric= 1633.3021
kappa_HS= 136029999999999984.00
omega   = 40.2530 deg
d_A     = 31.124159
Helm    = Other Renewables
Helm d  = -29.342858
DR      = 39.452 HLR
DR ratio= 136030000740543728.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]
```

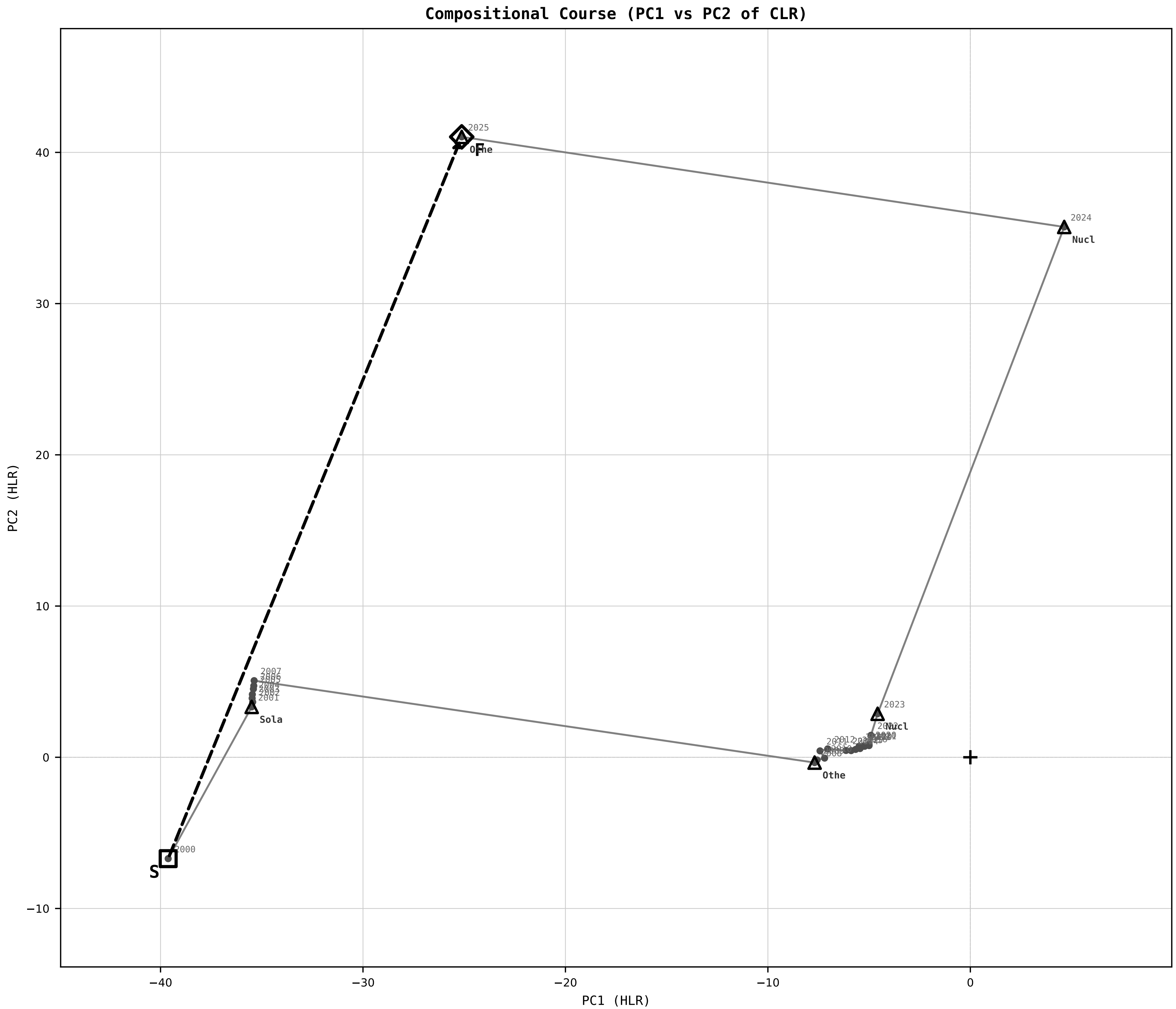
```
PAIR INDEX:
 0=Bioenergy-Coal
 1=Bioenergy-Gas
 2=Bioenergy-Hydro
 3=Bioenergy-Nuclear
 4=Bioenergy-Other Fossil
 5=Bioenergy-Other Renewables
 6=Bioenergy-Solar
 7=Bioenergy-Wind
 8=Coal-Gas
 9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind
```







SYSTEM COURSE PLOT | ember_DEU_Germany_generation_TWh.csv | D=9 N=26 | Years 2000–2025




```
COURSE METRICS

Scale Provenance:
  Y_j → x_j=Y_j/T → h_j=CLR
  → PC projection
  Unit: HLR (Higgins Log-Ratio)

Net distance   = 55.7717 HLR
Path length    = 136.5882 HLR
Directness     = 0.4083

DR start       = 40.231 HLR
  ratio        = 296679999472192320.0x
DR final       = 39.452 HLR
  ratio        = 136030000740543728.0x

E_metric start = 2262.9463
E_metric final = 2319.2562

PC1 loadings:
  Other Re +0.903
  Nuclear  -0.248
  Coal     -0.157

PC2 loadings:
  Nuclear  -0.881
  Solar    +0.310
  Other Re -0.180

Top displacement events:
  2024 Nuclear  d=34.458
  2025 Other Re d=31.124
  2001 Solar    d=30.338
  2008 Other Re d=28.812
  2023 Nuclear  d=1.534

The instrument reads.
The expert decides.
The loop stays open.
```

View B

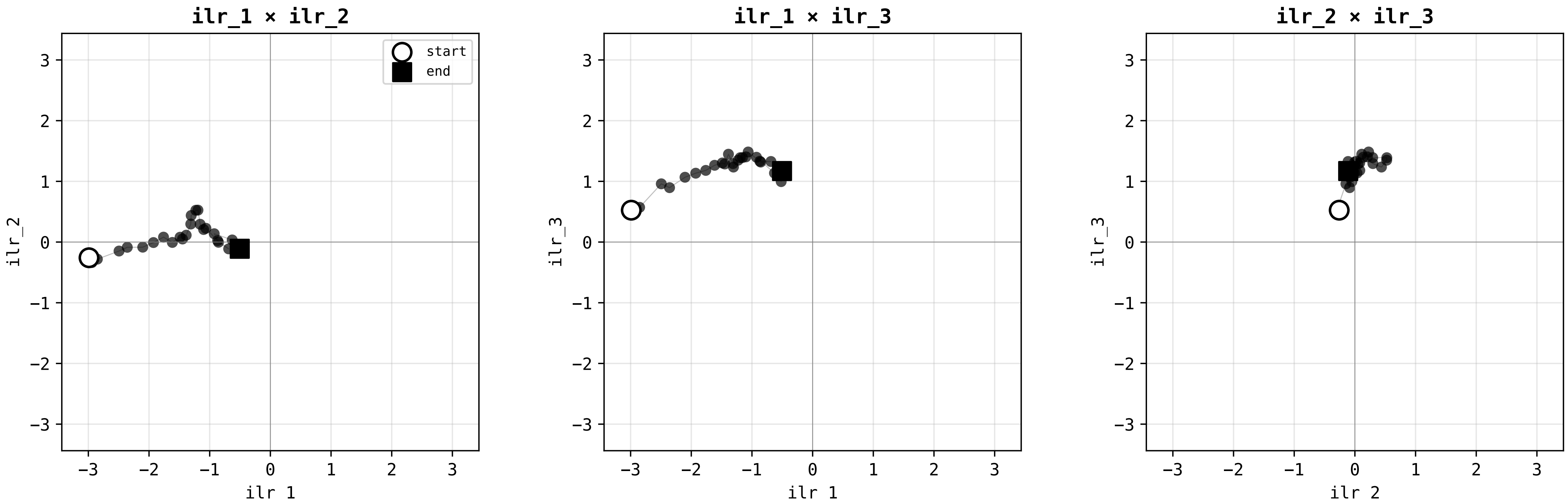
ILR-Helmert Orthogonal Triplet Plate

Orthonormal reading · three scatter faces of first three Helmert ILR axes

§ DEU Germany

ILR-Helmert Orthogonal Triplet — DEU Germany · Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes · N=26 years · D=9 carriers



Helmert basis loadings (which carriers each ILR axis contrasts):

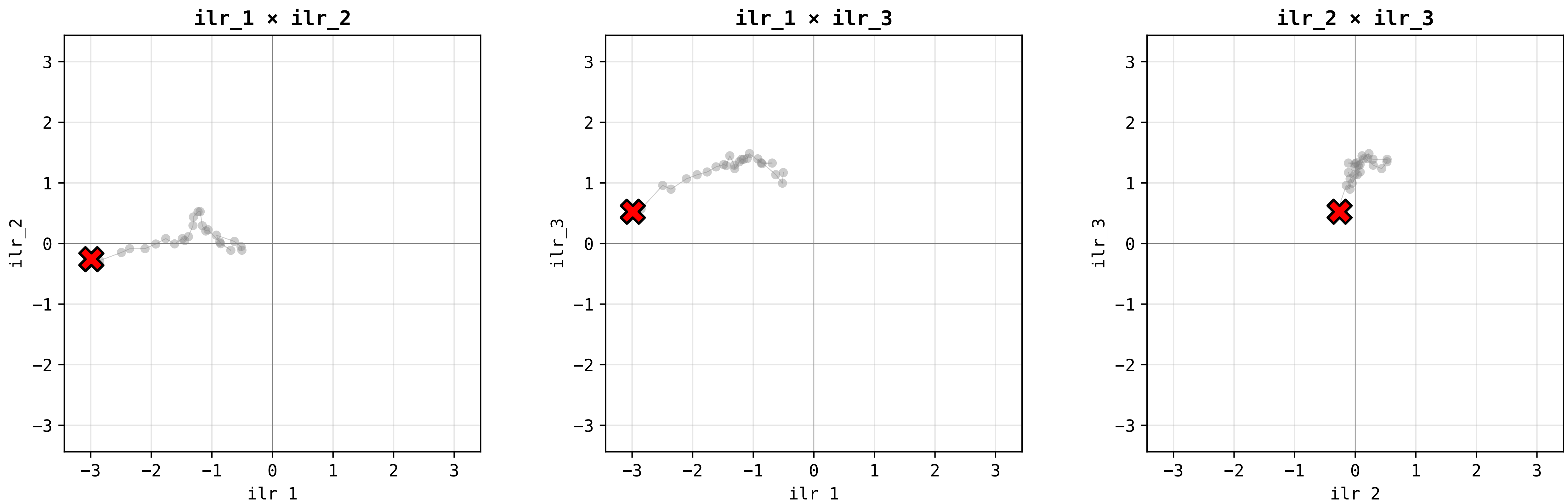
ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

Trajectory:
start year: 2000
end year: 2025
N readings: 26

Display:
fixed scale ±3.44 ILR
Order 1 (first principles)
per Output Doctrine v1.0

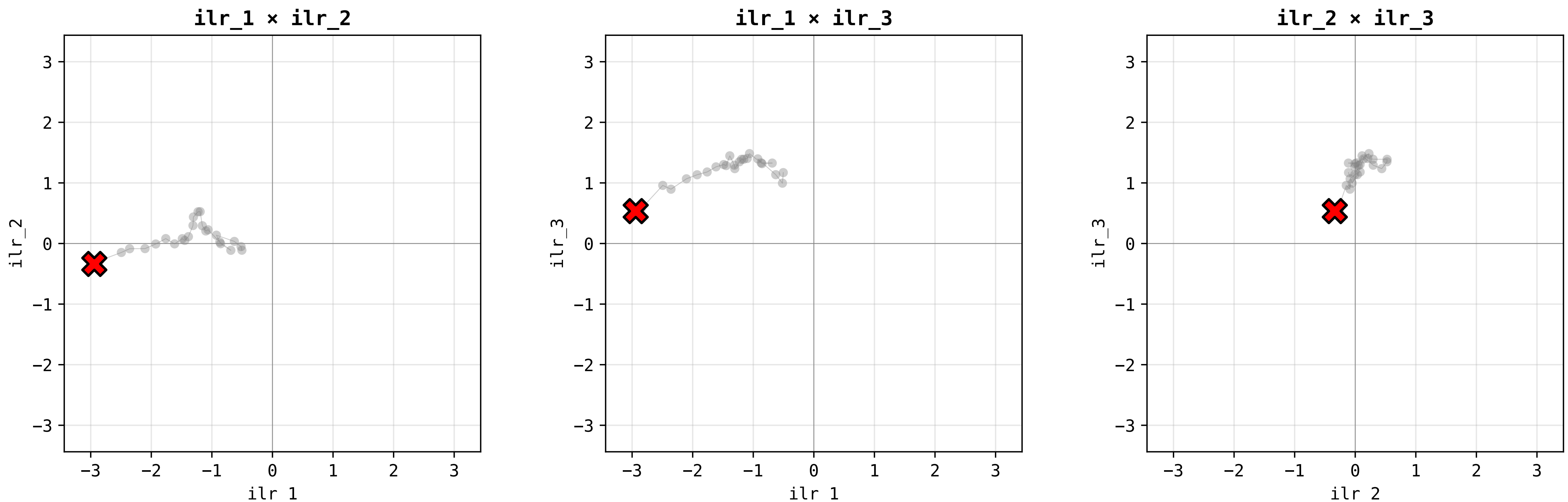
Symbology:
○ start · ▪ end
· intermediate years

ILR Triplet — DEU Germany · year 2000 (1/26)



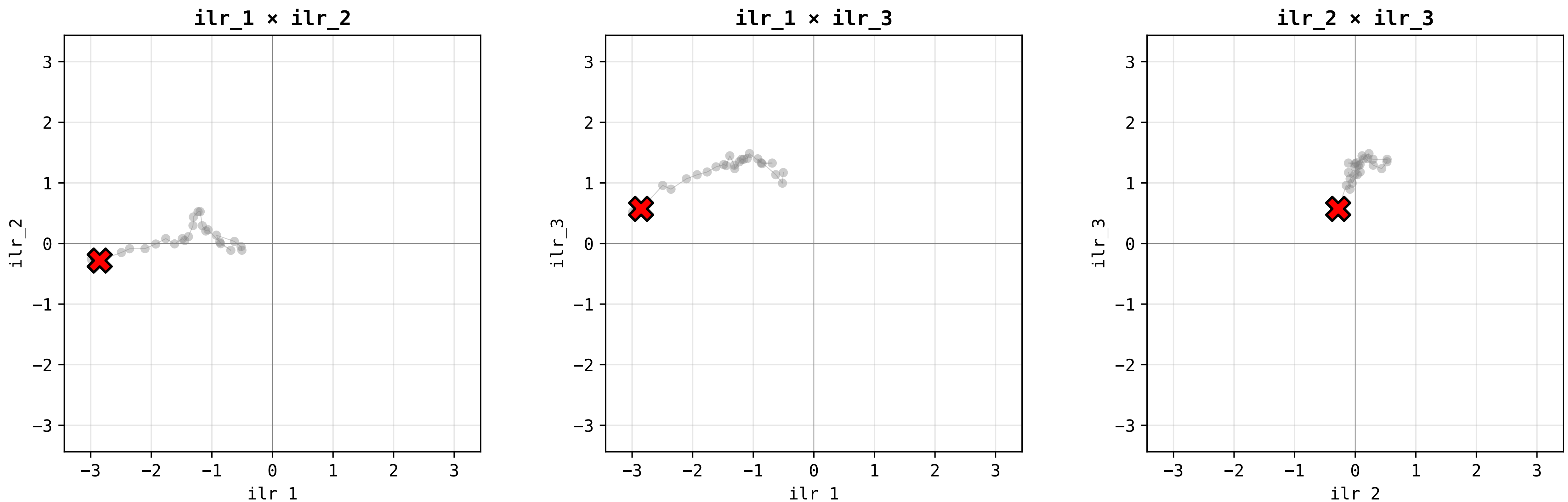
X = current year (2000) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2001 (2/26)



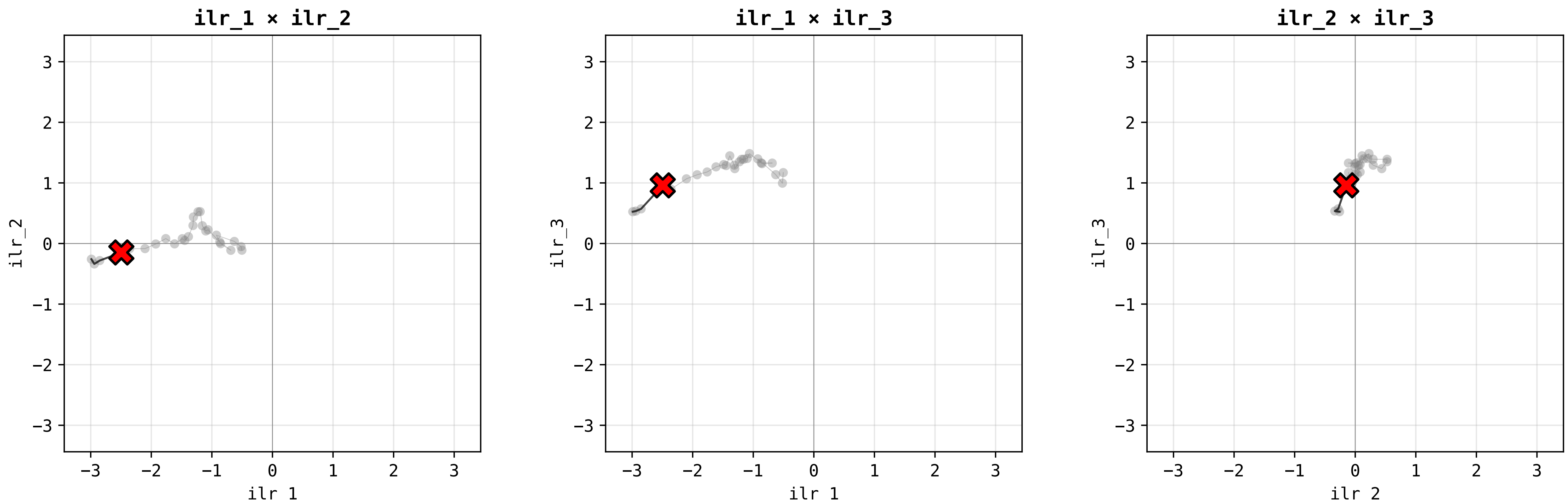
X = current year (2001) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2002 (3/26)



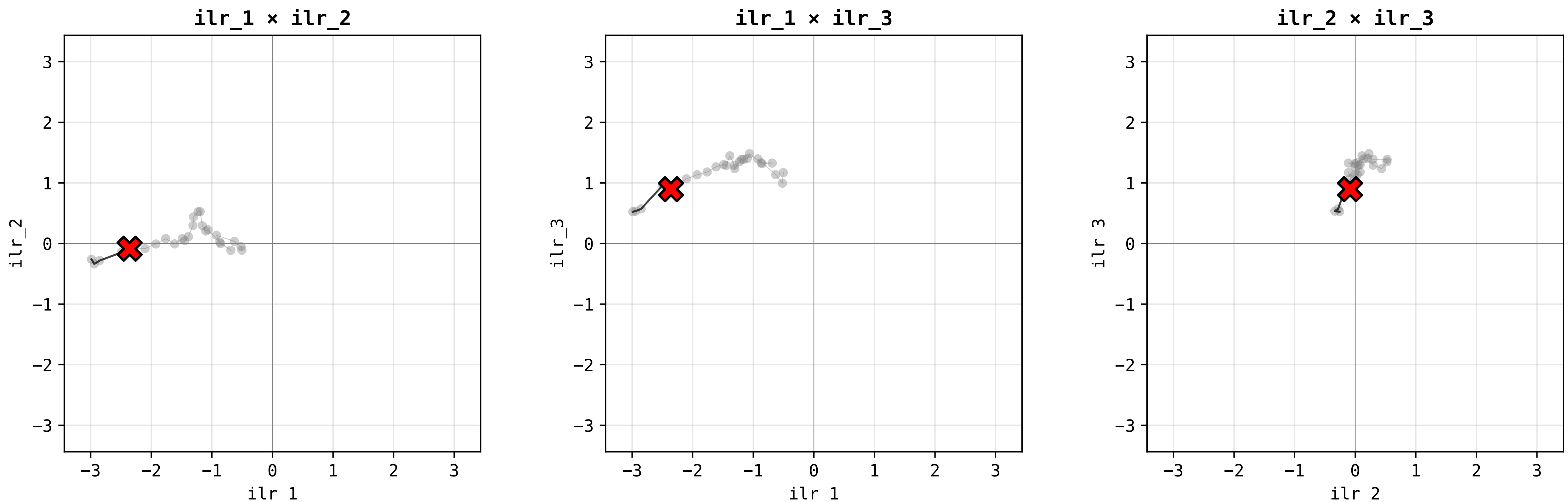
X = current year (2002) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2003 (4/26)



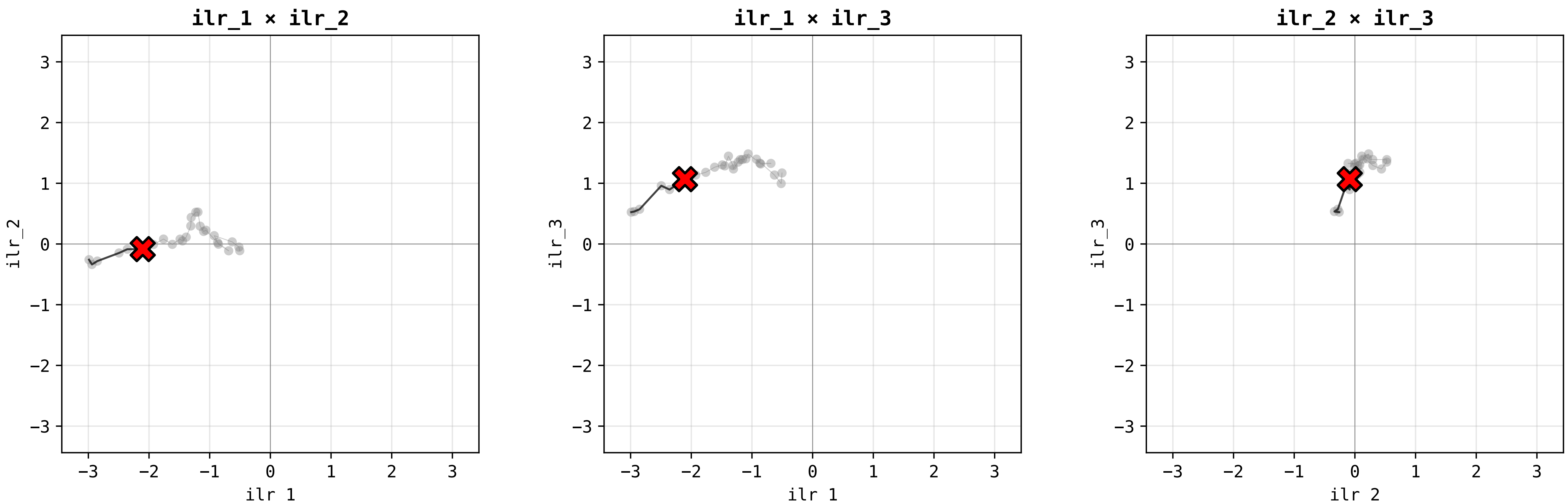
X = current year (2003) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2004 (5/26)



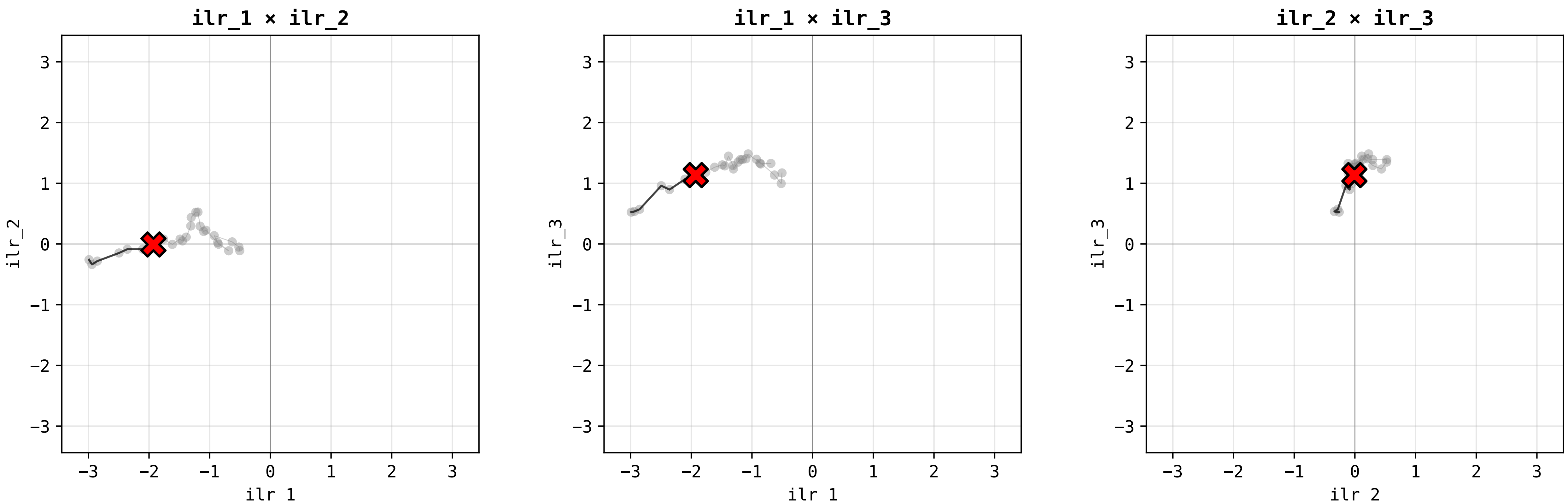
X = current year (2004) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2005 (6/26)



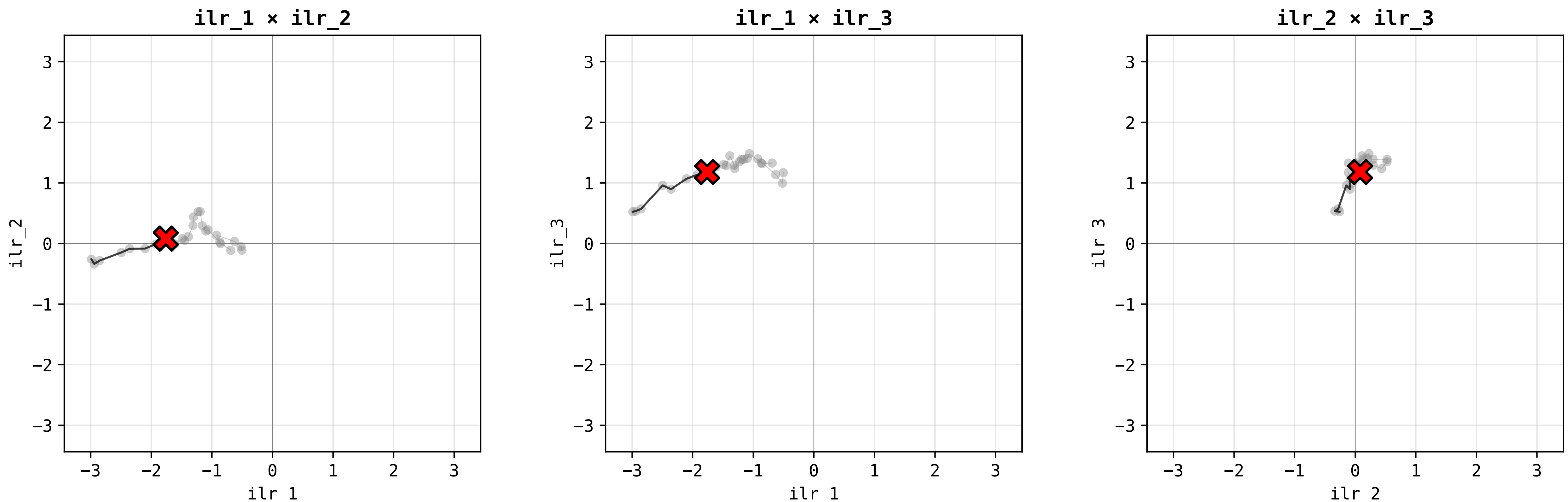
X = current year (2005) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2006 (7/26)



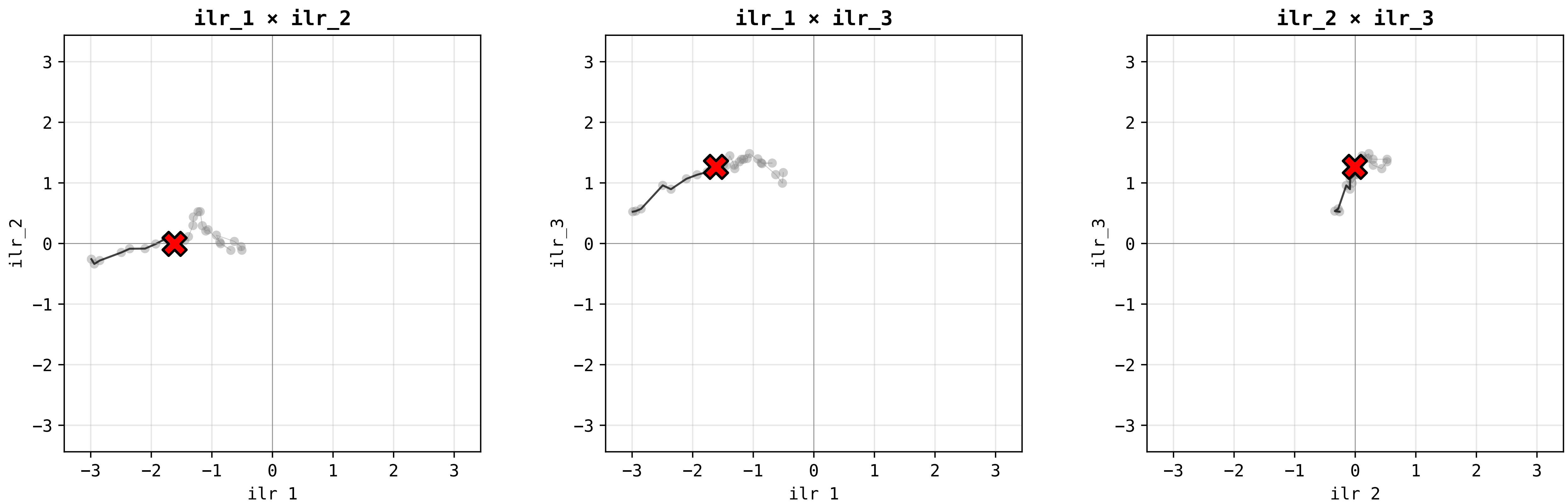
X = current year (2006) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2007 (8/26)



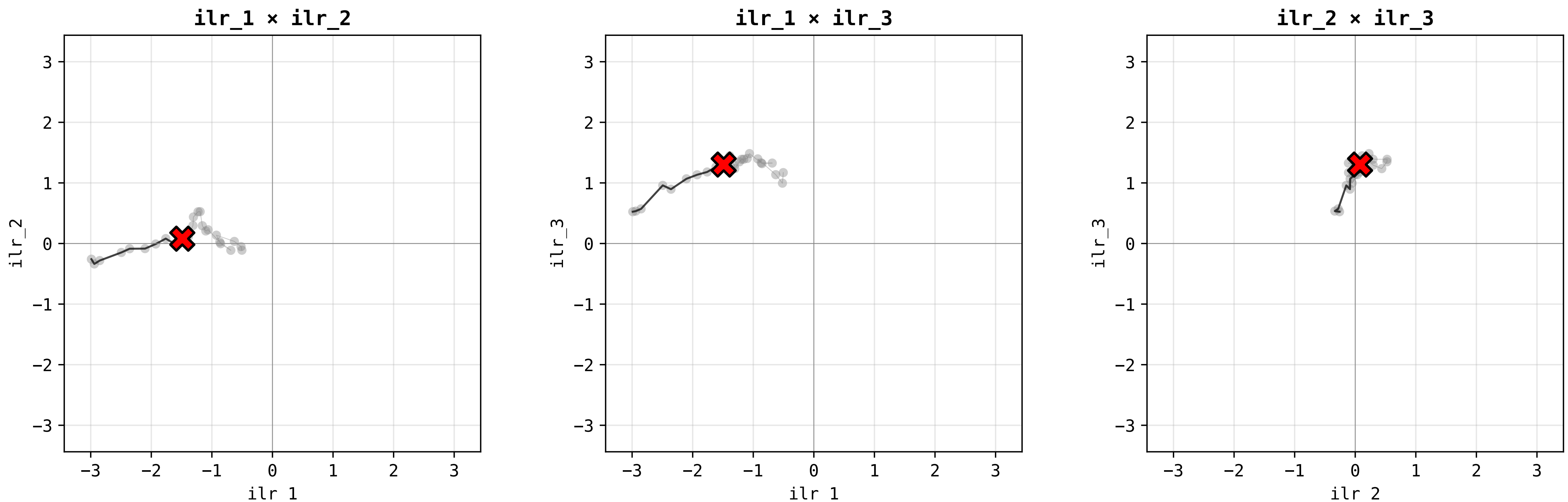
X = current year (2007) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2008 (9/26)



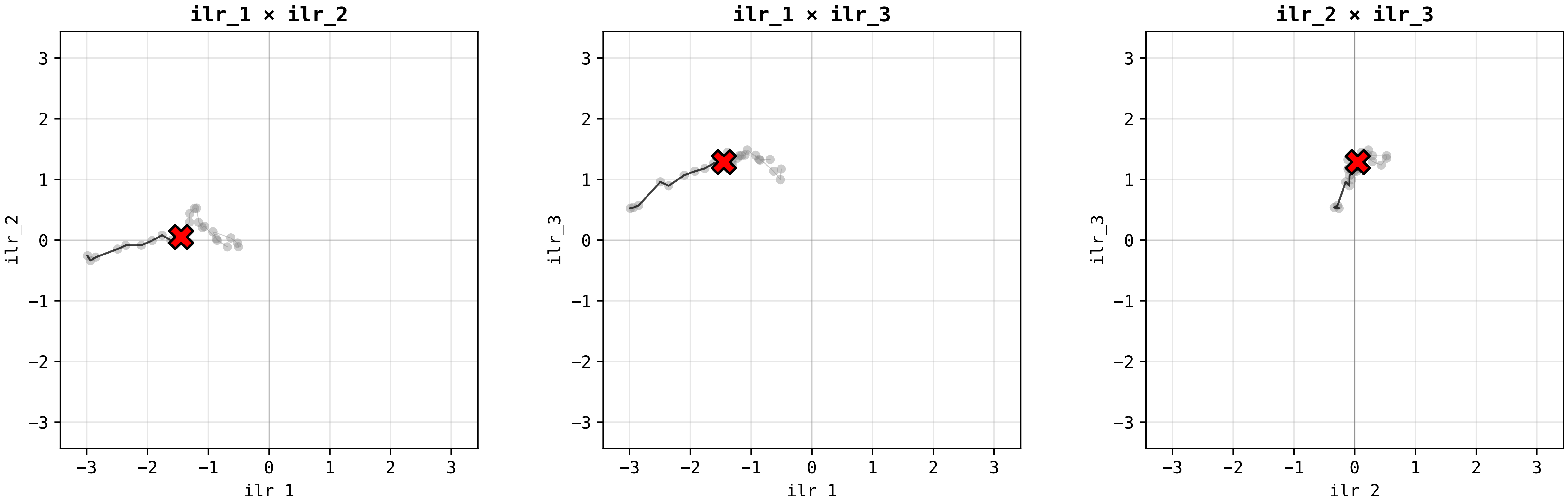
X = current year (2008) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2009 (10/26)



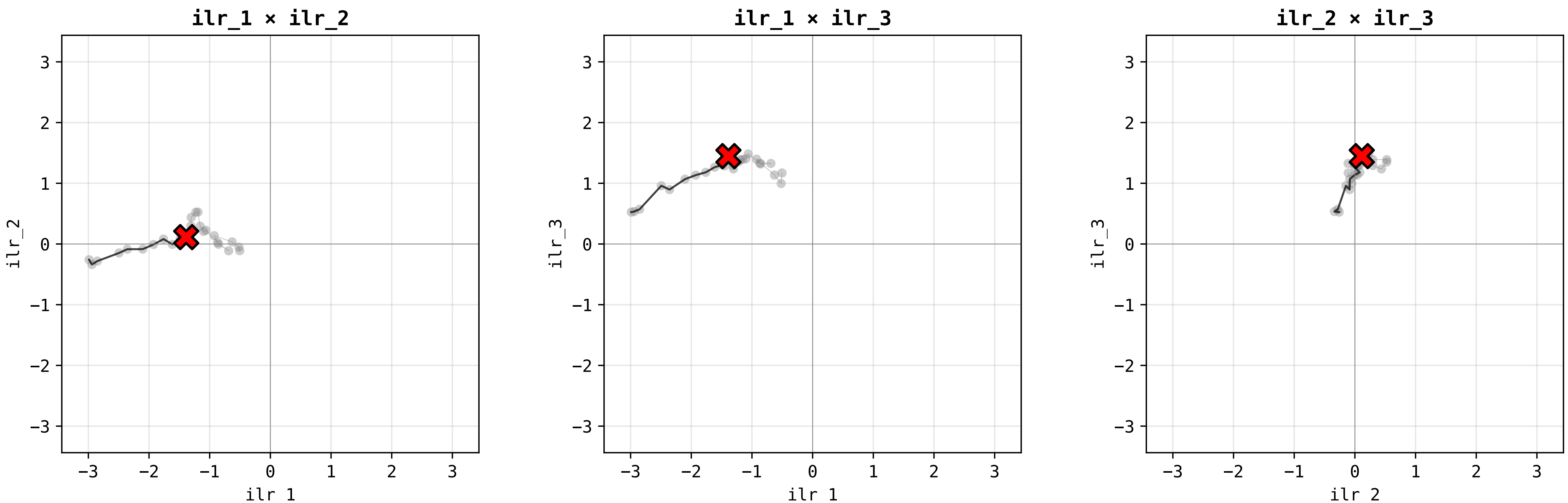
X = current year (2009) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2010 (11/26)



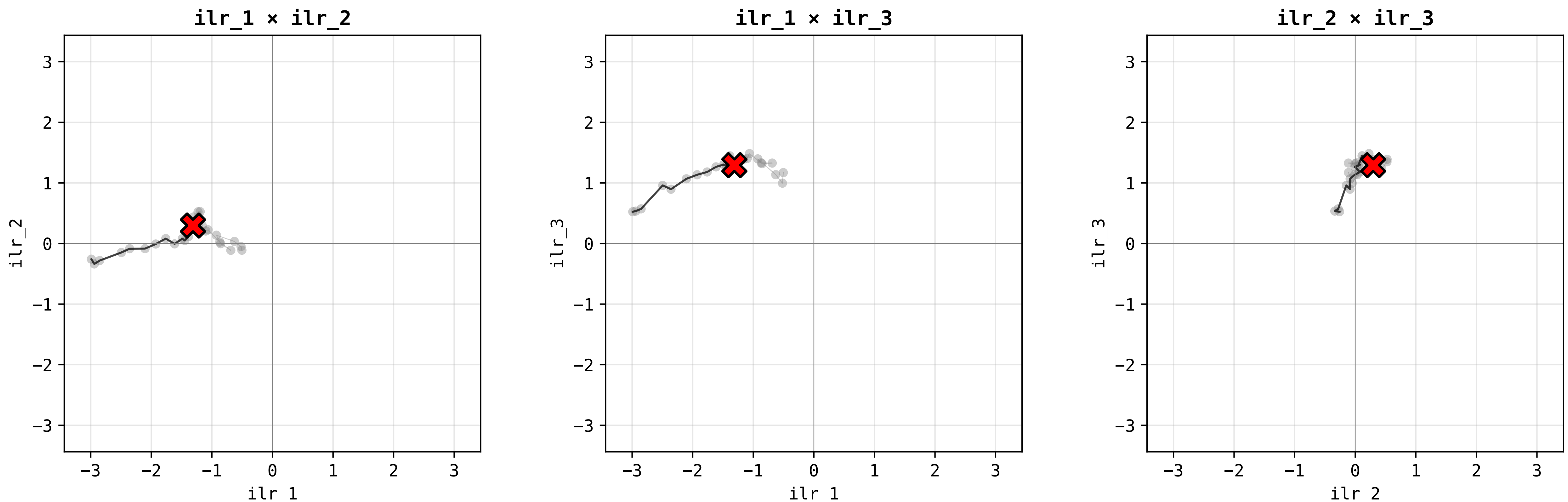
X = current year (2010) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2011 (12/26)



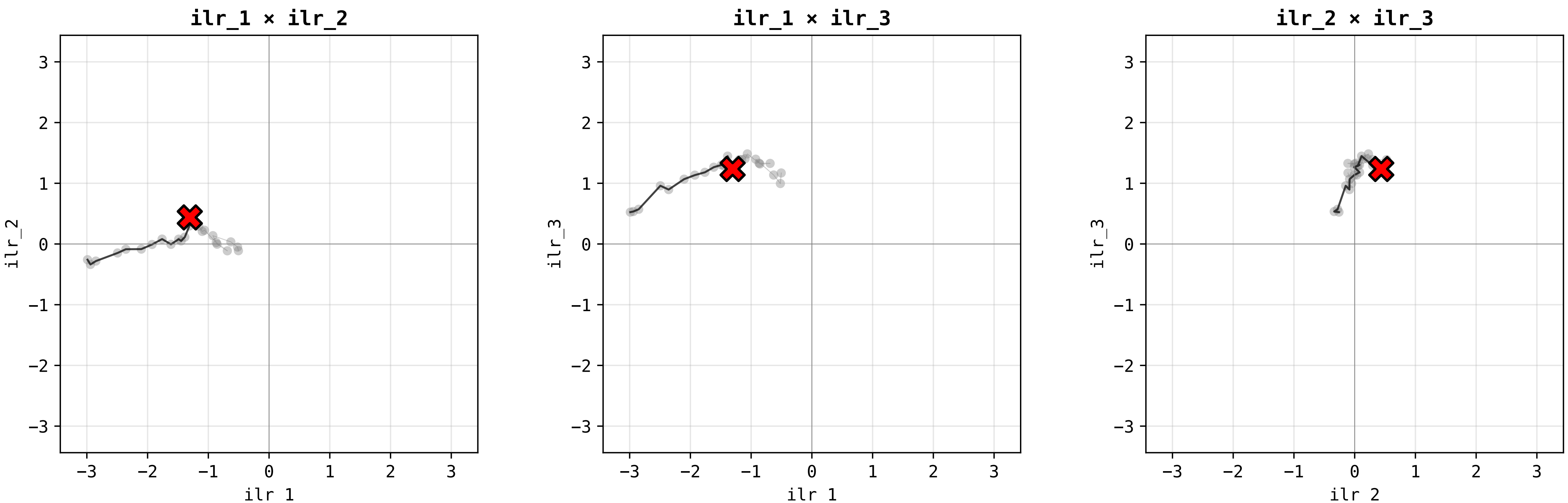
X = current year (2011) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2012 (13/26)



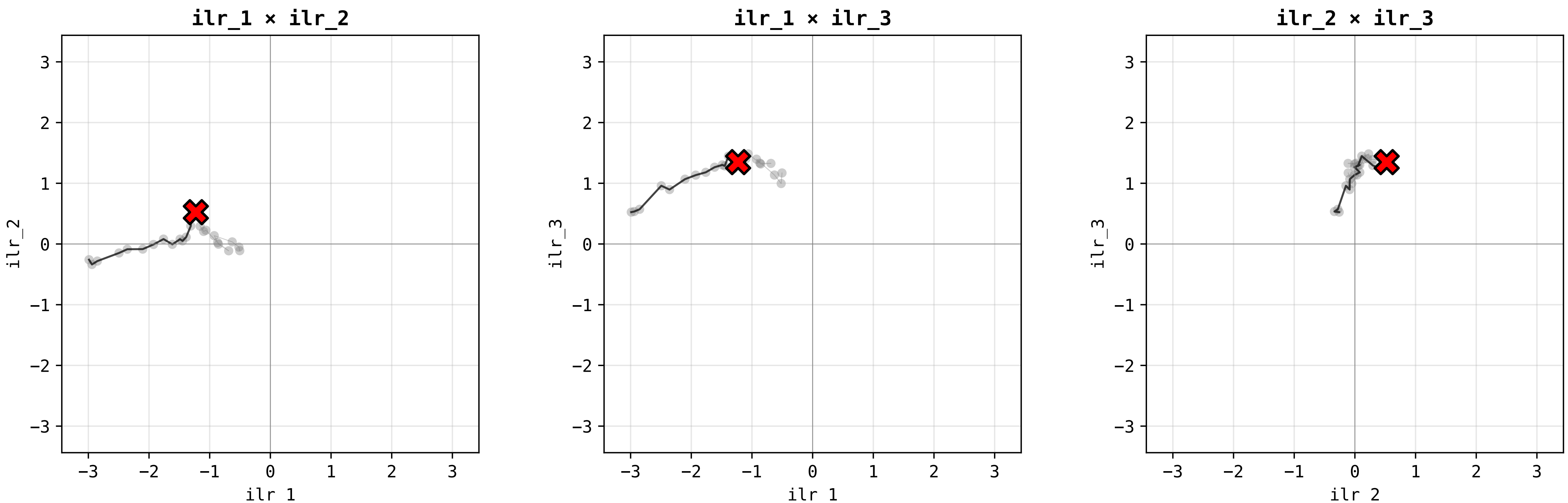
X = current year (2012) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2013 (14/26)



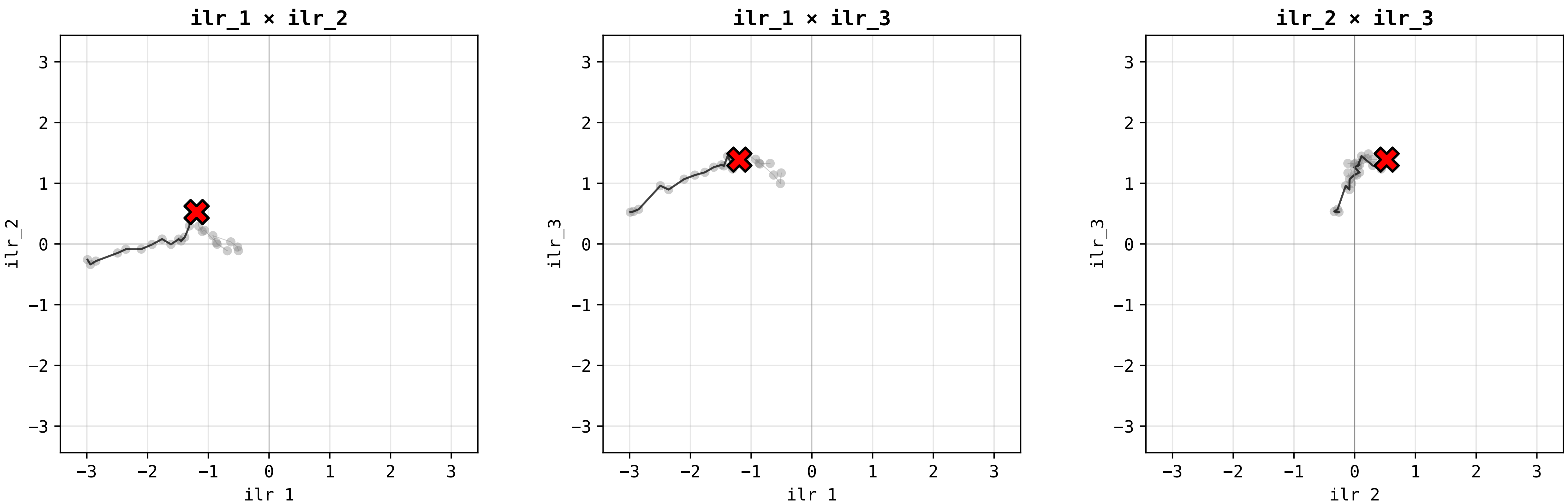
X = current year (2013) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2014 (15/26)



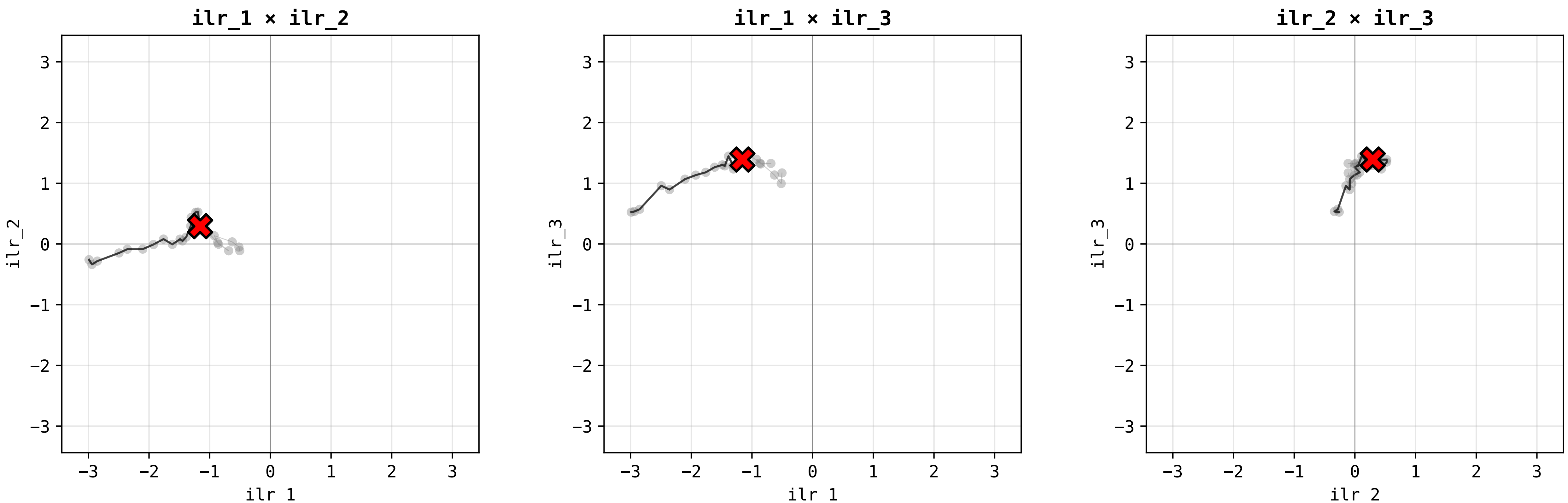
X = current year (2014) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2015 (16/26)



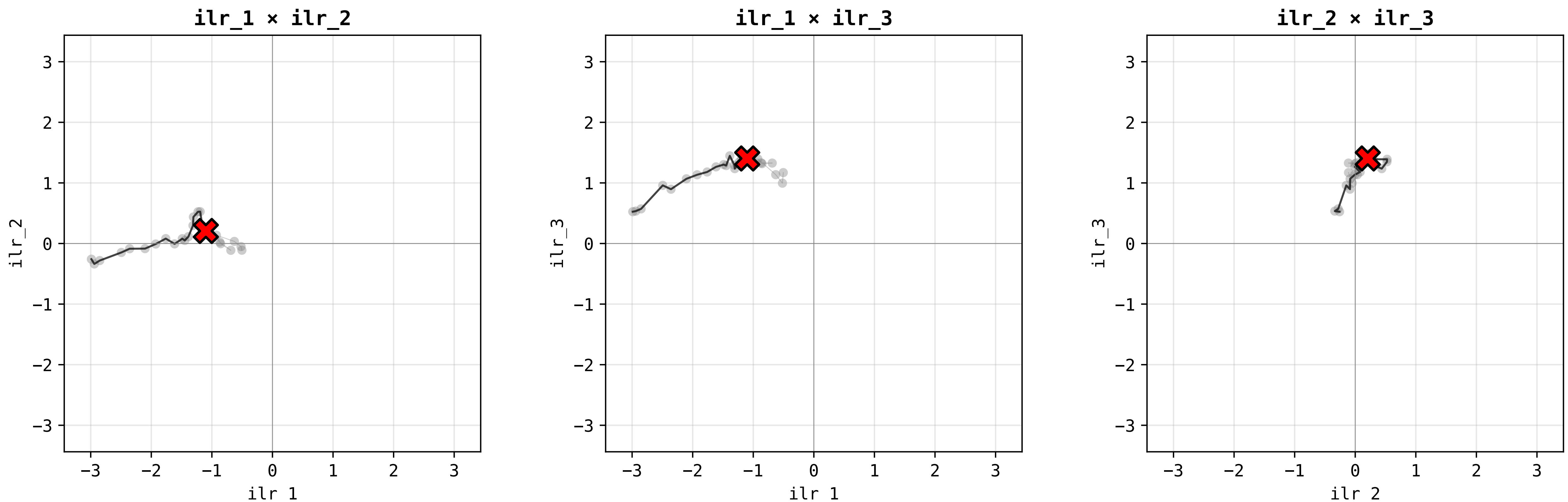
X = current year (2015) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2016 (17/26)



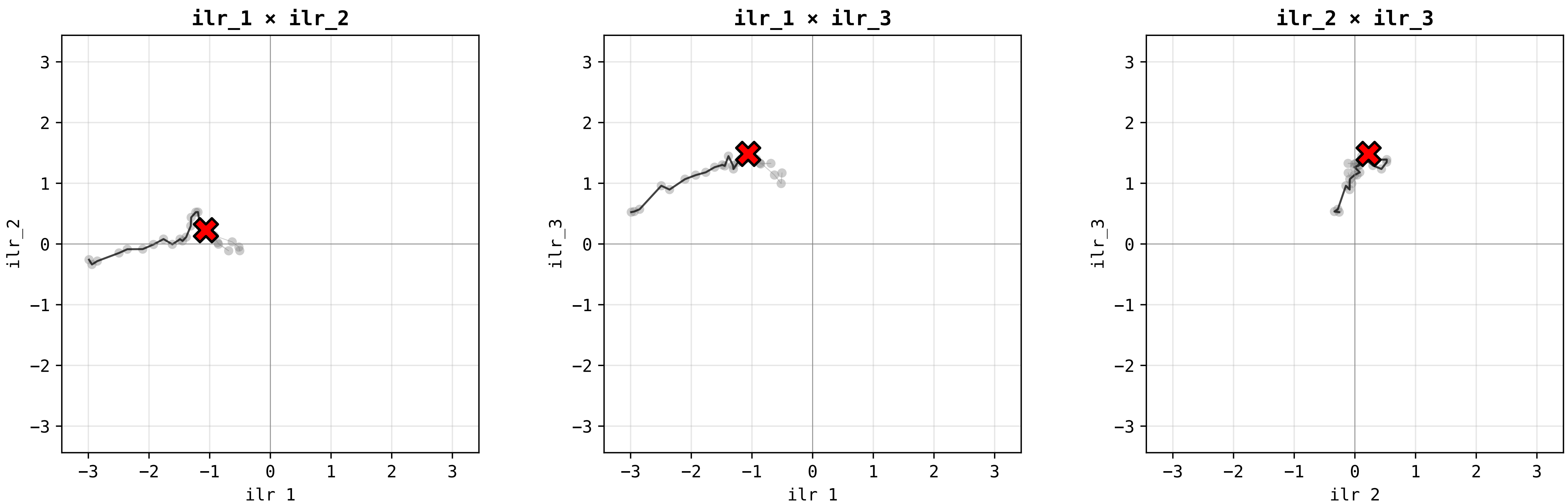
X = current year (2016) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2017 (18/26)



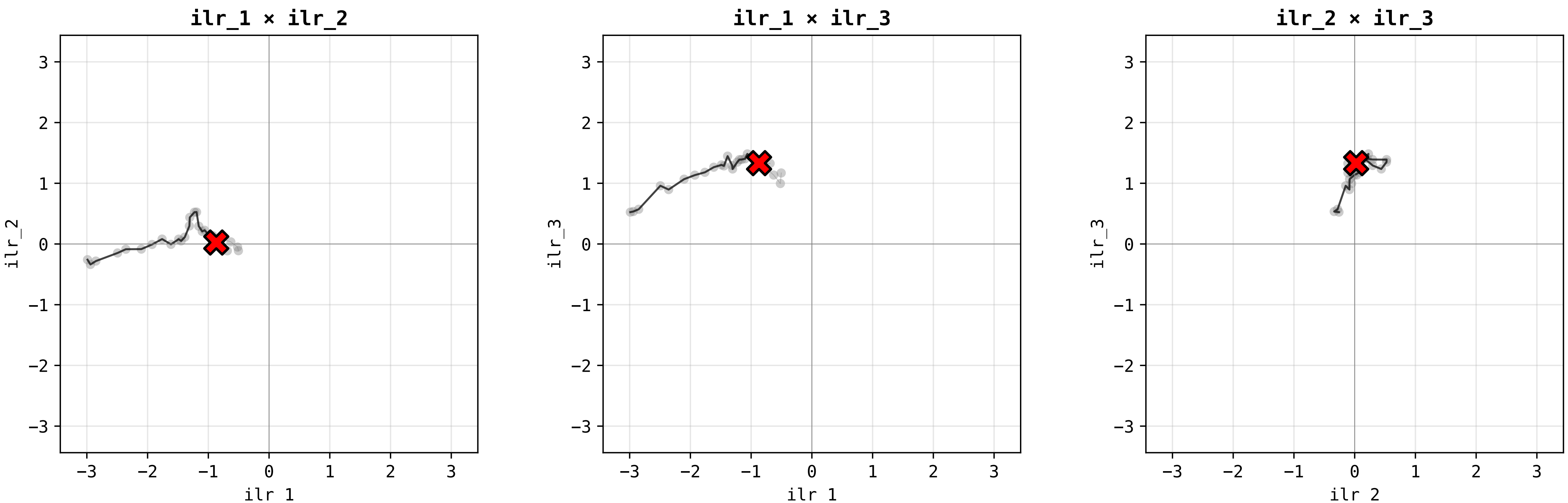
X = current year (2017) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2018 (19/26)



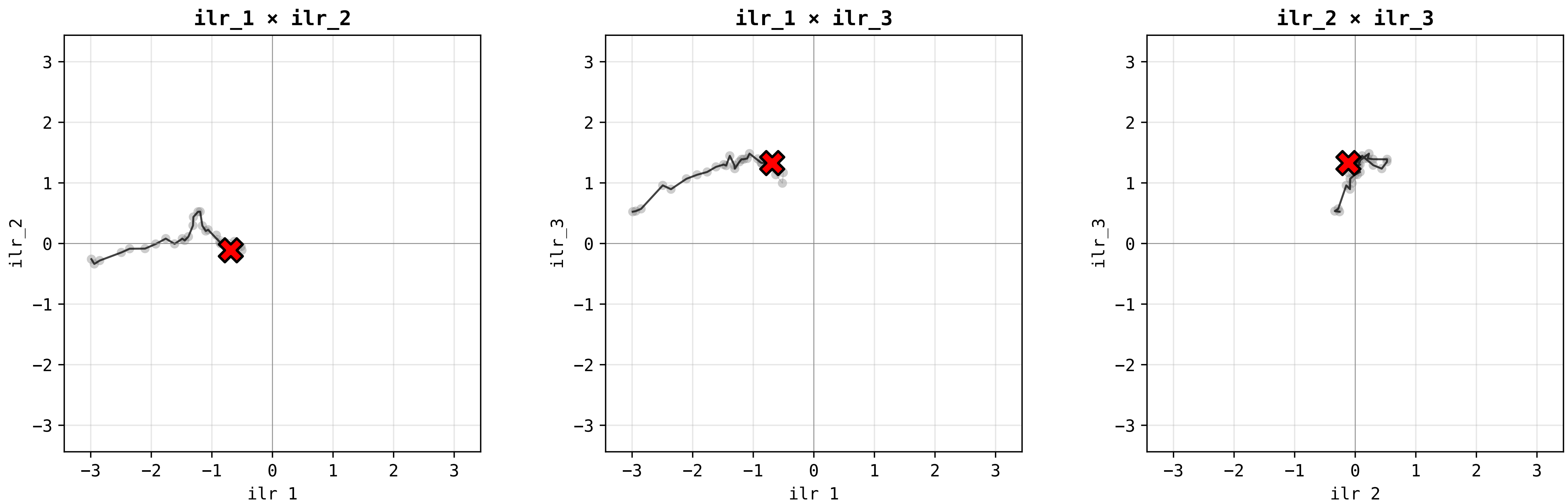
X = current year (2018) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2019 (20/26)



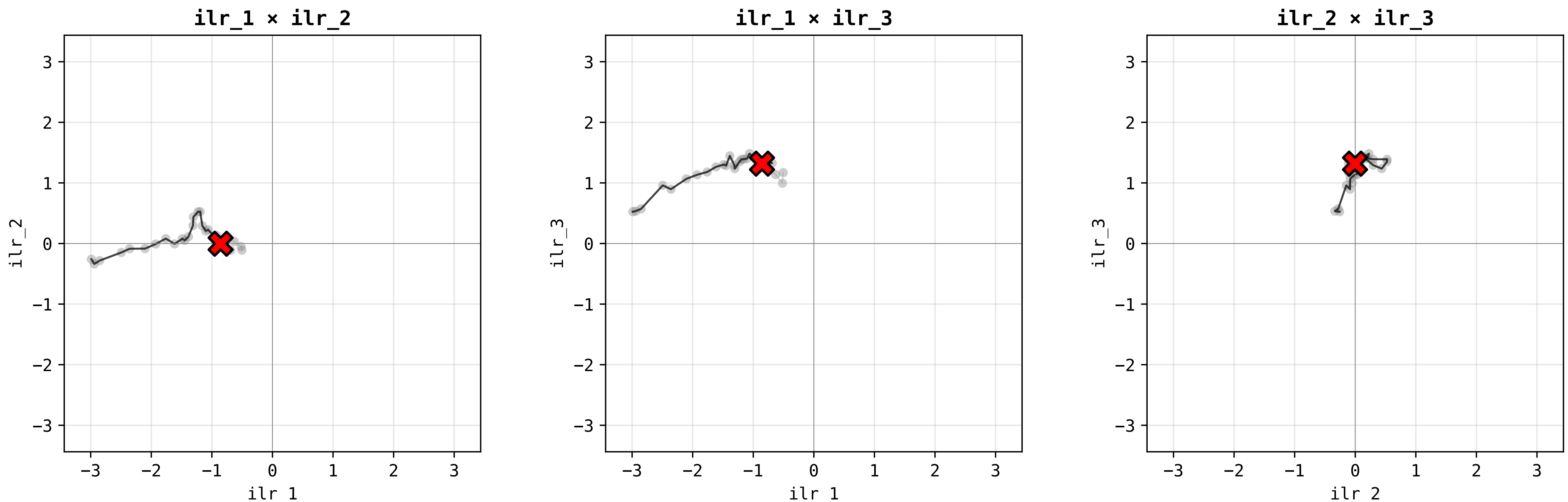
X = current year (2019) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2020 (21/26)



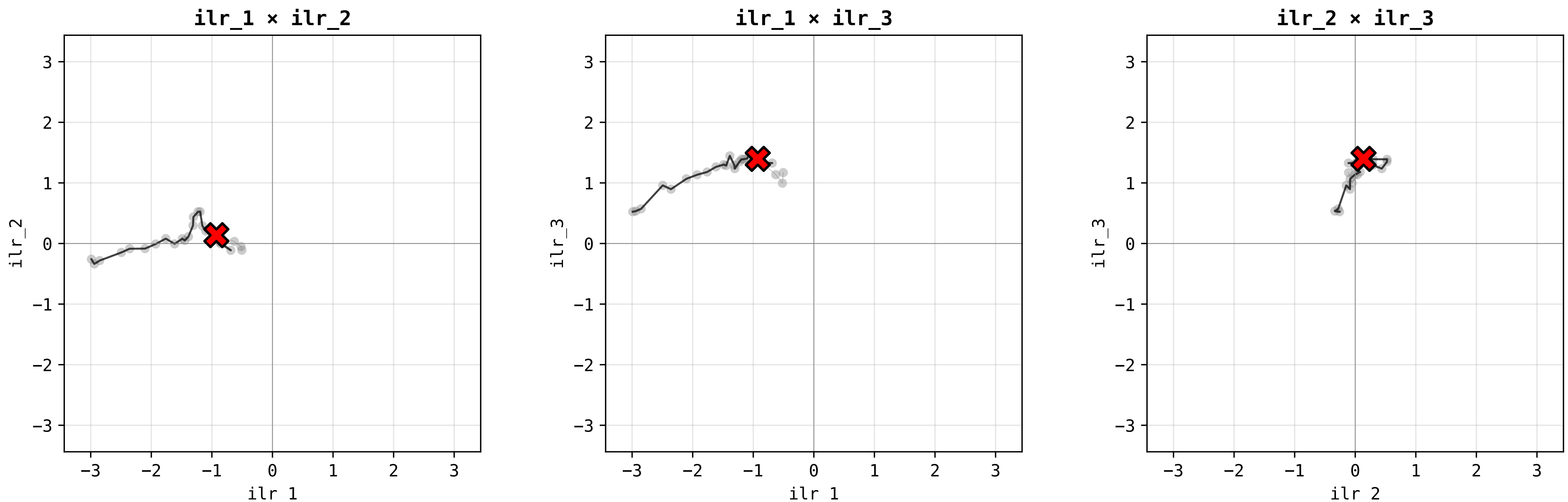
X = current year (2020) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2021 (22/26)



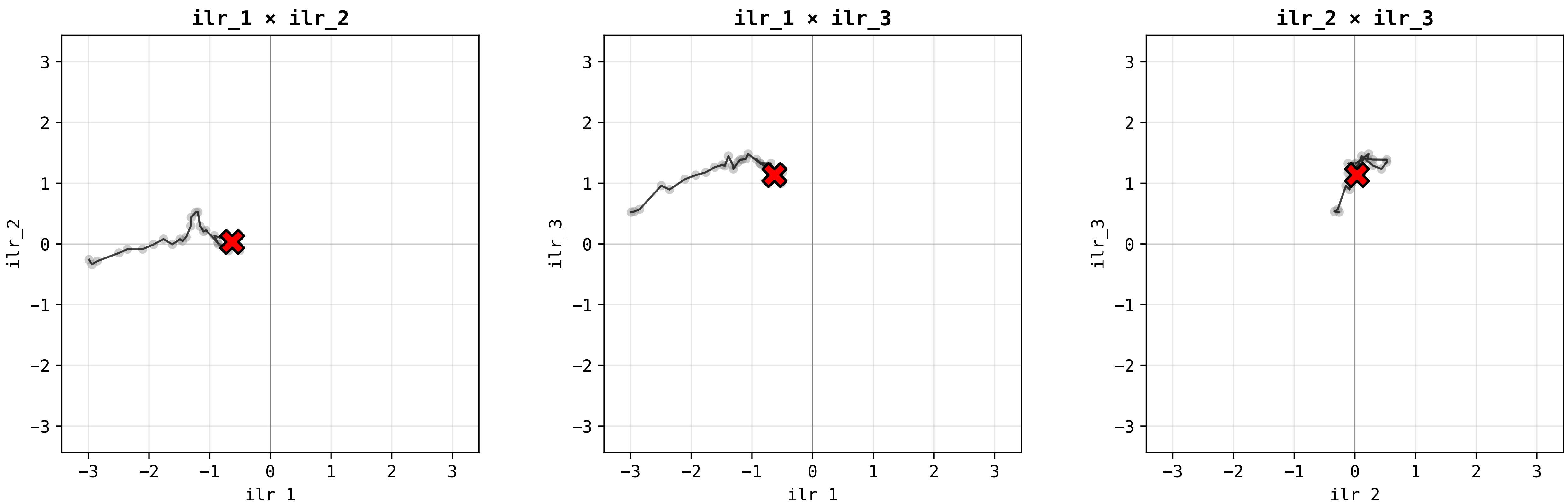
X = current year (2021) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2022 (23/26)



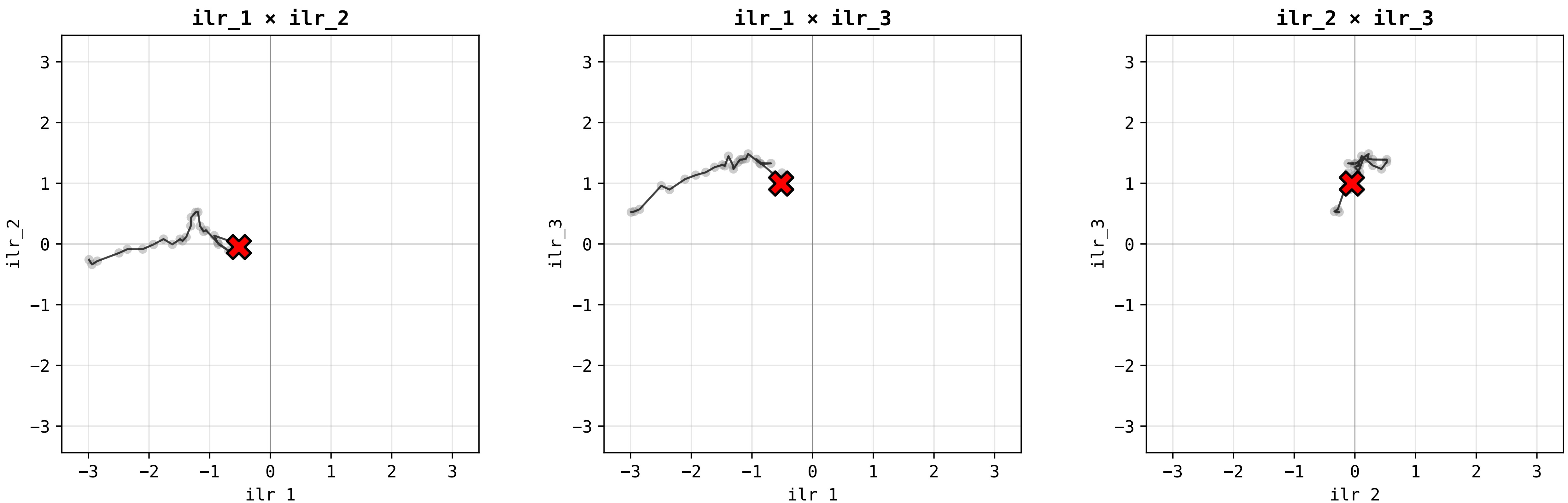
X = current year (2022) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2023 (24/26)



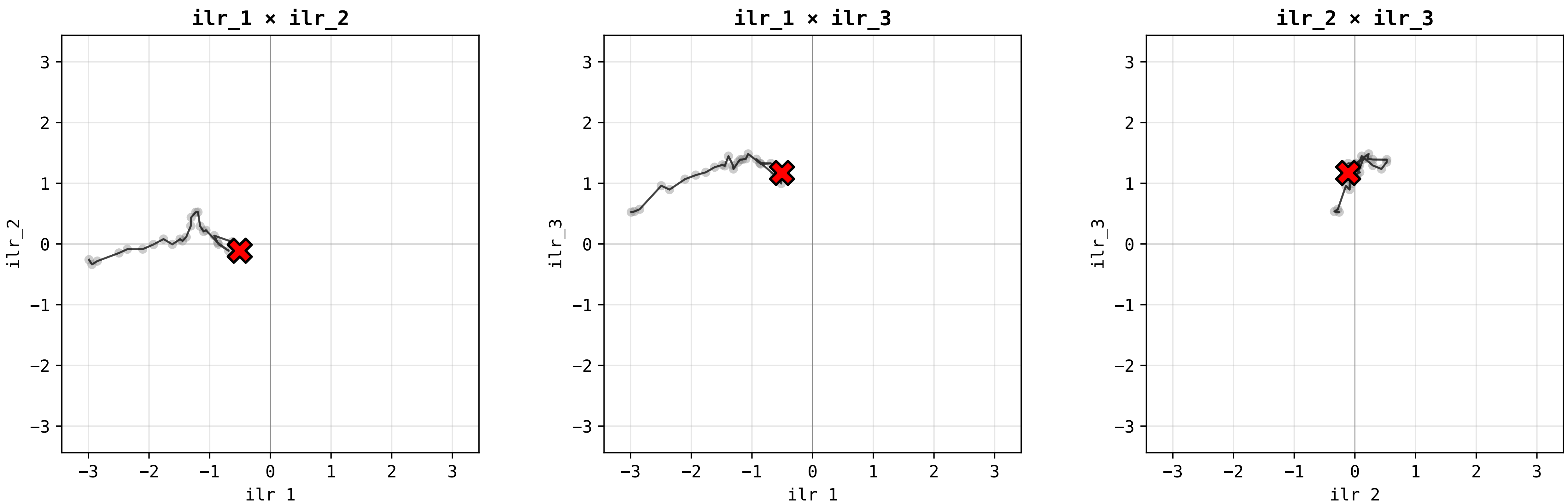
X = current year (2023) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2024 (25/26)



X = current year (2024) · black = past trajectory · gray = future · ilr range ±3.44

ILR Triplet — DEU Germany · year 2025 (26/26)



X = current year (2025) · black = past trajectory · gray = future · ilr range ±3.44